



Irregular Verb Errors Among International Students – A Study Based on the ICLE Corpus and Experimental Data

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To my professors, who fundamentally changed my perception of linguistics and shaped my
vision throughout my academic journey

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“Human relations are like the irregular verbs in a number of languages where nearly all verbs are irregular.”— Søren Kierkegaard

Abstract

This study offers a data-driven approach to the morphological instability and error patterns of irregular verbs among international learners of English. Based on written learner production data from the International Corpus of Learner English (ICLE) and supported by error analysis theory, the study utilizes Multiple Correspondence Analysis (MCA) to map these patterns through R Studio. It treats morphological acquisition as a complex system that needs to be tested and visualized through actual use.

Results reveal a clear trend in which irregular verb errors cluster around specific triggers, such as overgeneralization and stem-internal vowel changes, depending on the learner's profile. The integration of a multi-methods methodology, incorporating two questionnaires targeting past-tense usage in multi-format and spontaneous writing production formats, conducted with 22 undergraduate students from the L1 Anglais department at Université Bordeaux Montaigne, suggests that these tools may identify common challenges and highlights the discrepancy between theoretical recognition and active production.

While learner variables such as L1 background, age, and years of English at school help define the error patterns, the research shows that task type is a decisive factor, ranging from mixed-format to spontaneous written production, highly triggers the accuracy of irregular forms.

In conclusion, this study aims to provide a clearer picture of morphological development and also presents a replicable framework for exploring how forms shift in learner interlanguage, which can be further tested, extended, or challenged by future research.

Key Words: Irregular verbs, Error Analysis, R Studio, ICLE Corpus, Multiple Correspondence Analysis (MCA), Controlled Testing, Unguided Testing, ESL Learner, SLA Acquisition

Résumé

Cette étude propose une approche basée sur les données pour examiner l'instabilité morphologique et les schémas d'erreurs liés aux verbes irréguliers anglais chez les apprenants internationaux. S'appuyant sur les productions écrites du International Corpus of Learner English (ICLE) et sur la théorie de l'analyse des erreurs, l'étude utilise l'Analyse des Correspondances Multiples (ACM) pour cartographier ces schémas via R Studio. Elle traite l'acquisition morphologique comme un système complexe qui nécessite d'être testé et visualisé à travers l'usage réel.

Les résultats révèlent une tendance claire selon laquelle les erreurs de verbes irréguliers se regroupent autour de déclencheurs spécifiques, tels que la surgénéralisation et les changements vocaliques internes, selon le profil de l'apprenant. L'intégration d'une méthodologie multi-méthodes, comprenant deux questionnaires ciblant l'usage du passé sous des formats multi-choix et de production écrite spontanée, menée auprès de 22 étudiants de licence du département L1 Anglais de l'Université Bordeaux Montaigne, suggère que ces outils peuvent identifier des défis communs et met en évidence l'écart entre la reconnaissance théorique et la production active.

Alors que les variables liées à l'apprenant, telles que la langue maternelle (L1), l'âge et les années d'apprentissage de l'anglais à l'école, aident à définir les schémas d'erreurs, la recherche montre que le type de tâche est un facteur décisif ; allant d'un format mixte à une production écrite spontanée, celui-ci influence fortement l'exactitude des formes irrégulières.

En conclusion, cette étude vise à fournir une image plus claire du développement morphologique et présente également un cadre reproductible pour explorer la manière dont les formes évoluent dans l'interlangue de l'apprenant, ce qui pourra être testé, étendu ou remis en question par des recherches futures.

Mots-clés: Verbes irréguliers, Analyse d'erreurs, R Studio, Corpus ICLE, Analyse des Correspondances Multiples (ACM), Tests contrôlés, Tests non guidés, Apprenant d'anglais langue seconde (ESL), Acquisition des langues secondes (SLA).

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1. Introduction

The acquisition of English irregular verbs is one of the most difficult challenges in the field of Second Language Acquisition (SLA), as it remains a frequent cause of errors even among learners who have achieved native-like levels of lexical and syntactic proficiency. Unlike the predictable regularity of the dental suffix (-ed), the irregular verb system is a complicated mix of old language forms, based on internal vowel gradations (Ablaut), suppletion, and zero-marking. Within the learner's cognitive system, this separation leads to a constant struggle between two distinct memory systems: declarative memory, which is responsible for the storage and retrieval of special verb forms like *brought* or *went*, and procedural memory, which handles the regular grammar rules automatically.

In spontaneous language production, this struggle often leads to grammatical regression. Although advanced learners at the B2 and C1 levels of the Common European Framework of Reference for Languages (CEFR) frequently show a strong declarative knowledge of irregular patterns in guided, recognition-based tasks, they return to making common rule-based errors, such as *goed* or *arised*, when the cognitive load of spontaneous production is too high. This issue suggests that "advanced proficiency" should be defined not by their basic grammatical knowledge, but through morphological resilience: the capacity to maintain accuracy when dealing with irregular structures under processing pressure. By examining how these errors occur through both corpus-based and experimental approaches, the present study seeks to clarify the variables, typological, pedagogical, and cognitive, that affect the fossilization process within the advanced learner's interlanguage.

1.1. Background of the Study

The study of learner morphology has changed a lot from early behaviorist perspectives, which saw errors only as undesirable patterns to be eliminated. The cognitivist approach, introduced by Chomsky and later improved by Corder's seminal distinction between random mistakes and systematic errors, redefined learner errors as reflections of the internal cognitive structures of the language acquisition process. This shift prepared the ground for Interlanguage Theory, which claims that learners develop their own systematic linguistic system that exists on a continuum between the native language (L1) and the target language (L2).

Modern morphological research is no longer limited to just counting errors, but uses detailed cognitive theories, most importantly Ullman's (2015) Declarative/Procedural (DP) model and Pinker's (1999) Words-and-Rules theory. These models suggest that irregular verb acquisition is not a single, simple process, but a process based on frequency of input, where the stored memory of a form, supported by input frequency, must compete with the general "add -ed" rule. To observe these processes in real-life writing, researchers have started to use extensive learner corpora like the International Corpus of Learner English (ICLE), which provides data that

reflect real-world use by documenting the spontaneous written production of university-level students across different L1 backgrounds.

While the ICLE corpus provides a macro-analytic view of linguistic "outputs," this alone might be insufficient. Thus, it also needs a closer look supported by a micro-analytic perspective into how students recall these forms. To test this, we conducted two experimental tests within the 2LLIAM222 MATI - ORAL course at Université Bordeaux Montaigne. By analyzing how different tasks, ranging from recognition-based tasks to spontaneous production, affect performance, the research looks at the underlying cognitive strain, bridging the gap between written corpus data and real-time language performance.

1.2. Problem Statement

The main challenge for many advanced learners is the fossilization plateau. This refers to a stage where a student's progress stops, and specific errors become a permanent part of their speech and writing, despite continued instruction. The data from our research shows that even students who have studied English for 11 to 14 years still struggle with the overregularization and auxiliary selection errors during spontaneous production.

This occurs because current teaching methods prioritize building declarative knowledge, helping students memorize that the past form of "sing" is "sang", but they often struggle to support the development of procedural fluency. When students are under the "stress test" of spontaneous production, their brains tend to take a cognitive shortcut. They choose the easiest path, which is applying the regular rule *-ed* suffix, because retrieving the correct irregular form is too mentally demanding. This gap between classroom knowledge and real-world performance shows that even "advanced" students lack the resilience they need for spontaneous language use.

1.3. Research Questions

To examine this discrepancy between declarative knowledge and procedural production, this study focuses on the following research questions:

- Which learner profiles and verb types are most likely to show morphological instability during active production?
- How do cross-linguistic differences (Romance vs. Germanic L1) influence error strategies, such as auxiliary selection or participle forms?
- How do morphological classes (Strong, Weak, Suppletive, Invariable) interact with learner backgrounds to cause systematic challenges like Ablaut or suffixation errors?

- How does the transition from guided recognition (Test 1) to spontaneous production (Test 2) trigger morphological overgeneralizations?
- To what extent do processing limitations lead to accuracy trade-offs in real-time recall?
- Does morphological regularization persist as a stable feature of the learner's interlanguage despite extended exposure?
- Which language families demonstrate specific tendencies toward irregular verb errors?

1.4. Significance of the Study

The significance of this research offers a new way to measure advanced proficiency in English Language Teaching (ELT). By identifying areas where auxiliary and participle use fails under pressure, this research provides a clear guide for more effective teaching. It suggests focusing less on memorizing and more on tasks using "pushed output" to help students use language automatically.

The study strengthens research by integrating corpus and experimental data, offering a more complete view of how native language influence and cognitive limits affect learning. Finally, developing "morphological resilience" through Task-Based Language Teaching (TBLT) enables advanced learners to go beyond the fossilization plateau and achieve accuracy in spontaneous production in real-world settings.

1.5. Scopes and Delimitations

The scope of this research specifically targets university-level advanced learners of English, within the B2 to C1 levels of the CEFR. The research combines data from two main sources: the ICLEv3 corpus, consisting of argumentative essays from international students, and original experimental data collected from the 2LLIAM222 MATI-ORAL course at Université Bordeaux Montaigne. While the ICLE data offers a global cross-linguistic representation, including learners from 27 different countries, the micro-experimental section is focused on a specific participant group of 22 students, predominantly composed of native French speakers with some bilingual profiles.

The linguistic focus is strictly limited to the morphology of irregular past tense and past perfect forms, categorized into four operational verb types: Strong, Weak, Invariable, and Suppletive verbs. Errors are analyzed based on three main error type categories: overgeneralized -ed, auxiliary error, and incorrect participle. The study acknowledges the tests were timed (under 10 minutes), aiming to see how students use language naturally, rather than how they analyze grammar rules. Although the ICLE provides a massive dataset, this study excludes positive/negative interrogative forms due to technical extraction limitations within the interface.

2. Literature Review

This chapter provides the theoretical framework for investigating the morphological challenges encountered by L2 learners. It examines the development of Error Analysis, the cognitive mechanisms underlying interlanguage, and the structural properties of English irregular verbs. It also addresses the role of cross-linguistic influence, based on ICLE (version 3)¹ data and questionnaires conducted to students at Bordeaux Montaigne University, focusing on learners with diverse linguistic backgrounds, with particular analysis on Romance and Germanic L1 groups.

2.1. Second Language Acquisition and Error Analysis

The development of Second Language Acquisition (SLA) as a scientific field is closely related to changes in the perception and analysis of learner errors. Early approaches (Lado, 1957) considered errors to be negative phenomena that should be eliminated. In contrast, more recent perspectives consider them as an integral part of the language learning process. To understand the current framework of Error Analysis (EA) therefore requires examining the shift from behaviorist to cognitivist approaches, which redefined the role of the learner's internal processes.

In the mid-20th century, Behaviorism, represented by B. F. Skinner (1957), became dominant in theories of language acquisition. Through this perspective, the language learning process was seen as a habit formation based on stimulus, response, and reinforcement. Errors were interpreted as the result of incorrect habits or interference from the learner's first language (L1), and were seen as undesirable deviations that needed to be corrected as soon as possible.

Noam Chomsky (1959) later disagreed with this idea, stating that language can't be broken down into a set of learned behaviours; instead, it is controlled by internal cognitive structures. His suggestion of Universal Grammar changed the focus to the mental processes that happen when people learn a language. In this context, learner errors were viewed as evidence of rule formation rather than a simple failure. For example, the form of the verb "to go" as *goed* shows that the learner is actively building a linguistic system by using a productive past tense rule on an irregular verb.

Based on this cognitive perspective, Stephen Pit Corder (1974) was a key figure in the development of Error Analysis as a field of study. He highlighted that errors can provide insight a lot about how well the learner is learning the language. He made a clear distinction between mistakes and errors. **Mistakes** are related to performance and happen randomly, whereas **errors** are systematic and reveal how well the learner knows the target language. This distinction is particularly relevant in corpus-based research, in which the identification of common error patterns enables researchers to infer conclusions about the stages of language development.

¹ The International Corpus of Learner English (ICLE), version 3.3, available at <https://corpora.uclouvain.be/cecl/icle/>, contains 9,882 texts (5,990,101 words) and includes expanded sub-corpora for several L1 backgrounds, including the Romance and Germanic groups analyzed in this study.

Error Analysis has been standardized through methodological procedures that enable systematic investigation of a learner language. As described by Corder (1974) and further discussed by Ellis (1999), this methodology includes the gathering of learner data, the identification and categorization of errors, the explanation of their essential causes, and the evaluation of their effects on communication. This framework is especially important for studies utilizing learner corpora, like the ICLE corpus, as it guarantees consistency in the analysis of linguistic deviations. This methodology makes it possible not only to identify forms such as *chosed*, but also to interpret them in terms of rule-based processes such as overgeneralization in the context of English irregular verbs.

Another important part of Error Analysis is identifying the source of learner errors. As argued by Ellis (1999) and Corder (1974) that errors can be classified into two main groups: *interlingual* and *intralingual*. Interlingual errors result from the influence of the learner's native language, while intralingual errors are derived from the natural complexity of the target language. When it comes to English irregular verbs, intralingual processes such as overgeneralization are frequent, as learners apply regular rules to irregular forms. However, interlingual influence may also be a factor, depending on the learner's language background. This difference is determinant to this research, as it facilitates a more accurate interpretation of the error patterns found in the data.

Error Analysis, while offering valuable contributions, demonstrates certain limitations. It may miss correct forms if it only looks at learner errors. Correct forms can also give useful information about how people learn a language. Additionally, it does not consistently involve the dynamic and evolving characteristics of the learner language. These limitations have resulted in the formulation of more expansive theoretical frameworks, particularly interlanguage theory, which aims to characterize learner language as a systematic and evolving system in its own right. This point of view provides a more comprehensive account of the variability and instability seen in learner production and serves as the basis for the following section.

2.2. Theoretical Explanations for Irregular Verb Acquisition and Morphological Errors

The transition from the descriptive constraints of Error Analysis to a more explanatory and holistic understanding of learner language was marked by the emergence of several cognitive and social frameworks. Rather than viewing learner production as a mere collection of defective forms, modern Second Language Acquisition (SLA) research seeks to uncover the underlying mental mechanisms that govern these outputs. This section situates the morphological instability of English irregular verbs within seven major theoretical pillars, providing a wider perspective on why learners struggle to stabilize these forms.

2.2.1. Interlanguage Theory: The Learner's Independent System

The cornerstone of modern SLA, Interlanguage Theory, introduced by Larry Selinker (1972), redefined learner language as a legitimate, rule-governed linguistic system. Selinker argued that a learner's output is not a broken version of the target language, but a separate grammar, an interlanguage, that exists between the native language (L1) and the target language (L2). This system is driven by five central psycholinguistic processes: language transfer, transfer of training, strategies of L2 learning, strategies of L2 communication, and overgeneralization of target language material.

In the specific study of English irregular verbs, the interlanguage serves as a transitional stage where the learner's internal rules (such as the default application of the *-ed* suffix) compete with the idiosyncratic exceptions of the English language. This theory is vital for this research because it treats the variability found in the ICLE corpus not as failure, but as evidence of a learner's current development hypothesis regarding English morphology.

The internal systematicity of the interlanguage is clearly demonstrated when learners from distinct linguistic backgrounds produce identical non-target forms. To better understand, here are some examples found in the ICLE corpus regarding the overgeneralization of the verb **"arise"**:

(1) *The so-called professional army controversy has arised once again in the Spanish presidential campaign.* (SPM04055, L1: Spanish)

(2) *Political parties based only on environmental questions, have arised in countries all over the world.* (SWUL4022, L1: Swedish)

(3) *This situation has arised many polemics, especially by anti-abortion groups and conservative people who deny virgins the right to have babies.* (ITTO3030, L1: Italian)

The repeated use of the verb form **"arised"** by Spanish, Swedish, and Italian learners suggests that this isn't just a one-time error. Instead, it shows a common stage in the development of interlanguage where learners prioritize the productive regular rule of *-ed* suffixation over the idiosyncratic irregular form, *arisen*. This provides strong evidence for Selinker's (1972) claim that interlanguage operates as a rule-governed system; learners consistently utilize a logical, though non-target, morphological hypothesis to deal with the complexity of English irregular verbs.

2.2.2. The Dual-Route Model: Cognitive Competition of Words and Rules

Steven Pinker's (1999) Dual-Route Model provides an extremely important framework for explaining the neuro-cognitive explanations behind morphological errors. Pinker claims that the

human brain processes language via two distinct pathways: a combinatorial rule-based system that generates *Verb + -ed* and a declarative associative memory system for irregular forms (the lexicon).

Morphological instability occurs when there is a retrieval failure in the associative memory. If a learner cannot quickly find the irregular form, like *brought*, the procedural system wins the competition and applies the default rule. This may explain why even advanced learners produce errors like *runned* when their brains are working hard or under pressure. When the memory lookup for the irregular form is interrupted or weak, the brain falls back on the default rule.

Evidence of this cognitive defaulting is clearly visible in the following examples from the ICLE corpus²:

(4) *Obviously, Huck looks up to him ("He knowed how to do everything") and is impressed by the complicated but exciting plans he comes up with, e.g. the escape plan for Jim.* (DNNI4012, L1: Dutch)

(5) *In this way the child is leaved alone, does not have anyone that follow his growth and advise him on what is right or wrong.* (ITTO1034, L1: Italian)

(6) *But the land Japan took during the war was returned because Japan losed in the war.* (JPKO1007, L1: Japanese)

These errors are consistent with the predictions of the Dual-Route Model. In each instance, the students correctly identified the past tense category but failed the memory lookup for the specific irregular form (*knew*, *left*, *lost*). Since these are high-frequency verbs, it is unlikely that the learners don't know the right forms. Instead, the "rules" (procedural application) surpassed the "words" (memory retrieval) because of processing pressure. In the context of this study, this competition functions as a determinant variable for the Multiple Correspondence Analysis (MCA). Thus, the study aims to distinguish between rule-based defaults and lexical failures to identify which learner profiles and verb types are more likely to show morphological instability during active production (Li, Hiver, & Papi, 2022).

2.2.3. Fossilization and the Plateau of Stabilization

Fossilization is a major issue in language learning at the university level. Selinker (1972) and later Han (2004) characterize this phenomenon as the process in which specific linguistic

² The ICLE database uses a standardized coding system for file names to ensure privacy and provide categorization. The first two letters indicate the learner's native language/sub-corpus (e.g. JP for Japanese, FR for French), the next two letters represent the specific institution or sub-group, and the final digits serve as the unique identification number for each individual essay.

features stop evolving and become permanently embedded in the learner's interlanguage, regardless of further instruction (Li, Hiver, & Papi, 2022). In the case of recurring irregular verbs, fossilization is common because the incorrect forms, such as *drinked*, are communicatively sufficient, which reduces the internal pressure to revise the learner's mental representation. This leads to a developmental plateau where the morphological distinction between the past simple (V2) and the past participle (V3) often collapses. The following examples from the ICLE corpus provide evidence of this stabilization:

(7) *Other things, when I was a junior high school student, we were gave an assignment from my classroom teacher.* (JPSW1016, L1: Japanese)

(8) *Unfortunately, some people have took advantage of TV's influence on people's mind and use it in order to pass certain messages, often unacceptable ones, to the viewers.* (GRAR1019, L1: Greek)

(9) *According to this ideology, people do not need to know what is happening in the world about politic aspects of their government and their corrupt politicians, about conflicts between nations and what had been did to solve them.* (BRMG1276, L1: Portuguese)

(10) *Therefore, it has caused discussion and polemics because the censorship has became a cause to restriction of a human right.* (SPM03017, L1: Spanish)

The learners in these examples demonstrate a high level of lexical and syntactic complexity, but they consistently apply the wrong V3 form. The use of "**did**" as a past participle in a passive construction (example 9) or "**became**" in a perfect tense context, the example 10 indicates that the morphological distinction between V2 and V3 has merged into a singular, fossilized form. For these learners, the incorrect form has become a permanent part of their independent linguistic system, creating a plateau that persists though they have had a lot of academic exposure to English.

2.2.4. U-Shaped Learning: The Paradox of Progress

Morphological development is rarely a linear progression; it often follows a **U-Shaped trajectory** (Gass & Selinker, 2008; Williams et al., 2022). In Phase 1, students say the correct irregulars (like V2 *went*) without knowing the rule behind it. In Phase 2, when they learn the regular *-ed* rule, they regress and start to overgeneralize (for example, *goed* and *eated*). Finally, in Phase 3, they return to being accurate by making the general rule fit with certain lexical exceptions.

When university-level data has too many general errors, it usually means that the learner is undergoing this cognitive restructuring. Conversely, a non-target form such as *goed* or *buyed*

signifies a greater degree of rule-consciousness than a flawlessly imitated *went* as it demonstrates the active application of a grammatical hypothesis. The following examples from the ICLE corpus illustrate this creative backsliding phase:

(11) *In my opinion, this position should n't be taked in consideration.* (SPM04056, L1: Spanish)

(12) *So, some of them take the "Madrugadão" bus or they go walking to home and they can be assaulted, stealed or even dead.* (BRMG1294, L1: Portuguese)

(13) *These changes have also been accomplished and speeded up by organizations such as for example Greenpeace.* (SWUL4012, L1: Swedish)

(14) *People should not be forced to stay at home at night simply because they fear to be shooted at in the streets.* (ITTO2005, L1: Italian)

The appearance of forms such as "taked," "stealed," and "shooted" in the corpus signifies that the learners have progressed beyond simple imitation and are actively working to organize their morphological knowledge. From a U-Shaped Learning viewpoint, these errors don't mean failure; on the contrary, they mean that the interlanguage is going through a restructuring process. The learner shows that they are no longer treating each verb as a separate lexical item by adding the productive *-ed* suffix to verbs like "take" or "steal". Instead, they are trying to fit them into a coherent, rule-governed grammatical system. This paradox of progress is important for figuring out why morphological instability often remains the same or even increases at certain points in L2 development.

2.2.5. L1 Transfer: The Cross-Linguistic Influence

Language transfer, also known as cross-linguistic influence, is one of the factors shaping interlanguage patterns (Odlin, 2003). It refers to the way previously learned languages influence the acquisition of a new one, either through similarities or differences. For English irregular verbs, this influence can support learning through positive transfer or lead to difficulties through negative transfer, often described as interference (Odlin, 1989).

For many learners, what is considered as **irregular** in English is understood in relation to their first language. As Odlin (2003) points out, learners with Germanic L1s, such as Dutch or Swedish, may find English irregular verbs easier to acquire because these languages share historical roots and similar vowel-change patterns. In contrast, learners with Romance L1s, such as French, Spanish, or Italian, often face more difficulty, as their languages use different morphological markers and auxiliary structures to express the past tense. This difference in linguistic distance helps explain why certain types of errors, such as missing auxiliaries or

incorrect participle forms, are more common among learners from particular language backgrounds.

(15) *It is one of these movies where the suspension is build up and you don't even dare to go to the toilet in between because you are afraid you might miss an important part.*

(GEAU1087, L1: German)

(16) *Many things can influence the children life, many aspects contribute for to be a individual but the education always have be priority because without this the human beings are not.* (BRSS1331, L1: Portuguese)

(17) *As a general rule, volunteer soldiers are people that have few takings prospects in civil life, while military service is did by all.* (SPM04026, L1: Spanish)

(18) *Some people received their date of execution, were lead to the execution room and were finally taken back to their cell again.* (GESA4011, L1: German)

(19) *Very often we see family meetings which are hold by the fire of a TV set.* (SPM03018, L1: Spanish)

The distribution of these errors supports the role of cross-linguistic distance, especially for learners with Romance L1s. These learners often show instability in auxiliary–participle structures, producing forms such as “**have be**” or using the simple past instead of the past participle (e.g., “**is did**”). This suggests that their native language systems, which rely on different ways of marking tense and aspect, interfere with the English V2/V3 distinction. In many cases, learners fail to link the auxiliary verb to the correct irregular form, resulting in forms that are understandable but incorrect.

By contrast, learners with Germanic L1s show a different type of difficulty. Although German and English share structural similarities, these learners still produce non-target forms, often using the base form in passive or perfect contexts (e.g., “**is build**”, “**were lead**”). This suggests that similarities between the two languages do not always lead to accuracy; instead, learners may overuse the base form or assume it is enough in more complex tenses. These patterns indicate that L1 background is a significant predictor of the morphological challenges a learner will encounter when producing irregular verb forms.

2.2.6. The Input Hypothesis: Frequency and the “i+1” Gap

Stephen Krashen's (1985) Input Hypothesis states that people learn a language when they are exposed to “comprehensible input” that is just a little bit above their current level of skill. This is called **i+1**. When it comes to irregular verbs, input frequency is very important. High-frequency

irregular verbs (e.g., went, said, came) are often learned early and correctly because they show up a lot in the input, giving the learner a lot of chances to reinforce their mental images (Bybee, 2001).

However, many morphological errors at the university level can be linked to an inadequate exposure to lower-frequency irregular verbs. When a learner comes across or tries to use a verb that hasn't been reinforced with enough "i+1" input, they experience a gap in their interlanguage. To bridge this gap, the internal language processor uses the most useful rule by default: the regular *-ed* suffix. So, errors like *shrinked* or *flinged* don't always refer to lack of grammar knowledge. Instead, they could mean that the specific irregular form has not yet been converted from "input" to "intake" due to low exposure frequency.

(20) *The whole world was "shrunked" and we began to know each other better.*
(GRAR1007, L1: Greek)

(21) *Third, writing becomes a challenge when the time comes to decide, whether it is meaningful enough or complete to be "flinged to the public".* (LTVI1052, L1: Russian)

(22) *Revolutions are already coming when "the rest of the society", believe that their rights have been broken out, or blowed away.* (SPM04038, L1: Spanish)

The appearance of non-target forms such as "shrunked," "flinged," and "blowed" offers solid evidence for the frequency effect within the Input Hypothesis. Unlike highly frequent verbs, these particular lexical items (shrink, fling, blow) occur less frequently in standard pedagogical input, meaning that learners may not have acquired the necessary "i+1" exposure to stabilize the irregular forms (shrank/shrunk, flung/flung, blew/blown). Therefore, even advanced level university students are prone to overgeneralization as a compensatory mechanism. This shows that morphological accuracy is not solely dependent on rule acquisition, but is strongly influenced by the statistical frequency of the input encountered during the learner's development.

2.2.7. The Interaction Hypothesis: Noticing the Gap

Michael Long's Interaction Hypothesis (1996) discusses how important social interaction is for a second language acquisition. According to this perspective, language development is easier when learners engage in communication and negotiate meaning, especially when there are misunderstandings. During such interactions, learners are often exposed to feedback, such as recasts or clarification requests, which can help them modify their output.

Another concept related to this is the **Noticing Hypothesis**, proposed by Richard Schmidt (1990), which claims that "**noticing**" is a necessary condition for converting input into intake. This cognitive process allows learners to consciously attend to specific linguistic forms in the target language. Within this framework, learners may experience "**noticing the gap**," where they

recognize a mismatch between their current interlanguage and the target language. For instance, when a learner produces a form such as *catched* or *teached* and receives corrective feedback, this conscious awareness helps them perceive the linguistic deviation and adjust their understanding of the verb form.

However, the importance of the Interaction Hypothesis is more difficult to observe directly in written learner corpora like ICLE, where there are no chances for real-time interaction and feedback. Therefore, learners may continue to produce non-target forms without being encouraged to notice the gap between their production and the target form. In this sense, the persistence of such errors in corpus data may reflect the absence of interactive feedback rather than a lack of underlying linguistic competence. The subsequent examples highlight morphological errors that occur in written production, likely due to a lack of destabilizing feedback:

(23) pay → *payed*

He doesn't seem to be pleased anymore with being payed with words; he does want more for the different tasks he has carried out. (FRUC2023, L1: French)

(24) hear → *heard*

In this case claims of the people can be heard in the street against the death penalty, and others claiming revenge on the other side. (SPM04025, L1: Spanish)

(25) teach → *teached*

So, this theory should be taught very well because based on it the new professionals are going to follow their lives. (BRSM1404, L1: Portuguese)

Learner productions such as *payed* and *heard*, observed in the ICLE corpus, indicate that overgeneralized forms are still common in written work. These forms reflect rule-based processing in the learner's interlanguage system. However, the lack of interactional feedback in written contexts may facilitate their persistence as learners may not be encouraged to notice the difference between their production and the target form. This indicates that the stabilization of irregular verb morphology relies not only on exposure or rule acquisition but also on the availability of interactional feedback, which enables the restructuring of learners' interlanguage system.

2.3. Structural Categories of English Irregular Verbs

The structural layout of the irregular verbs is a major factor in the developmental path of Second Language Acquisition (SLA) for verb acquisition. Regular verbs follow a productive procedural rule, the addition of the suffix *-ed* to the base of the verb. Irregular verbs are stored as lexical exceptions. But linguistic research shows that these exceptions are not a mass of chaotic forms,

but are paradigmatically structured. In his research, Pinker (1999) argues that the human brain tries to find patterns of regularity even in irregular systems in order to improve memory retrieval.

A strict taxonomic baseline must be built in order to analyze the morphological instability found in learner corpora. Thus, this section proceeds from the general linguistic category of irregular verbs to the specific structural paradigms used as variables in the results of this study.

2.3.1. Quirk's Seven Classes

English irregular verbs are classified into seven classes based on the framework provided by Quirk et al. (1985). The taxonomy they use relies on the "Pattern of Identity", which describes the relationship between the base form (V1), the past simple (V2) and the past participle (V3). Quirk et al. list these classes, from total identity of all parts to complex distinctions of both vowel gradation (ablaut) and suffixation summarized in Table 1.

Table 1. Linguistic Classification of Irregular Verb Forms (Adapted from Quirk et al., 1985).

Class	Morphologic Characteristics	V1/V2/V3 Forms
1	V2 is identical with V3; identical vowels; optional suffixation.	<i>learn – learnt/learned – learnt/learned</i>
2	V3 has two suffixes; no change of base vowel for V2.	<i>mow – mowed – mown</i>
3	V2 is identical with V3; change of base vowel; suffixation used.	<i>mean – meant – meant</i>
4	V2 and V3 are different; range of vowel changes; -en suffix in V3.	<i>break – broke – broken</i>
5	All three parts are identical; no change of base vowel.	<i>cut – cut – cut</i>
6	V2 is identical with V3; no suffixation; change of base vowel.	<i>meet – met – met</i>
7	V2 and V3 are different; no suffixation; change of base vowel.	<i>swim – swam – swum</i>

Table 1 shows that the morphological complexity of irregular verbs increases from Class 5 to Class 7. The highest degree of formal identity is Class 5, since there are no inflectional markers to differentiate a tense or aspect. Classes 4 and 7, on the other hand, are the most complex paradigms and require the extensive internal modification of the verb stem, commonly called Ablaut (vowel gradation). It is emphasized that these seven classes are a very accurate

linguistic classification but they are not equal in terms of frequency of membership; some classes have a lot of common verbs like Class 1 and 3, while others like Class 2 are more limited (Quirk et al., 1985).

Cognitively this seven-class division points out the different degrees of memory retrieval required. Learners must differentiate between verbs that involve suffixation (Classes 1, 2, 3, 4) and those that depend only on internal changes (Classes 6 and 7). Such a detailed linguistic mapping constitutes the necessary basis to identify specific structural points of failure in L2 interlanguage development. To fill the gap between this theoretical classification and the practical needs of the corpus analysis, the following section reorganizes these classes into operational frameworks, according to their patterns of identity.

2.3.2. Operational Categorization for the Present Study

The structural complexity of English irregular verbs, as typically outlined in Quirk et al.'s (1985) seven-class model, needs a more simplified categorization for the purposes of actual error analysis. While Quirk's classification is comprehensive from a linguistic point of view, it often causes difficulties for corpus-based research, due to the possible absence of data in less populated classes. Moreover, the teaching and cognitive acquisition of irregular verbs often relies on the perceived structural symmetry of the verb's principal parts, rather than on its historical class.

Therefore, irregular verbs are grouped into four distinct paradigms based on their formal patterns: **No Change (A-A-A)**, **One Change (A-B-B)**, **Two Change (A-B-C / A-B-A)**, and the **-en Ending Paradigm**. This approach can be considered in terms of Cognitive Load Theory (Sweller, 1988)³ because the number of allomorphic forms associated with a verb may increase the cognitive demands on the learners during the retrieval. By grouping verbs into these operational clusters, this study aims to find out whether specific structural patterns can be used to predict specific error types in L2 written production. The operational model is detailed in Table 2, which provides a summary of the structural patterns used in the following analysis in relation to Quirk's classes.

Table 2. Structural Paradigms and Predicted Error Patterns

Category	Quirk's Class	Pattern	V1-V2-V3
No Change :	Class 5	A-A-A	<i>cut-cut-cut</i>
One Change :	Class 1, 3, 6	A-B-B	<i>leave-left-left</i>

³ Cognitive Load Theory, proposed by Sweller (1988), suggests that because working memory is limited, the quantity of mental effort needed to complete a task affects instructional design and linguistic processing. In the case of irregular verbs, high cognitive load is experienced during the simultaneous retrieval of non-productive morphological forms and complex syntactic structures, which may lead to performance errors in L2 learners.

Two Change :	Class 2, 4, 7	A-B-C / A-B-A	<i>sing-sang-sung</i>
-en Ending :	Class 2, 4	Mixed	<i>take-took-taken</i>

Table 2 is the main taxonomic reference for the qualitative and quantitative analysis of this study. By changing the focus from historical classification to pattern-based clustering, the framework enables a more precise investigation of the “Interlanguage” (Selinker, 1972) developed by advanced L2 learners.

Table 2 indicates a clear hierarchy of morphological difficulty based on the number of forms that a learner has to memorize. The categories of “No Change” and “One Change” are more “economical” in a cognitive sense in that the learner only has to deal with one or two different forms for each verb (e.g. cut-cut or leave-left). On the other hand, the “Two Change” and “-en Ending” categories are more challenging, requiring a tripartite distinction where all three forms (V1, V2 and V3) could be different.

In the Competition Model (MacWhinney, 1987), the cue validity and the frequency of linguistic structures determine the detection and retrieval of correct forms by the learner. When the cognitive cost of retrieving a complex irregular form is too high, the brain often employs a “**winner-take-all**” approach, allowing a more productive analogic process to win. This leads to predictable deviations: for example, learners may overregularize complex morphological systems (e.g. adding *-ed* to irregular stems). On the other hand, this cognitive “**shortcut**” resembles paradigm levelling, where a previously complex tripartite⁴ verbal system is reduced to a more efficient bipartite⁵ system, minimizing processing competition (Hock, 1991).

2.3.2.1. No Change Verbs (A-A-A)

One of the particular types of morphological phenomena is the No Change paradigm, which is equivalent to Quirk et al. (1985)’s **Class 5**, where V1, V2 and V3 are phonologically and orthographically identical. The main representatives of this category are high frequency verbs, such as *cut*, *hit*, *cast*, *put* and *cost*. The A-A-A pattern has been seen as highly unstable in different formats, although it might be thought to be easier to acquire because of its formal economy. In No Change verbs, the past tense cue is a “zero-morpheme”⁶ creating a “cognitive vacuum.” The expected error of overregularization often comes from a communicative need on

⁴ While Hock (1991) focuses on the “elimination of alternations,” I use the term “tripartite” here to refer to irregular verbs that traditionally maintain three distinct principal parts (e.g., break-broke-broken).

⁵ “Bipartite” refers to the reduction of a verb paradigm to two distinct forms (e.g., keep-kept-kept). This represents the structural outcome of the “winner-take-all” strategy described in the Competition Model (MacWhinney, 1987).

⁶ In morphology, a zero morph is a placeholder used to account for homophonous forms in a paradigm that carry grammatical meaning without a change in form (Bauer, 1988). Pinker (1999) describes this as a “**silent bit of nothing, -∅,**” which marks the past tense in “no-change” verbs like *cut* or *hit*.

the part of learners to explicitly mark the past tense, according to the **default mechanism** of the **words-and-rules theory** (Pinker, 1999)⁷.

This development strategy persists, which is evident in the preliminary results of the two experimental questionnaires⁸ conducted for the purposes of this study. Non-target forms such as *cutted* and *puted* appear even in advanced learners in written production. These results suggest that the cognitive power of the regular /-ed/ suffix remains a strong force in the absence of a visible marker from the irregular paradigm. The data from the questionnaires, and ICLE Corpus, show a secondary level of instability in the auxiliary verbs. The use of unnecessary or incorrect markers such as *had put* or *didn't cut* in simple past contexts suggests that the lack of formal change in the verb itself may lead to confusion in the learner's syntactic choices.

Another specific issue in this group is the verb *read*, which is a challenging one across three levels:

- Phonologically (/ri:d/ to /red/)
- Categorically One Change (A-B-B) verb
- Orthographically an A-A-A verb.

The conflict between the "visible" and the "audible" often leads learners to produce *readed*, as a way of addressing the phonological shift and the written regularity. As will be shown in the experimental data of results, **No Change** verbs are therefore not only simple forms, but rather a result of intense competition between the lexical memory and the learner's tendency towards consistency in the target language.

2.3.2.2. One Change Verbs (A-B-B)

The category (A-B-B) is the most numerically dominant in the English irregular system, grouping Quirk et al.'s (1985) **Classes 1, 3 and 6**. For these verbs, like *leave-left-left*, *keep-kept-kept*, *bring-brought-brought*, there is one morphological change, where the past simple (V2) and the past participle (V3) are the same. These verbs signal tense morphologically, unlike the No Change paradigm, but this is generally an internal vowel gradation (e.g. *meet* → *met* / *feed* → *fed*), or a combination of vowel shift and consonant change (e.g. *buy* → *bought* / *catch* → *caught*).

⁷ Pinker (1999) proposes the words-and-rules theory, which claims that the human brain processes language as a dual-system: a lexicon of memorized words and a grammar of combinatorial rules (Ellis, 2019). This theory is based on a default mechanism by which a productive rule (like adding *-ed*) is added to any item whose particular irregular form is not retrieved from memory. When the memory lookup fails for an irregular form, this default rule comes to the rescue to ensure the grammatical requirement of tense is met.

⁸ The two experimental questionnaires were conducted between March 2026 and April 2026 with three groups of total 22 students from Bordeaux Montaigne University enrolled in the 2LLIAM222 MATI - ORAL course during the 2025-2026 academic year. The questionnaires specifically targeted the morphological retrieval of irregular verbs under timed conditions to minimize the effects of explicit observation (Ellis, 2019).

The main cognitive challenge in this category is the conflict between the stability of the base stem and the retrieval of the irregular past form. Bybee's (2007) Frequency Effect implies that irregular verbs are stored as individual rote-learned units whose accessibility depends on their lexical strength. These stored forms must constantly compete with the highly productive regular rule of adding /-ed/. When a learner can't remember the specific irregular form from memory, the high type frequency of the regular rule leads the mental grammar to use the base stem instead. The data, collected from questionnaires, indicates that even when learners recognize the need for a past tense marker, they often fail to retrieve the internal phonological shift, resulting in overregularization errors like “*leaved*” or “*keeped*.” This suggests that the productivity of the regular pattern often suppresses the memory of the irregular form when that form's lexical representation is not sufficiently strengthened by frequency.

Within the synchronic taxonomy, verbs such as *sleep-slept-slept* and *feel-felt-felt* are classified as One-Change (A-B-B) verbs, characterized by formal identity between the past simple (V2) and the past participle (V3). These “**hybrid**” forms present a unique dual complexity because, unlike just internal vowel shifts or regular suffixation, they require both an internal stem modification (/i:/ → /e/) and the addition of a dental suffix (/t/).

For the L2 learner, this creates a “halfway irregularity” where the verb appears to follow a rule by ending in a dental sound but violates the principle of stem consistency, often leading to transition errors such as *sleeped* or double-marked forms like *felted*.

2.3.2.3. Two Change Verbs (A-B-C / A-B-A)

The Two Change category is a fusion of the complex structures in **Classes 2, 4, and 7** in Quirk et al. (1985). It represents the highest level of morphological difficulty within the English irregular system. These paradigms involve handling a tripartite distinction (A-B-C, e.g., *sing-sang-sung*) or a circular retrieval process (A-B-A, e.g., *become-became-become*). Unlike the former categories, the learner has to store and retrieve three different phonological representations for one lemma, resulting in a considerably higher cognitive load during real-time language production.

A major phenomenon in this category is Paradigm Leveling. Learners often attempt to reduce the three separate distinctions to a more manageable bipartite (A-B-B) system to reduce the load on their associative memory. This happens in the Incorrect Participle error, where the past simple (V2) form is overgeneralized as the past participle (V3). For example, a learner might produce non-target constructions such as *has sang* or *has wrote*, instead of the required tripartite forms. This is not, in fact, necessarily an overregularization error but rather a strategy of morphological economy, where the noticeable or recently activated irregular form is used in all past-time contexts.

The A-B-A pattern (e.g. *run-ran-run* or *come-came-come*) presents a unique circularity problem. In this case the learner has already moved on to a different form for the past simple and must revert to the base form for the participle. Results from the questionnaire suggest that this

reversal is often difficult for learners, often with the V2 form kept in the participle slots such as, *has became* or *has ran*. It is argued that the activation of a past-tense form in the mental lexicon can effectively block the retrieval of the identical base-participle form. Consequently, the Two Change paradigm is an important measure of the learner's ability to maintain complex morphological distinctions under syntactic pressure.

2.3.2.4. The "-en" Ending Paradigm

Unlike the purely internal changes discussed in previous sections, the “-en” Ending Paradigm requires two morphological changes:

- An internal vowel change in the stem
- An external suffix (e.g., speak-spoke-spoken or take-took-taken).

This category is particularly susceptible to a phenomenon known as Suffix Omission. The complexities of this tripartite system often lead learners to perform the required vowel change but not to attach the final nasal suffix (Anderwald, 2009). This results in the production of V2 forms in V3 places and in the production of non-target constructions such as “*has broke*” and “*has spoke*”.

This tendency to drop the suffix is frequently a matter of phonological efficiency. The Words-and-Rules theory assumes that the brain employs a “blocking principle” (Pinker, 1999): once a specific irregular form is retrieved from memory, it typically blocks the application of morphological rules. For a learner who has successfully changed the internal stem (e.g., *break* → *broke*), this retrieved V2 form may block the perceived need for further marking. Thus, the learner may interpret the past-time meaning as already adequately marked and view the -en suffix as redundant in their developing grammar. The results of my questionnaires reflect this situation, with a high occurrence of these “abandoned” forms in contexts of the perfect tense. As MacWhinney (1987) notes, the complexity of managing this tripartite system often leads to processing hesitation, causing learners to select auxiliaries inappropriately or omit them entirely while dealing with this complex tripartite system. Finally, the consistent use of the -en ending is a sign that the learner has moved from simple communication to advanced morphological accuracy.

2.4. Taxonomy of Irregular Verbs

While in the previous sections we have focused on the synchronic layout of irregular verb paradigms, a complete taxonomy requires a perspective that can uncover the systemic roots of these patterns. As Pinker (1999) observes, irregularity is not a monolithic category but a spectrum, or continuum, of productivity and generality. This continuum ranges from the highly productive regular -ed rule at one end to “total suppletion” (e.g., *go/went*) at the other, with

various families of systematic vowel shifts situated in between. As Finegan (2012) notes, these forms are not errors but **relics** of past regularities. Treating irregular verbs as a spectrum, rather than minimal exceptions, clarifies the historical and cognitive processes that shaped the modern system.

In order to build a strong theoretical framework, this section first looks at the historical evolution of irregular verbs, following their development from the seven strong classes of Old English, through the morphological loss and analogical levelling of Middle English, to their current equivalents in Modern English. It is important to understand this change in time as it shows how Germanic patterns were transformed into the fossilized irregularities of today over a millennium of language development.

Following this historical background, the next subsections aim to provide a detailed look at four categories of irregular verbs: the very systematic **Strong Verbs**, the **Weak Verbs**, and the unique **Suppletive** and **Invariable** forms. This classification sets the theoretical base for the error analysis in Chapter 4, highlighting the correlation between a verb's historical characteristics and the specific cognitive challenges it presents to L2 learners (Pinker & Ullman, 2002).

2.4.1. Historical Background of Irregular Verbs

The irregular verb system of modern English is not a collection of linguistic errors, but a fossilized copy of an old and highly structured Germanic verb architecture (Hogg & Denison, 2006). To understand the present state of irregularity, we must follow its development through three distinct phases: **Old English (c. 450–1100)**, **Middle English (c. 1100–1500)**, and the transition to **Modern English (1500–Present)**. From a diachronic perspective, this shows how a previously dominant system of vowel gradation was increasingly pushed to the margins by a more productive suffix-based morphology.

Old English was synthetic but significantly less complex than its Proto-Indo-European ancestor. The Germanic transition replaced complex aspectual categories with a binary tense opposition between the present and preterite. While Strong verbs kept Ablaut (internal vowel gradation) to indicate these tenses, the category of aspect became secondary to this new two-way tense system (Hogg & Fulk, 2011).

Unlike Modern English, which generally has one form for the past tense, like *wrote*, Old English strong verbs were characterized by a distinction between the preterite singular and the preterite plural vowels.

- For instance, the class 1 verb *drīfan*, → *drive*, had the singular past *drāf* and the past plural *drifon*.

Within this framework, the past participle (e.g., *drifen*) functioned as a deverbal adjective, specifying not just past time but a completed state.

The transition to Middle English (ME) after the Norman Conquest in 1066 triggered a massive destabilization of the verbal system (Baugh & Cable, 2013; Hogg & Denison, 2006). With the loss of English as a prestigious language and the rise of Anglo-Norman French, the formal pressures to maintain the complex seven-class strong system decreased. Two major changes occurred during this period:

- **Levelling of Inflections:** The vowel markers for singular and plural past forms began to merge (level), usually in favor of the singular vowel. This changed the ancient quadripartite⁹ paradigm to the tripartite system we know today (Three different forms of V1-V2-V3 such as *drive-drove-driven*).
- **The Rise of the Weak Conjugation:** The “Weak” system, utilizing a dental suffix (*/-de/* or */-te/*), was far more productive. The new verbs which came into the language from French and Norse were almost all of the weak patterns. Many strong verbs were “adopted” into the weak system (e.g. OE *helpan* became *helped*), a process known as **regularization**.

By the beginning of the Early Modern English period, the Great Vowel Shift was actively in process, basically changing the pronunciation of all Middle English long vowels (Nevalainen, 2006).

However, due to the fact that emerging printing press technology, introduced to England by William Caxton in 1476, had already begun to freeze orthographic conventions before these phonological changes were completed, a permanent divergence was created between spelling and sound (Baugh & Cable, 2013). This explains the “unorthodox” vowel symbols in modern English, where identical spellings often conceal varying pronunciations in irregular paradigms (Baugh & Cable, 2013). A main example of this is the V1-read (*/ri:d/*) and V2/V3-read (*/red/*) distinction. In this instance, the orthography remains a frozen record of the period before the shift affected the vowels in different parts of the verb paradigm differently (Anderwald, 2009).

The irregular verbs that are still around today are those with high **Token Frequency** (Bybee, 2007). These verbs like *go*, *be*, *see*, and *take* were in constant use in informal speech, their irregular forms became deeply ingrained in the collective memory, allowing them to invoke the “Conserving Effect” to resist the analogical levelling¹⁰ of the productive */-ed/* rule. Our modern world’s irregular verbs are real survivors of the Old English strong classes that were just too frequent to be forgotten.

⁹ While Baugh and Cable (2013) and Hogg and Denison (2006) refer to “four principal parts” I use “quadripartite” to describe the original Germanic strong verb system consisting of distinct forms for the infinitive, preterite singular, preterite plural, and past participle.

¹⁰ Analogical (or morpho-phonemic) levelling is a process in which morpho-phonemic alternations, which Bybee (1985) argues are often arbitrary from a synchronic perspective, are eliminated to align the verbal paradigm with its semantic structure. This typically results in a one-to-one correspondence between form and meaning.

The following tables show the transition of verb forms from the highly inflected Old English classes to the simplified Modern English irregular system.

Table 3a. The Diachronic Evolution of Class 1 Strong Verbs: *Vowel Gradation (Ablaut) – To Drive*

Period	Verb Class	Infinitive	Preterite Singular	Preterite Plural	Past Participle	Morphological Result
Old English	Class 1	<i>drīfan</i>	<i>drāf</i>	<i>drifon</i>	<i>drifen</i>	Quadripartite distinction
Middle English	Class 1	<i>driven</i>	<i>droof</i>	<i>droven</i>	<i>driven</i>	Leveling begins
Modern English	SV: Vowel Gradation	<i>drive</i>	<i>drove</i>	<i>drove</i>	<i>driven</i>	Tripartite (A-B-C)

Table 3b. The Diachronic Evolution of Class 3 Strong Verbs: *Full Vowel Change – To Fight*

Period	Verb Class	Infinitive	Preterite Singular	Preterite Plural	Past Participle	Morphological Result
Old English	Class 3	<i>feohtan</i>	<i>feaht</i>	<i>fuhton</i>	<i>fohten</i>	Quadripartite distinction
Middle English	Class 3	<i>fihten</i>	<i>faght / fought</i>	<i>foughten</i>	<i>foughten</i>	Radical Phonetic Shift
Modern English	SV: Full Vowel Change	<i>fight</i>	<i>fought</i>	<i>fought</i>	<i>fought</i>	One-Change (A-B-B)

Table 3c. The Diachronic Evolution of Class 2 Strong Verbs: *Weak Morphology – To Shoot*

Period	Verb Class	Infinitive	Preterite Singular	Preterite Plural	Past Participle	Morphological Result
Old English	Class 2	<i>scēotan</i>	<i>scēat</i>	<i>scuton</i>	<i>scoten</i>	Quadripartite distinction
Middle English	Class 2	<i>shoten</i>	<i>shotte</i>	<i>shotten</i>	<i>shotten</i>	Suffix incorporation

Modern English	Weak Verb	<i>shot</i>	<i>shot</i>	<i>shot</i>	<i>shot</i>	One-Change (A-B-B)
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The drastic reduction in morphological complexity characterizing the transition from Old English to the modern system is illustrated in Tables 3a–3c. The common trend across these periods is analogical levelling. Bybee (1985) notes that such levelling typically occurs among closely related forms to achieve a more economical correspondence between sound and meaning, thereby reducing the cognitive load required to store and retrieve irregular allomorphs from the mental lexicon.

For Class 1 verbs (e.g. *drīfan*) the Old English system required the speaker to manage four different vowel grades (**quadripartite distinction**). The difference between the preterite singular (*drāf*) and the preterite plural (*drifon*) has a special importance; it shows that the language once used number-based Ablaut. However, as English developed through the Middle English period, these subtle phonological features began to merge. The preterite plural had been fully absorbed into the singular by the time of the Modern English period, and the modern-day tripartite (A-B-C) paradigm was formed. The transition from a four-part to a three-part system greatly reduced the “perceptual salience” required to signal tense marking (Hogg & Fulk, 2011).

Moreover, the tables show the Strong-to-Weak shift, as in shoot. It was originally a strong verb with a complex vowel-shifting pattern, but eventually formed into a One Change (A-B-B) pattern. This change reflects the productive power of the dental suffix, which worked as a default rule that ended up replacing the older internal changes of all but the most common verbs (Hare & Elman, 1995). For the L2 learner, this historical loss is the source of today’s errors; the irregular verbs they struggle with today are essentially “incomplete” regularizations—forms that have lost their original systemic logic but have not yet fully adapted to the /-ed/ rule.

2.4.2. Strong Verbs

Strong Verbs are the most linguistically strict layer of the English verbal system. Their distinctive feature is that they signify tense and aspect by internal vocalic alternation, a process commonly referred to as Ablaut. The difference between strong and weak verbs is that, while the weak verb attaches a suffix to the external part of the root, the strong verb uses the phonetic space within the root. So, grammatical information is expressed in the stem, not added to it. This category is a direct inheritance from the verbal system of Proto-Indo-European (PIE) in which grammatical functions were indicated by the pitch and stress of vowels (Mailhammer, 2007).

Strong verbs are characterized by systematic changes of root vowels in historically predictable patterns. Such systematic irregularities are seen in Modern English verbs like *drink–drank–drunk* (Vowel Gradation/Ablaut)¹¹ or *write–wrote–written* (Full Vowel Change).

¹¹ In the context of Class 3 strong verbs, the Ablaut series follows a systematic shift: Infinitive /i/ (e.g., drink), Past Simple /æ/ (e.g., drank), and Past Participle /ʌ/ (e.g., drunk). These vowel patterns originate from the accent and stress rules of Proto-Indo-European.

The primary cognitive difficulty involved here is the phonological proximity of the V2 and V3 forms (e.g., *drank* vs. *drunk*), which frequently causes learners to simplify the tripartite (A-B-C) system to a bipartite (A-B-B) one, a process known as Paradigm Levelling. As Pinker (1999) points out, these verbs are not generated by rules but are retrieved from memory as stored lexical forms. This instability explains the frequent non-target forms such as *has drank* which appear in learner interlanguage.

The following table summarizes the morphological development of Strong Verbs:

Table 4a. Morphological Patterns in Strong Verbs

Feature	Infinitive (V1)	Past (V2)	Participle (V3)	Historical Root
Ablaut	<i>drink</i>	<i>drank</i>	<i>drunk</i>	Class 3 Strong
Ablaut	<i>begin</i>	<i>began</i>	<i>begun</i>	Class 3 Strong
Full Vowel Change	<i>write</i>	<i>wrote</i>	<i>written</i>	Class 1 Strong
Full Vowel Change	<i>rise</i>	<i>rose</i>	<i>risen</i>	Class 1 Strong

As shown in Table 4a, the modification of the verb stem often hides the phonetic relationship between the infinitive and the past form. The weak phonetic link between these forms makes it difficult for learners to retrieve the irregular past tense form from their associative memory, often leading to errors of stem preservation where the productive /-ed/ rule is inappropriately applied to the base stem in more extreme cases of paradigm collapse.

2.4.3. Weak Verbs

Weak verbs can be identified by the suffixation of a dental suffix (/ -d/, / -t/ or / -ed/) to indicate the past tense. Although most weak verbs in Modern English are regular, there is a large minority of irregular verbs, mainly because of historical phonetic processes such as vowel shortening and consonant cluster simplification. According to Minkova (2014), vowel shortening was a compensatory phenomenon in which the vowel length was reduced when a dental suffix was added to a long vowel root, thereby maintaining the prosodic balance of the syllable. This is applicable to *sleep–slept*, *feel–felt*, and *have–had*. These are examples of a category of verbs that make internal phonetic changes. The dental suffix causes the internal change.

This category also encompasses “Complex” weak verbs—those that exhibit both internal vowel modification and dental suffixation (e.g., *think–thought*, *catch–caught*, *sell–sold*). These forms display the influence of the voiceless velar fricative (the old / -x/ sound)¹², which was lost

¹² The voiceless velar fricative [x] is produced by turbulent airflow through a narrow constriction between the back of the tongue and the soft palate (velum) (Ladefoged & Johnson, 2015). In Middle English this sound was usually

in Middle English, leaving a complex vowel rime (Hogg & Fulk, 2011). From an L2 learner perspective, these verbs present a "dual complexity" challenge, often leading to "hybrid errors" such as *selled* or *solded*, where learners struggle to stabilize the combined representation of internal vowel shifts and external dental suffixation.

Table 4b. Morphological Patterns in Weak Verbs

Feature	Infinitive (V1)	Past (V2)	Participle (V3)	Historical Root
Vowel Shortening + Suffix	<i>sleep</i>	<i>slept</i>	<i>slept</i>	Weak (Shortened)
Vowel Shortening + Suffix	<i>have</i>	<i>had</i>	<i>had</i>	Weak (Shortened)
Vowel Change + Suffix	<i>think</i>	<i>thought</i>	<i>thought</i>	Weak (with Vowel Change)
Vowel Change + Suffix	<i>sell</i>	<i>sold</i>	<i>sold</i>	Weak (with Vowel Change)

Table 4b shows that the change in the verb stem often masks the phonetic connection between the infinitive and the past form. Therefore, the Old English long vowel in *slēpan* (to sleep) was shortened before the past tense suffix in *slēpte*, which resulted in the modern variation in vowel seen in *sleep/slept* (Hogg & Fulk, 2011). These irregularities are historical “fossilizations” in which the productive power of the dental suffix first functioned as a default rule, and eventually replaced older internal changes for all but the most frequent verbs (Hogg & Denison, 2006).

2.4.4. Suppletive Verbs

Spencer (1991) characterizes suppletion as the "**limit to phonological explanation**" in linguistic theory. He defines "total suppletion" as an alternation where there is absolutely no phonological connection between the two forms in a paradigm. The most famous instance in English is the verb *go*, which derives its past form *went* from the ancient verb *wenden* (to wend). However, these verbs are among the most stable in the language, despite their radical structural differences, because of their very high frequency of use.

written with the digraph -gh- (as thought or night) and, in Modern English, it eventually became silent or turned into a [f] sound (Hogg & Denison, 2006).

Table 4c. Suppletion: *Lexical Substitution Across Verbal Paradigms*

Feature	Infinitive (V1)	Past (V2)	Participle (V3)	Historical Root
Suppletion	<i>go</i>	<i>went</i>	<i>gone</i>	Multiple Roots
Suppletion	<i>be</i>	<i>was/were</i>	<i>been</i>	Multiple Roots

As Table 4c shows, suppletive verbs do not have phonetic transparency between their infinitive and past forms. Interestingly, these forms are logically the most difficult to predict, but L2 learners often acquire them earlier than other irregulars. This suppletion paradox is explained by **high token frequency**, which results from the fact that these verbs are so frequent that they are memorized as independent lexical chunks rather than being processed through the use of morphological rules (Bybee, 2007).

2.4.5. Invariable Verbs

Invariable verbs, or zero-marking verbs, are formed as **A-A-A** pattern where the infinitive, past, and past participle are identical. Pinker (1999) observes that all verbs in this category end in the dental consonants /t/ or /d/. Historically, this identity resulted from the phonological reduction of earlier inflected versions (such as Middle English *hitte* becoming *hit*), where the stem effectively **absorbed** its suffix over time. From a cognitive perspective, Pinker argues that because these stems already end in a dental sound, the learner's mental grammar may mistakenly assume the form is already inflected, which blocks the regular rule. However, for L2 learners, this lack of visible change often leads to overregularization errors like *hitted*, as they attempt to apply the productive /-ed/ rule to a stem they do not yet recognize as being effectively inflected.

Table 4d. Invariable (A-A-A) Verbs and Suffix Merging

Feature	Infinitive (V1)	Past (V2)	Participle (V3)	Historical Root
Invariable	<i>hit</i>	<i>hit</i>	<i>hit</i>	Weak (Suffix Merged)
Invariable	<i>put</i>	<i>put</i>	<i>put</i>	Weak (Suffix Merged)
Invariable	<i>cost</i>	<i>cost</i>	<i>cost</i>	Weak (Suffix Merged)

Invariable verbs, as indicated in Table 4d, are verbs with only one form in their entire paradigm. The main difficulty for the learner is the absence of a visual marker of tense, which is opposite to the general rule of English past tense morphology. This often leads to

overregularization errors such as *hitted* or *costed*, as learners try to make the past tense visible by adding the productive /-ed/ suffix to the infinitive form.

2.5. Categorization of Interlanguage Errors in Irregular Verbs

Learner errors are important to Second Language Acquisition (SLA) because they tell us certain things about the structure of the learner's developing interlanguage system. Williams et al. (2022) argued that errors were not random deviations but systematic results of cognitive and linguistic processes. This point of view is especially relevant in the area of English irregular verbs, where learners have to deal with a predominantly rule-based morphological system with a set of exceptions stored lexically.

Irregular verbs are a unique challenge because there is not a productive rule. Regular verbs are formed with the addition of the suffix *-ed*, whereas irregular verbs have to be remembered in their stored forms, often requiring changes to internal vowels, suppletion or non-transparent patterns (Pinker, 1999). Learners commonly rely on analogical reasoning and default strategies. This results in systematic and recurring types of error.

Previous SLA research has shown that the errors with irregular verbs follow logical rules rather than being random. These errors help identify specific stages of interlanguage development as a learner's internal grammar evolves over time (Gass & Selinker, 2008; Lightbown & Spada, 2013). Consequently, this study organizes error types by looking at how verbs are structured and the mental effort required for learners to process them.

2.5.1. Overgeneralization and Morphological Distortion

Overgeneralization, the use of a productive grammatical rule in contexts where it is not appropriate, is a well-known mechanism in SLA (Lightbown & Spada, 2013). In English irregular verbs, this is mainly because the regular *-ed* pattern is so common that learners tend to apply the regular form when it is not readily available.

One of the most frequent indicators of this development is what we call stem-based overregularization. Essentially, learners often tend to stick to the *-ed* suffix, applying it even where it doesn't belong (e.g. *fly* → *flied*, *run* → *runned*). We see a similar pattern with no change verbs like *hit* or *put*. Because there is no visible change in these words to mark the past tense, learners seem to feel the form is incomplete and feel the need to add an explicit marker, leading them to produce misconjugated verbs like *hitted* or *putted* just to be safe.

Double marking is a more complex pattern where learners sometimes mix irregular and regular morphology within the same verb (e.g., *choose* → *chose* → *chosed*, *make* → *made* → *maded*, *wake* → *woke* → *woked*). Thanks to these hybrid forms, we can infer that learners have partial access to the irregular paradigm, but they continue to use the regular rule as a backup strategy. Analogical processes of the same kind are observed in forms like *founded* instead of

found, which suggests that learners may refer to phonological similarity rather than to the retrieval of the correct form (Marcus et al., 1992; Pinker, 1999).

This analogical reasoning often leads learners to look for consistency even within the irregular system itself, based on existing phonological patterns. For example, a learner who has adopted the *take-took* (/teɪk/ - /tʊk/) paradigm may develop a pattern of producing *mook* (/mʊk/) for the verb *make* (/meɪk/) following the same morphological logic, which is logical but incorrect. On the contrary, the existence of *make-made* (/meɪk/ - /meɪd/) might trigger the production of *tade* (/teɪd/) instead of *took*. These errors show a “competition” between different irregular patterns, depending on their phonetic similarity (Pinker, 1999; Pinker & Ullman, 2002).

From the same aspect, learners might have issues with the incorrect pairing of phonologically similar pairs such as *sneeze/freeze* (/sniːz/ - /friːz/) or *match/catch* (/mætʃ/ - /kætʃ/). The learner may generalize the *froze* (/froz/) form to produce *snoze* (/snoʊz/) for *sneeze* or apply the regular *sneezed* (/sniːzd/) pattern to produce the mistaken *friezed* (/friːzd/). The same goes for the *catch-caught* (/kætʃ/ - /kɔːt/) model which might lead learners to produce *maught* (/mɔːt/) for *match*. The regular *matched* (/mætʃt/) might result in the overregularized *catchted* (/kætʃt/). These examples illustrate that morphological distortion is not simply a failure to remember a rule, but rather an active, if misapplied, search for linguistic symmetry within associative networks (Bybee, 2001, 2007). While these specific errors are theoretically predicted based on analogical networks, they reflect the learner's systematic attempt to impose consistency on the irregular system.

Another important subclass is the overextension of the suffix, particularly the *-n* or *-en* marker which is associated with some irregular participles. Forms produced by learners such as *rosen*, *readen*, *sinken*, and *losen* show that this suffix is seen as a general marker of irregularity. This may lead sometimes to overused forms (e.g. *growned*, *knowned*), or to an attempt of reconstructing an incomplete paradigm. The absence of this suffix (e.g. *flown* → *flow*) strengthens instability in the representation of participial morphology.

Overgeneralization is also found at the paradigm level, where learners apply transformations to forms that do not belong to the verb paradigm. Learners do not always distinguish between V1, V2 and V3 as shown in the addition of *-ed* to participial forms (*growned*, *knowned*) or the changing of past forms (*chose* → *chosed*). In the same way, forms like “*rise* → *rosen*” demonstrate attempts to regularize or to reconstruct patterns based on partial knowledge.

Orthographic effects are especially relevant for morphological distortion in written production. The high-frequency errors like *payed* and *sayed* reveal that learners usually focus more on transparent spelling rules than on irregular forms. Also, the confusion of semantically or morphologically related verbs such as *lie* (in the sense of “to recline”) and *lay* reveals the lexical interference in learner output.

Finally, it is worth mentioning that these overgeneralized forms are often found within larger grammatical structures, such as combinations with auxiliaries (e.g. *didn't went*, *has went*). This suggests that overgeneralization is not restricted to isolated verb forms, but is linked with

more general aspects of grammatical encoding. These patterns generally propose that the morphological distortion in irregular verbs is a systematic attempt to regularize a complex and relatively unpredictable system, as predicted by usage-based accounts of language acquisition (Bybee, 2007).

2.5.2. Participle Selection and V2/V3 Confusion

L2 learners often face challenges in distinguishing between different past tense forms. This confusion often happens because some verbs use the same form for two different functions (e.g. *slept-slept*) (Lightbown & Spada, 2013). When learners see these identical forms, they often assume a single, uniform pattern applies across the entire verb system (Gass & Selinker, 2008).

One common learner strategy is to construct the V3 form by adding an *-en/-n* suffix (e.g. *write* → *writen*, *hide* → *hiden*). This shows that the learners are aware of irregular morphology, but also are uncertain about the correct form. In other words, learners overgeneralize the participle suffix where it does not belong.

Another common fallacy is the use of the wrong verb form in perfect tense structures, such as *have went* or *has saw*. In these kinds of instances, learners are unable to differentiate between tense and aspect and treat V2 as interchangeable with V3. In a similar way, constructions such as *have see* or *had go* indicate that the participle form is either not retrieved or not recognized as necessary. These errors show an instability in the mapping of auxiliary verbs to their actual morphological forms.

In passive structures, we encounter errors when learners use base or past forms instead of the participle (e.g. *was broke*, *is tell*, *are take*). These patterns suggest that the functional role of V3 in grammatical constructions is not fully perceived in the learner's interlanguage system.

From a processing perspective, these errors might also reflect the accessibility of verb forms. Irregular forms are stored in learners' memory, and under pressure learners may rely on the more accessible V2 form. This aligns with the role of frequency-based approaches to language acquisition, which emphasize the importance of frequency and retrieval in affecting learner performance (Bybee, 2007).

2.5.3. Auxiliary and Periphrastic Errors

Auxiliary and periphrastic structures represent a great challenge for learners dealing with English irregular verbs, particularly when it comes to coordinating auxiliaries with lexical forms. Many errors suggest that learners struggle to properly distribute grammatical information across the verb phrase.

A classic example is the misapplication of *do-support* in past tense phrases such as *didn't went* or *didn't saw*, where the tense marking is wrongly preserved on the main verb. This results in double marking, showing that the learner hasn't yet comprehended that once *did* is used, the main verb must return to its base form.

Another persistent issue is related to the verb *be*. Learners sometimes treat *be* as a standard verb that requires *do-support* (e.g., *didn't be* instead of *wasn't*), effectively overgeneralizing auxiliary patterns while neglecting the unique, suppletive nature of the verb *be*.

Lastly, while not systematically observed in this study's dataset, previous studies (Dulay et al., 1982) have highlighted the complete omission of auxiliaries, leading to "**bare V3**" structures (e.g. *He gone*, *The window broken*). These occurrences point to an attempt to lower cognitive load by dropping function words, while keeping the lexical meaning.

These errors can be related to a heavy cognitive demand of multi-component verb phrases, which often leads learners to adopt simplified or generalized strategies (Mackey, 2012).

2.5.4. Cognitive Load and Performance Variability

Learner performance with irregular verbs is not only shaped by morphological knowledge, but has deep ties to processing constraints. As tasks become more complex, the simultaneous management of tense, auxiliaries and specific verb forms becomes a heavy cognitive load (Mackey, 2012).

Tense-aspect misalignment is a noticeable trend in the questionnaire data. Interestingly, even when learners know the correct irregular forms, they often struggle to pair them to the correct tense-aspect combinations—for example, choosing the simple past when the past perfect is required. This suggests that the mental effort required to retrieve less frequent forms (such as *swum* or *flown*) might actually interfere with grammatical processing. In these moments, learners seem to prioritize the retrieval of the word itself, leaving little cognitive room to ensure the correct tense is used.

This variability also shows up in phonological and lexical confusion. Errors including *flough*, *flight* and *flow* for *fly*, and the classic confusion of *lie* and *lay* (*laid* for *lied* or vice versa) reveal that learners often switch to phonological similarity when a form isn't readily accessible. This shows some instability in how these words are stored and retrieved in the mental lexicon (Bybee, 2007).

These patterns demonstrate that errors in irregular verbs are ultimately not just a matter of not knowing the rules. Instead, they are often the result of processing limitations, thus supporting the view that cognitive barriers are important in interlanguage performance (Mackey, 2012).

2.6. Cross-linguistic Influence: Past Tense Morphology in Romance and Germanic Languages

Cross-linguistic Influence (CLI), as known as linguistic transfer, plays a critical role in how a learner's first language (L1) shapes the acquisition and use of a second language (L2). As Odlin (1989) observed, the structural distance between these two languages can either smooth the path or create significant complication in morphological development. When it comes to English irregular verbs, there are two main challenges: not only learners have to learn by heart

idiosyncratic forms, but also navigate a syntactic landscape for negation and questioning that may seem entirely alien, based on the typological rules of their native language.

Table 5. Typological Comparison of Past Tense and Past Perfect Structures: *Romance and Germanic vs. English*

Language Family	Language	Affirmative Past	Negative Past	Past Perfect (Aff./Neg.)	Functional Conflict with English
Target	English	V2	Aux + Not + V1	Aux + V3 / Aux + Not + V3	–
Romance	French	Aux + V3	Aux + Not + V3	Aux + V3 / Aux + Not + V3	Reliance on Aux+V3 for simple past (Passé Composé).
	Italian	Aux + V3	Not + Aux + V3	Aux + V3 / Not + Aux + V3	Absence of Do-Support; auxiliary-driven negation.
	Spanish	V2	Not + V2	Aux + V3 / Not + Aux + V3	Synthetic negation (No+V2); no auxiliary change.
	Portuguese	V2	Not + V2	Aux + V3 / Not + Aux + V3	Synthetic negation (No+V2); no auxiliary change.
Germanic	German	Aux + V3	Aux + Not + V3	Aux + V3 / Aux + Not + V3	Confusing the German <i>Perfekt</i> with the English Simple Past.
	Dutch	Aux + V3	Aux + Not + V3	Aux + V3 / Aux + Not + V3	Tending to use auxiliary verbs to express the past
	Swedish	V2	V2 + Not	Aux + V3 / Aux + Not + V3	V2 negation without auxiliary (Do-Support).
	Norwegian	V2	V2 + Not	Aux + V3 / Aux + Not + V3	V2 negation without auxiliary (Do-Support).

Table 5 displays the structural variations between English and the learners' L1s. As the typological diagram indicates, the primary cause of challenge for learners of all eight languages analyzed is English periphrastic negation (*did not* + *V1*). This structure is not found in their native language mechanisms at all, creating a functional conflict where learners have no choice but to abandon their L1 logic in exchange for a complex, layered auxiliary system.

2.6.1. Romance Languages

The morphological distance between Romance languages and English creates a strong negative transfer effect. English has a periphrastic **do-support** system (*did not* + *VI*) for negation, while Romance languages use auxiliary-participle combinations or synthetic inflections, as categorized in Table 5.

2.6.1.1. French

For most past actions, French employs the **passé composé** (auxiliary + past participle). Because this structure is "structurally identical" to the English present perfect, it can lead some Francophone learners to overuse the present perfect in contexts where the simple past is required (Ayoun & Salaberry, 2008). Their findings suggest that while this transfer is not widespread across all learners, some individuals may systematically rely on this periphrastic "*auxiliary* + *participle*" pattern as an interlanguage strategy.

- **Affirmative:** J'ai parlé (*Aux* + *V3*). / **Negative:** Je n'ai pas parlé (*Aux* + *Not* + *V3*).

The Plus-que-parfait is similar to the English structure (*Aux* + *V3*) where the auxiliary is in the imperfect (*avais*). The conflict is that the negative marker (*ne...pas*) surrounds the auxiliary, while in English it comes after.

- **Affirmative:** J'avais parlé (*Aux* + *V3*). / **Negative:** Je n'avais pas parlé (*Aux* + *Not* + *V3*).

2.6.1.2. Italian

Italian has the "*Passato Prossimo* (*Aux* + *V3*)". Swan & Smith (2001) claim that Italian learners struggle with English negation because their L1 negation (*non*) is simply placed before the auxiliary.

- **Affirmative:** Ho parlato (*Aux* + *V3*). / **Negative:** Non ho parlato (*Not* + *Aux* + *V3*).

The "*Trapassato Prossimo*" is formed with an auxiliary plus a participle. The main issue is the lack of *do-support* in English negatives; Italian learners often attempt to negate English past tenses by putting "*not*" in front of the verb, in relation to the pattern in their L1.

- **Affirmative:** Avevo parlato (*Aux* + *V3*). / **Negative:** Non avevo parlato (*Not* + *Aux* + *V3*).

2.6.1.3. Spanish

Spanish has a synthetic Preterite (V2) system. But, according to Swan & Smith (2001) Spanish negation, like in the example below, does not require an auxiliary change and double-marking errors such as "*I didn't spoke*" are possible.

- **Affirmative:** Hablé (V2). / **Negative:** No hablé (*Not* + V2).

The Pretérito Pluscuamperfecto follows the Aux + V3. Spanish learners often confuse English auxiliary had, sometimes replacing it with have because of phonetic similarity.

- **Affirmative:** Había hablado (Aux + V3). / **Negative:** No había hablado (*Not* + Aux + V3).

2.6.1.4. Portuguese

The Portuguese affirmative past is a synthetic example (V2). The use of the English "*did not + VI*" structure is syntactically far from being close to Portuguese speakers, as there is no dummy auxiliary like did (Swan & Smith, 2001).

- **Affirmative:** Eu falei (V2). / **Negative:** Eu não falei (*Not* + V2).

Portuguese often has the auxiliary *tinha* (*had*) + V3. The functional conflict appears in negation, in which the negative particle is before the auxiliary (*Not* + Aux + V3), unlike the English Aux + *Not* + V3.

- **Affirmative:** Eu tinha falado (Aux + V3). **Negative:** Eu não tinha falado (*Not* + Aux + V3).

2.6.2. Germanic Languages

For native speakers of Germanic languages, learning the English past tense is easier because the languages are closely related—a concept known as “**typological proximity**.” These learners recognize how strong verbs change internally (like *sing* → *sang*) because they share the same historical roots. However, Ringbom (2007) points out this similarity helps learners in identifying the irregular verb forms (**positive transfer**), but often deceives them in applying native grammar rules in forming complex or negative sentences in English (**negative transfer**).

2.6.2.1. German

In speech, German uses *Perfekt* (*Aux* + *V3*). According to Swan & Smith (2001), this structure forces learners to use periphrastic *have* + *V3* forms rather than the synthetic simple past required by English.

- **Affirmative:** Ich habe gesprochen (*Aux* + *V3*). / **Negative:** Ich habe nicht gesprochen (*Aux* + *Not* + *V3*).

The *Plusquamperfekt* is built like the English but has a different placement of negatives. German learners may place not at the end of the sentence, imitating L1 syntax.

- **Affirmative:** Ich hatte gesprochen (*Aux* + *V3*). / **Negative:** Ich hatte nicht gesprochen (*Aux* + *Not* + *V3*).

2.6.2.2. Dutch

Dutch uses the *Voltooid Tegenwoordige Tijd* (*Aux* + *V3*). The periphrastic past is functionally predominant and consequently there is an over-reliance on auxiliary verbs in Dutch-English interlanguage (Swan & Smith, 2001).

- **Affirmative:** Ik heb gesproken (*Aux* + *V3*). / **Negative:** Ik heb niet gesproken (*Aux* + *Not* + *V3*).

The *Voltooid Verleden Tijd* (*Aux* + *V3*) is structurally comparable to English, but the placement of negation is still a point of negative transfer.

- **Affirmative:** Ik had gesproken (*Aux* + *V3*). / **Negative:** Ik had niet gesproken (*Aux* + *Not* + *V3*).

2.6.2.3. Norwegian

Norwegian has a synthetic *Preteritum* (*V2*). Swan & Smith (2001) highlight that the main problem is the word order in Norwegian, where negation follows the inflected verb without an auxiliary.

- **Affirmative:** Jeg snakket (*V2*). / **Negative:** Jeg snakket ikke (*V2* + *Not*).

The *Preteritum Perfekt* is a pattern of *Aux* + *V3*. The particle *ikke*, however, follows the auxiliary in negation, the only Germanic structure that is close to English.

- **Affirmative:** Jeg hadde snakket (*Aux + V3*). / **Negative:** Jeg hadde ikke snakket (*Aux + Not + V3*).

2.6.2.4. Swedish

Swedish syntax is like Norwegian. Swedish learners struggle to move from V2-negation to the English Aux-negation (did not), as highlighted by Swan & Smith (2001).

- **Affirmative:** Jag pratade (*V2*). / **Negative:** Jag pratade inte (*V2 + Not*).

Aux + V3 = pluskvamperfekt Swedish negation, like Norwegian, comes after the auxiliary, but the cognitive pressure of switching from synthetic past to periphrastic past perfect is still a factor of error.

- **Affirmative:** Jag hade pratat (*Aux + V3*). / **Negative:** Jag hade inte pratat (*Aux + Not + V3*).

3. Methodology

This chapter defines the methodological framework designed to systematically investigate the morphological and syntactic errors produced by second language (L2) learners of English regarding irregular past tense and past perfect morphology. This study adopts a mixed-methods approach, combining corpus-based linguistic analysis with controlled experimental testing, to identify the cognitive and contextual variation in interlanguage performance. The study also combines natural learner data with targeted experimental tasks, including two Google Forms-based irregular verb tests, to examine how first language (L1) background, verb typology, and changing task demands interact during the processing and production of non-standard irregular verb forms.

3.1. Data Sources

To get a complete picture of learner behavior, this study combines two different types of data: a large learner corpus and two controlled grammar tests. It is necessary to draw a clear distinction regarding the methodological status of these data sources: while the ICLE corpus provides naturalistic, observational data that documents the linguistic product of spontaneous writing, the questionnaires offer controlled experimental data designed to interrogate underlying cognitive processes under manipulated task demands. These two methods together make it possible to overcome the limitations of using just one tool, providing us a way to verify if the same error patterns appear in both environments.

3.1.1. International Corpus of Learner English (ICLE)

The corpus-based part of this study used data collected from the International Corpus of Learner English (ICLEv3) (Granger et al., 2020). The ICLE is a highly organized and internationally known learner corpus, which gives a unique insight into the interlanguage profiles of advanced second language (L2) learners of English. The corpus is composed mostly of argumentative essays written by third- and fourth-year university students, the majority of whom are majoring in English language, literature, or modern foreign languages. As these learners are at a higher-intermediate to advanced level of proficiency (CEFR B2 to C1), their errors are rarely due to a complete lack of grammatical knowledge, but rather indicate deeper cognitive persistence, L1 transfer patterns, and fossilized morphological overgeneralizations (Li, Hiver, & Papi, 2022). The use of the ICLEv3 is motivated methodologically in terms of ecological validity (Ellis, 2019). Natural tasks of essay writing trigger spontaneous language performance, while controlled testing environments require students to focus on grammatical form. In this context, which focuses on meaning, learners must be able to manage multiple cognitive levels at the same time, for example, rhetorical organization, lexical selection and syntactic planning. As a result, the cognitive load increases, and learners are more likely to experience morphological levelling, paradigm confusion, and structural errors when retrieving irregular past tense and past perfect forms.

The ICLEv3 user interface offers two primary ways of finding data: the **Simple Search** and the **Advanced Search** functions. The corpus's Advanced Search allows users to search texts for exact Part-of-Speech (POS) tags, where lexical verbs in the past tense are tagged as *VVD*, and past participles are tagged as *VVN* (Granger et al., 2020). The auxiliary forms for the past tense and perfective aspects involve different categories: *VDD* for the periphrastic auxiliary *did*, and *VHD/VHN* for the perfective aspect auxiliary *had*. However, an initial review of the automatic POS tagging within the corpus has shown some minor limitations associated with non-standard, inaccurate morphological forms. Automatic parsing algorithms can incorrectly classify highly irregular interlanguage tokens produced by learners because of their idiosyncratic orthographic shapes, which can result in missing a systematic error if the analysis depends only on the POS tags. To avoid this technical barrier and ensure complete data integrity, this study adopted a strict, exhaustive manual extraction strategy entirely through the Simple Search interface, as shown in the interface layout in Figure 1.

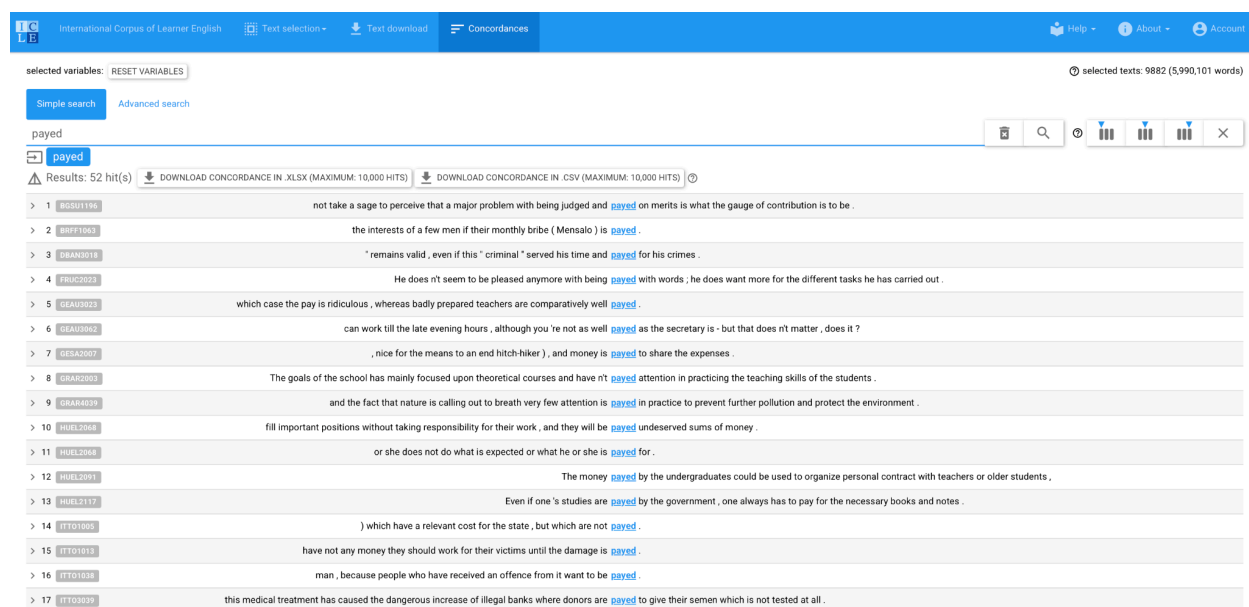


Figure 1. The ICLeV3 Simple Search Interface Used for Structural Query Entry.

The Advanced Search interface, unlike the open query input in the Simple Search interface, requires the user to build queries following these pre-set automated grammatical criteria. This alternative search mode, instead of allowing free-text entry, restricts queries to the structural configuration and parameter selection fields shown in Figure 2.

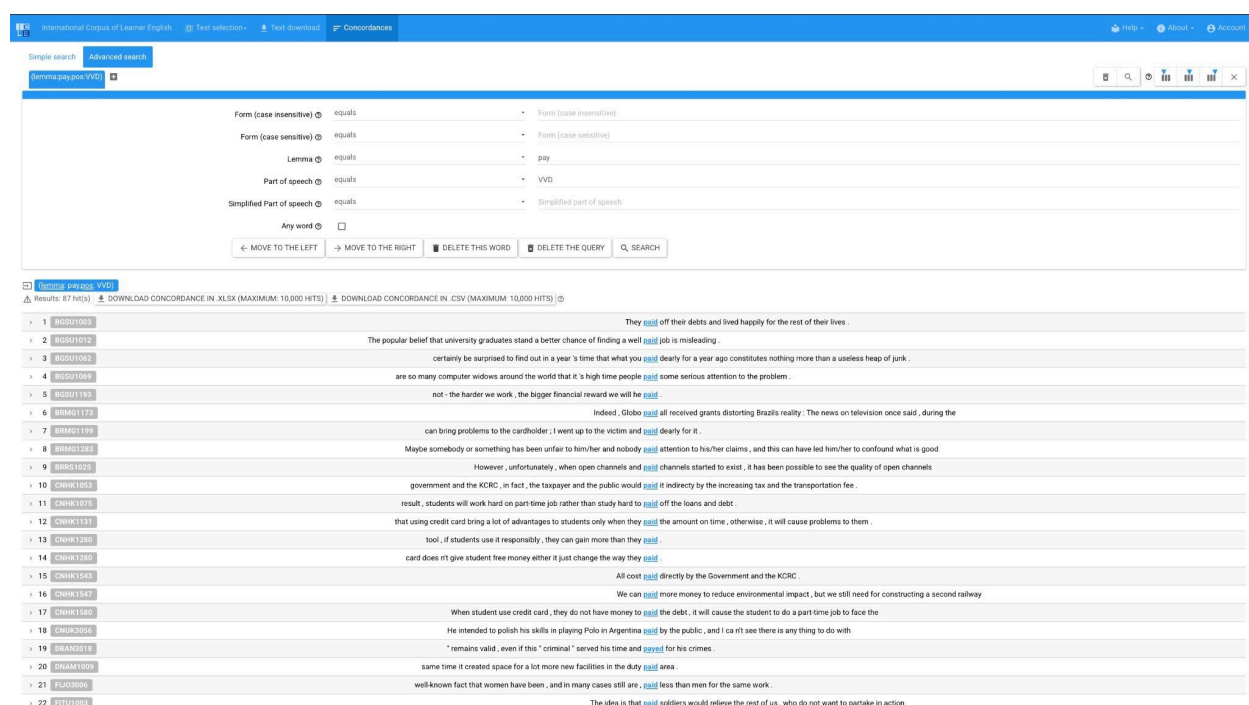


Figure 2. The ICLeV3 Advanced Search Interface and Tag-Based Query Demonstration.

To illustrate the limitations of relying on these grammatical tags alone, a query comparison was made within the corpus using the non-standard form “payed”. The query for the automated *VVD* tag in the Advanced Search interface (Figure 2) returned 87 concordance lines containing both standard productions (paid) and non-standard forms (payed). A thorough manual parsing of the 87 automated results found only 12 instances of the target error, *payed*. In contrast, it is worth comparing this result to the 52 clean error tokens directly retrieved using the Simple Search string query (Figure 1), which suggests that the automated parser could not manage to accurately detect or tag 40 instances of the same morphosyntactic form. These missing tokens were completely absent from the *VVD* dataset, demonstrating that automated parsing algorithms often misclassify highly irregular interlanguage tokens into unrelated grammatical categories due to their idiosyncratic orthographic shapes. Therefore, it can lead to overlooking systematic errors if the analysis depends only on automated tags.

Instead of using only the automated grammatical tags, a detailed lexical database of English irregular verbs was compiled based on Quirk’s seven structural classes. For each target irregular verb, all theoretically possible non-standard morphological variations, such as overgeneralization, incorrect stem modification, hypercorrection, and historical Germanic suffixation, were systematically entered as explicit search queries in the interface. For example, to observe how learners produce internal vowel gradation alongside systemic regularizations within the Class Three paradigm, the interface was queried for multiple non-standard versions of the verb sing, including “*singed*”, “*sanged*”, “*sunged*”, “*singen*”, “*sangen*”, and “*sungen*”. Similarly, to identify error patterns in Class Seven verbs that require both vowel change and irregular past participle markers, queries were generated to search for corrupted structures of the verb eat, such as “*eated*”, “*ated*”, and “*eatened*”. These orthographic variations were identified for all target irregular verbs and a search strategy was preferred over filtering by automated POS tags to reduce the risk of false negatives that might have been mistagged by the automated parsers. The exact formulation of these extensive keyword queries and the application of manual parameters in the system can be seen documented across Figure 1 and Figure 2 respectively.

After the thorough extraction, cleaning, and filtering processes, the final dataset was composed of **516** validated morphosyntactic error occurrences from **80** different irregular verb paradigms. To keep the linguistic and contextual integrity of every non-standard production, metadata for each error token was methodically linked to the individual learner profile sheets built in the ICLEv3 framework. The typology of the contributing student group is quite diverse in terms of cultural and linguistic background and includes learners from a very wide range of geographical and first language backgrounds. In particular, the database gathers interlanguage productions of learners from Austria (German), Belgium (French and Dutch L1 Groups), Bosnia-Herzegovina (Serbian), Botswana (Tswana), Brazil (Portuguese), Bulgaria (Bulgarian), Finland (Finnish and Swedish L1 Groups), Germany (German), Greece (Greek), Hungary (Hungarian), Iran (Persian), Italy (Italian and French L1 Groups), Japan (Japanese), Lithuania (Lithuanian and Russian L1 Groups), Macedonia (Macedonian), Norway (Norwegian), Pakistan (Urdu and Punjabi L1 Groups), Poland (Polish), Russia (Russian), Serbia (Serbian), South Africa

(Tswana), South Korea (Korean), Spain (Spanish and German L1 Groups), Sweden (Swedish), Switzerland (German), the Czech Republic (Czech), Turkey (Turkish). This rich cross-linguistic representation provides a strong testing ground. The next statistical analysis can then separate universal L2 developmental patterns from specific crosslinguistic interference.

Furthermore, demographic and pedagogical variables extracted from these learner profiles indicate that the formal, institutional exposure to English is highly heterogeneous across the participant pool. Most essay writers (83.82%) have had 6 or more years of formal English schooling, but the duration ranges from as little as 0 to 2 years in some cases, to as many as 13 years for some populations, such as Macedonian learners (Granger et al., 2020). Even with this large educational gap, the overall proficiency level is still within the higher-intermediate to advanced level (B2 to C1 CEFR), providing an excellent analytical scope into the connection between morphological fossilization and long-term classroom instruction. In order to maintain a full structural organization during the variable selection phase, all extracted data points were compiled in a separate database organized on nine systematic analytical dimensions. This CSV structure is a complete record of the structural and demographic metadata for each token under the following standard titles: **Verb, Error Form, Irregular Verb Type, Country, Native Language, Error Type, Language Family, Region, and Years of English at School**. This precise taxonomy construction was used to categorize the collected dataset and every observed irregular verb error was linked back to its precise structural, typological and socio-pedagogical source in the results chapter.

3.1.2 Experimental Questionnaires

To support the natural learner data extracted from the ICLEv3 corpus, controlled experimental questionnaires were conducted at **Bordeaux Montaigne University** between **March and April 2026**. The experiment was conducted with a sample of **22 international students** distributed across 3 different class groups, aged between **18 and 22** years old, all enrolled in degree programs within the English Department. These participants were the students enrolled in the undergraduate course **2LILAM222 MATI - ORAL**, which was held under a dual-teaching framework co-directed by **Ms. Alison May Lagarde** and myself.

The main methodological aim of using these testing instruments was to extend the scope of analysis beyond the pre-existing ICLE data by incorporating two distinct experimental approaches: guided multi-format assessments and spontaneous production tasks. The present study therefore aimed at directly linking spontaneous written production with controlled task performance, under experimental conditions that would determine the degree of variation of morphological forms in natural essay writing and in explicitly grammatical contexts.

The experimental data collection was carried out via two different tests based on Google Forms that were conducted in order. The general design of these instruments was based on the idea of modifying task demands to see if different cognitive loads and contextual limitations trigger or prevent morphological overgeneralizations. Both questionnaires collected important

sociolinguistic and pedagogical background information such as the participant's age, country of birth, native language, and the exact length of time that the participant had received formal English instruction. These are described in more detail in the next section on participant profiles and demographic data. All 22 participants completed each test in under 10 minutes, which shows that their responses reflected spontaneous retrieval of morphological forms and not long-term thinking or the use of external reference materials.

Irregular Verbs Usage Test 1 functions as the initial multi-format evaluation tool consisting of exactly 25 structured questions, intentionally balanced to guide the students from explicit target recognition tasks toward structured morphological insertions. In contrast to this guided layout, **Irregular Verbs Usage Test 2** is designed as a production-focused, context-based evaluation framework consisting of exactly 20 open-ended, non-directed questions. By combining a guided instrument (Test 1) with an unguided format (Test 2), this dual-test methodology provides a solid analytical baseline for connecting and comparing corpus data with experimental performances, as demonstrated throughout the following sections and chapters.

3.2. Participant Profile and Demographic Data

This study is based on a small group of 22 international undergraduate students enrolled in the MATI - ORAL 2LILAM222 course in the Department of English at Bordeaux Montaigne University. The participants range in age from 18 to 22 years and represent an advanced L2 interlanguage framework. However, the demographic and sociolinguistic profile of the sample shows variation in terms of country of origin, L1 configurations and multilingual backgrounds.

The participant group is predominantly composed of monolingual native French speakers, with some bilingual and multilingual profiles. There is an important subgroup of five participants with mixed bilingual or heritage language backgrounds (including cross-linguistic exposure to Algerian Arabic, Spanish, German, Portuguese and Russian). In addition, the experiment involves one participant born in England who is an L1 native English speaker, which provides a reference point for a targeted retrieval of irregular verbs in the overall group performance.

To offer a clearer overview of these linguistic and demographic characteristics, Table 6 below presents the profiles of all 22 participants:

Table 6. Sociolinguistic and Demographic Distribution of the Participant Students

Country of Birth	Native Language/s (L1)	Number of Learners	Linguistic Profile Group
France	French	16	Monolingual L2 Learners
France	French & Algerian (Arabic)	1	Bilingual L2 Learner
France	French & Spanish	1	Bilingual L2 Learners

Germany	German & French	1	Bilingual L2 Learners
Portugal	Portuguese & French	1	Bilingual L2 Learners
Luxembourg	Russian & French	1	Multilingual L2 Learner
England	English	1	Native L1 Control Baseline

One methodological limitation concerning data tracking should be noted. To ensure ethical compliance and to reduce evaluation anxiety during the rapid testing procedure, total anonymity was preserved and no unique learner IDs were assigned throughout the two sessions. All students signed a formal research form, granting their consent regarding their participation in the experimental questionnaires via QR codes. The purpose and voluntariness of the study were clearly explained to the participants in the consent form, and the confidentiality of the data was ensured at all stages of data processing and publication. Therefore, the responses in Test 1 and Test 2 were not compared individually, but the data were analyzed at the group level to examine the patterns in the learners' interlanguage system.

Furthermore, self-reported estimates of the duration of institutional instruction (Years of English at School) were compared across the two online questionnaires. Initial observations exposed certain variability and potential inconsistencies in participant reporting. To minimize recall bias and ensure reliable learner profiles, we avoided using this continuous variable in its numerical form. Instead, we recoded the data into two different groupings: **0–10 years** and **11–14 years** of instructional exposure. This strategy provides a more stable basis for analysis, allowing us to systematically evaluate the influence of language exposure on morphosyntactic accuracy while reducing the impact of unreliable subjective reporting.

3.3. Design of the Experimental Tasks

The experimental tasks were arranged systematically in a psychological and linguistic structure to manipulate cognitive load and track changing task demands. The two testing instruments were intended to provoke the operational limits of interlanguage performance, rather than to measure static grammatical knowledge. This was accomplished by changing from highly structured recognition directives to unguided, contextually driven morphological production. The progressive architecture of these tasks allows for a strategic tracking of how advanced L2 learners deal with standard and non-standard irregular representations under various degrees of explicit surveillance.

Irregular Verbs Usage Test 1 was created as a multi-format diagnostic test. It consists of 25 questions divided into four different sections. The main methodological goal of this task was to observe learner behaviour when structural boundaries were provided and to see if explicit orthographic distractors triggered basic regularization patterns:

- **Section 1 & 2 (Multiple-Choice Selectors - V2 & V3):** Section 1 (10 items) and Section 2 (4 items) are respectively the paradigms of the Past Simple (V2) and Past Participle (V3). Multiple-choice tasks are linguistically the least demanding retrieval tasks, since they depend on recognition memory rather than spontaneous recall. However, the cognitive load was intentionally manipulated by inserting specific non-standard morphological innovations as distractors next to the target forms (e.g. forcing a choice between the standard *made* and the hyper-regularized *made* or periphrastic *didn't made*). This design tests whether the learner's mental lexicon has stable morphological representations or tends to show paradigm confusion with rule-based regularised alternatives.
- **Section 3 (Acceptability Judgement / Error Recognition):** This section contains an explicit meta-linguistic challenge on 4 questions. Participants are required to make explicit acceptability judgements on whole sentences with respect to syntax and morphology. The presence of major interlanguage forms (e.g., “*She has beguned the task*” or “*I swum yesterday*”) changes the task from a simple lexical selection to a systemic error parsing. It requires the learner to apply their conscious pedagogical grammatical rules to evaluate non-standard inflectional markers.
- **Section 4 (Lemma-Based Insertion):** The final section of Test 1 has 7 questions in a guided fill-in-the-blank format and serves as a transition bridge. Participants were asked to provide the correct past form of the base-form lemmas given in brackets. This task increases cognitive demand by removing several answer choices, while maintaining the sentence structure short and direct, thus concentrating the participant’s attention fully to the formal inflectional accuracy.

Irregular Verbs Usage Test 2 is an unguided, production-based task with 20 open-ended non-directed questions, in contrast to the guided structure of the first instrument. By discarding all orthographic cues, recognition choices and metalinguistic options, this instrument forces participants to actively retrieve and construct the correct morphological shape entirely from memory under a much higher cognitive load:

- **Section 1 (Past Simple Production):** This section has 10 questions and requires the building of Past Simple (V2) forms in continuous sentence contexts. The sentence contexts require the use of both positive and negative forms (e.g. *see, not/pay, write*) to increase task difficulty and avoid over-reliance on automatic suffixation. The presence of negative markers (*not/pay*) forces the student to handle auxiliary insertion (*did + not*) and to avoid double-marking the past tense on the lexical stem (*didn't paid*), revealing possible fossilized speech patterns.
- **Section 2 (Perfect Tense Selection and Production – V3):** The last 10 questions increase the cognitive expectation by using the Perfect Tenses (V3). In contrast to the earlier sections which explicitly asked for the target tense, this section requires

participants to choose and form the Present Perfect (*have/has* + *V3*) or the Past Perfect (*had* + *V3*) based on complex, temporal aspect markers within the clauses (e.g. “*By the time Katy graduated, she (become) fluent...*”). This involves a two-step cognitive process: the learner must first analyze the timeline to select the correct auxiliary, and second, retrieve the correct irregular verb form (*become* vs. *became*).

A major design parameter in the both tasks was the absence of any external monitoring and the use of a rapid execution design. Most importantly, all 22 international students were able to complete each test in under 10 minutes. In psycholinguistic test designs, such rapid completion rates reflect that the operational tasks successfully bypassed extended, metalinguistic reflection, effectively minimizing the activation of conscious monitoring strategies (Ellis et al., 2009). On average, the participant students spent less than 30 seconds per question. Under the same time conditions, a performance difference is expected between the two tests, with participants likely to show higher accuracy in Test 1. This difference is due to the lower cognitive load of the multiple-choice format, where orthographic cues support recognition memory and make items easier to process under time pressure compared to the unguided, recall-based production demands of Test 2 (Hulstijn, 2001). Therefore, their responses reflect their natural, spontaneous grammar instead of analytical reasoning or a reliance on reference materials.

3.4. Data Coding and Variable Selection for MCA

The main goal of this thesis is to determine systemic relations between specific learner profiles (e.g., native language backgrounds, typological language families, institutional learning length) and the structural nature of morphosyntactic deviations in English irregular verb morphology. To achieve this goal, the study uses a dual quantitative approach. While the experimental questionnaires (Test 1 and Test 2) are evaluated using descriptive frequency tracking, Multiple Correspondence Analysis (MCA) is used to identify patterns within the categorical variables of the dataset that are difficult to examine using standard statistical techniques (Greenacre, 2017). However, compiled from the International Corpus of Learner English (ICLE) was at first displayed in a “wide” format, with a focus on the participants (Granger et al., 2020). Transforming this raw informational input into an academically structured statistical matrix needed a comprehensive data reshaping pipeline and a theoretically motivated variable selection procedure (Desagulier, 2017).

Data Reshaping: Wide to Long Categorical Restructuring: In the first stage of data collection, each row of the spreadsheet database represented a single real student participant with subsequent columns representing their unique responses in 20 to 25 different morphosyntactic slots. This wide format is suitable for descriptive tracking or individual scoring but numerically incompatible with the categorical demands of correspondence analysis (Le Roux & Rouanet, 2010). Multiple Correspondence Analysis calculates geometric proximities from a contingency

matrix of categorical tokens, rather than from a sequence of syntactic tokens calculated based on a single participant profile (Greenacre, 2017).

Therefore, a data restructuring algorithm was performed through R Studio based on tidy-data principles (Wickham et al., 2019). The wide-format participant matrices were melted and split programmatically into a structurally cohesive “long” format. In this long configuration, the basic unit of statistical observation moves from the individual student to the individual morphosyntactic error token. Each discrete morphological variance produced or evaluated by a learner is defined as an independent observation row. For example, one French learner of English exposed to English in an institution for 12 years made five errors on items and four irregular verb errors, and thus contributed nine different structural rows to the long matrix. In each individual row, the background characteristics of the respective learner were systematically maintained, while the exact linguistic properties of the corresponding error token were captured satisfying the necessary distributional assumptions of the structural exploratory analysis (Husson et al., 2017).

The Coding Engine: Categorical Variable Taxonomy: To supply the data analysis with structurally valid categories a multi-layered linguistic taxonomy was developed in the R programming language according to common morphosyntactic classification schemes in SLA research (Lyster & Ranta, 1997). The long dataset has two main types of qualitative variables: **Linguistic Variables** (related to the grammar and morphosyntactic properties of the error) and **Extralinguistic/Typological Variables** (related to the socio-educational and genetic lines of the learners). The levels of structure that have been specified for each variable are outlined as follows:

Linguistic Variable: Type of Irregular Verb (Active Syntax): This variable identifies the underlying structural morphology of the target verb, enabling the model to capture whether certain groups of learners have difficulties with certain inflectional classes (Pinker, 1999). This variable is divided into four distinct categories, as presented below:

- **Strong Verb:** Verbs that form their past simple (V2) or past participle (V3) through internal stem vowel changes, including both systematic ablaut patterns (e.g., *drink-drank-drunk*) and morphologically restructured stems that historically originated from strong classes (e.g., *find-found-found*, *beat-beat-beaten*).
- **Weak Verb:** Verbs¹³ whose irregularity is not based on classic ablaut, but on minor dental suffixes or voicing changes without standard regular processing (e.g., *build-built*, *pay-paid*).

¹³ This category includes “hybrid” verbs, which combine vowel modification with dental suffixation. In this study, they are classified as **Weak Verbs** to ensure morphological consistency, as the dental suffix serves as their primary inflectional marker.

- **Invariable Verb:** Verbs having perfect syncretism of the base form, past simple and past participle. They have an unchanged phonetic and orthographic form (e.g. *put-put-put*, *cut-cut-cut*).
- **Suppletive Verb:** Verbs that use entirely different etymological roots to fill out their inflectional paradigm, displaying maximal irregularity, mainly evidenced by the copula *be* (*was/were/been*).

Linguistic Variable: Type of Error (Active Typology): This variable stands for the specific morphosyntactic and cognitive nature of the error produced by the learner. Errors are categorized across descriptive qualitative levels to observe interlanguage development (Selinker, 1972) rather than only classifying responses as binary correct or incorrect markers:

- **Overgeneralized_ed:** The regular use of the regular Germanic dental suffix (-ed) on an irregular verbal base as a clear cognitive default mechanism (e.g., *goed*, *putted*, *arised*) due to rule-based overextension (Pinker & Ullman, 2002).
- **Incorrect_Particiiple:** Structural syncretism errors where a learner confuses or misapplies irregular slots, such as using a past simple form in a perfect aspect frame (e.g. *had flew*, *has became*) or producing a structurally distorted vowel change. This category is used as the analytical label in my CSV data for the pattern commonly known in SLA research as “V2/V3 confusion”.
- **Auxiliary_Error:** Errors occurring in the secondary periphrastic layer of the verb phrase, such as the omission, misselection, or structural change of the auxiliary verbs (*have/be*) needed for aspectual or passive constructions.

Extralinguistic & Supplementary Variables (Learner Profiles): To find connections between these linguistic errors and learner identities, a set of demographic, educational, and cross-linguistic categories were set up in the R environment:

- **Country:** Investigates institutional and geopolitical educational backgrounds (e.g., France, Germany, Luxembourg, Turkey, Brazil).
- **Native Language (L1):** The actual native language of the participant (e.g., French, German, Turkish, Spanish, Portuguese) to evaluate cross-lingual influence (Odlin, 1989).
- **Language Family & Group:** A macro-level typological variable designed particularly to connect the micro-experimental data to the ICLE dataset, distinguishing learners according to genetic linguistic language families, with a particular focus on Romance versus Germanic structures (Jarvis & Pavlenko, 2008).
- **Years of English at School:** Gathers the structural duration of institutional SLA exposure, in independent ranges (e.g., 4 years, 8 years, 12 years, 6 years or more).

Data Reshaping: Wide to Long Categorical Restructuring: At the corpus level, this phase involved a key methodological decision: explicitly defining the roles of variables in the MCA

procedure. The linguistic categories (Irregular Verb Type and Error Type) were configured as Active Qualitative Variables according to the established structural linguistic methodologies (Desagulier, 2017). However, the learner-based background categories (Country, Native Language, Language Family, and Years of English) were labelled as Supplementary/Illustrative Qualitative Variables (`quali.sup`).

There is no mathematical or theoretical reason for configuring it differently. If background variables (e.g., country or language family) were included as active variables, the geometric algorithms of MCA would optimize the spatial distribution of the map to maximize the distances between human populations (Le Roux & Rouanet, 2010). This would add massive demographic variance to the core calculation, thereby obscuring the subtle structural variations that exist between the linguistic error patterns themselves. Only *Irregular Verb Type* and *Error Type* are in the active pool. The geometric axes of the spatial continuum are constructed as a pure, unaltered “Linguistic Error Space” (Desagulier, 2017). The categories of the learner background follow this phase, they are projected onto the pre-established structural grid, as passive vectors. This elegant distinction allows a researcher to be able to identify which distinct learner profiles are attracted to which independent clusters of irregular error patterns. A direct, neutral factual answer to the core psycholinguistic research questions is therefore provided. The resulting geometric axes do not serve to validate a specific hypothesis; rather, MCA is employed here as an exploratory tool to visualize the potential associations between categorical variables, such as learner profiles and error types.

Code Provenance and Computational Disclosure: In line with academic standards for computational research, the source of the data analysis scripts used in the current study is reported here. The foundational execution frameworks and the core statistical computing scripts for the Multiple Correspondence Analysis (MCA) were written by the thesis supervisor, **Professor Guillaume Desagulier**. In addition, **the Gemini artificial intelligence** assistance was used under careful human supervision to support extensive data cleaning procedures, the wide-to-long restructuring transformations, and the automated exploratory data plotting pipelines. The final integration, verification of algorithmic assumptions, script deployment in R Studio, and subsequent linguistic analytical modifications were executed entirely by the researcher.

3.5. Statistical Framework

This study employs a dual statistical approach based on the specific distributional properties of each data source to examine interactions between learner profiles and irregular verb deviations. Accordingly, the quantitative analysis is divided into descriptive frequency modeling for the questionnaire data and multidimensional exploratory geometric analysis for the ICLE corpus data. This reflects the structural characteristics of the corpus data and the response format of the questionnaire data.

The questionnaire data, involving a targeted group of 22 international participants, is analyzed using descriptive statistical methods within the R programming environment (Field, 2012). In general, non-linear dimensional reduction techniques are considered to be unsuitable for low-N questionnaire datasets due to the risk of artificial variance inflation and overfitting. Therefore, the intra-test productions and evaluations of the tests are presented in terms of absolute frequencies (f), relative accuracy percentages, and standardized error rates for the targeted structures. These descriptive metrics are then analyzed using bivariate cross-tabulations (contingency matrices) to track error types across learner background variables, such as Native Language (L1) groups and typological language families. For this experimental block, visualizations avoid complex spatial projections and instead use categorical bar plots and grouped percentage charts created with the `ggplot2` package. These provide a clear and transparent representation of interlanguage variation observed in the experiments (Wickham, 2016).

On the other hand, the corpus data at the macro-level collected from the International Corpus of Learner English (ICLE) contains high-dimensional categorical strings and requires an advanced exploratory multivariate method to explore and visualize the underlying associations within large contingency tables of nominal categorical variables (Desagulier, 2017). The extraction of this large-scale corpus is performed using Multiple Correspondence Analysis (MCA) to reduce the nominal contingency profiles into a low-dimensional Euclidean space. MCA can be considered as a generalization of Principal Component Analysis (PCA) for qualitative data (Greenacre, 2017). The calculation is carried out on an indicator matrix or a Burt matrix ($B = X^T X$) containing all dual cross-tabulations of the active linguistic variables. The total inertia ($\mathcal{I}$), corresponding to the overall variance or structural heterogeneity of the corpus error distribution, is obtained with the chi-squared (χ^2) distance metric as follows:

$$\mathcal{I} = \sum_i \sum_j \frac{(f_{ij} - e_{ij})^2}{n \cdot e_{ij}} \quad (1)$$

where f_{ij} is the observed frequency for a specific error-to-verb profile combination, e_{ij} is the expected frequency assuming independence, and n is the total number of corpus tokens recorded. By calculating eigenvalues and principal inertias, the high-dimensional matrix is reduced to a two-dimensional geometric continuum, establishing a baseline "Linguistic Error Space" (Greenacre, 2017).

The last layer of the statistical framework is the post-hoc mapping of extralinguistic learner variables onto the pre-calculated linguistic space. In order to avoid demographic imbalances or asymmetrical country-based sample sizes from influencing the structure of the core linguistic model, learner profiles such as L1, Country, and Language Family are assigned a strictly passive supplementary status (`quali.sup`) within the `FactoMineR` analysis

workflow (Husson et al., 2017). These illustrative variables do not take part in the initial eigenvalue extraction or the principal coordinate calculations. Instead, their categorical sublevels are represented on the fixed coordinates as supplementary coordinates (G_s), defined as the barycenter (center of gravity) of the active profiles sharing those extralinguistic features:

$$G_s = \frac{1}{n_s} \sum_{i \in I_s} F_i \quad (2)$$

where n_s is the total number of error tokens belonging to supplementary category s , and F_i is the coordinate vector of the active individual row i . This layout allows the study to visualize and evaluate the structural influence of different learner backgrounds in terms of point trajectories and confidence ellipses, while maintaining the geometric space as a neutral representation of grammatical and morphosyntactic error dynamics (Desagulier, 2017).

4. Results

The results of this study are divided into two main areas: an analysis of **corpus data** from the International Corpus of Learner English (ICLE) and **experimental data** from a specific learner group at Bordeaux Montaigne University. This section presents the frequency distributions, statistical significance tests, and visual mapping used to identify how learners acquire English irregular verb morphology.

4.1. Analysis of Irregular Verb Errors: The ICLE Dataset

This section of the results analyzes a dataset of 516 verified irregular verb errors extracted from the ICLE corpus. The analysis explores the relationship between a learner's native language (L1) and their morphological performance.

4.1.1. Global Analysis of Irregular Verb Errors

This section provides a high-level overview of the 516 irregular verb errors found in the ICLE corpus. Before diving into complex statistical modeling or specific linguistic comparisons, we will first look at the overall distribution of these errors. This baseline analysis identifies broad trends and patterns across learners from many different language backgrounds.

The data reveals a strong cognitive preference for paradigm regularization. Across the entire corpus, regardless of the learner's native language, morphosyntactic errors are mainly caused by one structural strategy: **Overgeneralization** (*overgeneralized_ed*). This makes up

66.28% ($n=342$) of the verified dataset, providing strong evidence for the psychological dominance of regular inflectional rules. When non-native speakers experience retrieval difficulties or performance slips during spontaneous writing, their usual strategy is to apply the highly productive past tense suffix *-ed*. In contrast, **Incorrect Participle** errors (*incorrect_participle*) form the second-largest category, representing 28.10% ($n=145$) of the total dataset. This error type indicates a specific developmental stage where learners acknowledge irregular past-reference markers but struggle to correctly use the precise phonological forms or vowel changes (ablaut) necessary for perfective or passive structures. This category includes all instances of “suffix omission” (e.g., *has broke* for *has broken*), where the learner fails to attach the required *-en* suffix as discussed in the theoretical framework (see Section 2.3.2.4). Another category, **Auxiliary Errors** (*auxiliary_error*), which involve non-standard structural combinations or tense/aspect mismatches, represents a smaller portion at 5.04% ($n=26$). Finally, the uncategorized errors (**Other**)¹⁴ account for a minor 0.58% ($n=3$) of the overall dataset.

When the dataset is cross-tabulated by the learners' native language families, the relationship between L1 structural typology and L2 performance becomes highly visible. Table 7a presents the frequency distribution, while Table 7b outlines the row-normalized percentage distributions across different language groups.

Table 7a. Cross-Tabulation of Language Families by Specific Error Types

Language Family	Auxiliary Error	Incorrect Participle	Other	Overgeneralized_ed	Total (n)
Germanic	5	7	0	45	57
Indo-European (Other) ¹⁵	9	50	1	100	160
Japonic	1	5	1	18	25
Koreanic	5	10	0	3	18
Niger-Congo	2	18	0	26	46
Romance	4	26	0	70	100
Sino-Tibetan	0	18	0	52	70
Turkic	0	2	0	13	15
Uralic	0	9	1	15	25

¹⁴ The "Other" category consists of three unclassified error tokens ($n=3$) that did not align with established morphological paradigms (*overgeneralized_ed*, *incorrect_participle*, *auxiliary_error*), specifically including two instances of “lay → laid” and one instance of “lie → laid”.

¹⁵ The category "Indo-European (Other)" includes all language backgrounds within the Indo-European family except Germanic or Romance languages.

Table 7b. Normalized Percentage Distribution of Error Types Across Language Families

Language Family	Auxiliary Error	Incorrect Participle	Other	Overgeneralized_ed
Germanic	8.77%	12.28%	0.00%	78.95%
Indo-European (Other)	5.62%	31.25%	0.62%	62.50%
Japonic	4.00%	20.00%	4.00%	72.00%
Koreanic	27.78%	55.56%	0.00%	16.67%
Niger-Congo	4.35%	39.13%	0.00%	56.52%
Romance	4.00%	26.00%	0.00%	70.00%
Sino-Tibetan	0.00%	25.71%	0.00%	74.29%
Turkic	0.00%	13.33%	0.00%	86.67%
Uralic	0.00%	36.00%	4.00%	60.00%

Examining these proportions reveals variations in how different learners process morphology. The highest rate of rule overgeneralization is documented among Turkic L1 speakers, where 86.67% of errors involve the incorrect addition of the standard regular suffix. This is closely followed by Germanic speakers at 78.95% and Sino-Tibetan learners at 74.29%. The high frequency within the Turkic L1 group may reflect the structural influence of a highly regular, agglutinative native language, where straightforward suffixation is typologically common.

On the other hand, Koreanic L1 speakers differ greatly from this global regularizing trend. For Koreanic learners, overgeneralization drops to a minimum of 16.67%, whereas Incorrect Participle increases to a dominant 55.56%, along with a 27.78% rate of Auxiliary Errors. This clear shift implies a unique pedagogical or typological difficulty: rather than relying on standard regular forms, these learners actively produce non-standard irregular variants or experience profound structural challenges when dealing with English auxiliary and aspectual structures. Meanwhile, Romance learners occupy a highly stable, intermediate cross-linguistic position, with 70.00% of their errors stemming from overgeneralization and 26.00% from improper participle selection, reflecting a balanced distribution between regular rule extension and irregular phonological confusion.

Beyond the error types themselves, a systematic global analysis requires looking at how these language families interact with specific morphological classes of irregular verbs (Strong, Weak, Invariable, and Suppletive). Table 8 presents this interaction.

Table 8. Distribution of Targeted Irregular Verb Types Across Language Families

Language Family	Invariable	Invariable / Weak ¹⁶	Strong	Suppletive	Weak	Total
Germanic	13	0	21	0	23	57
Indo-European (Other)	29	0	69	10	52	160
Japonic	6	0	9	1	9	25
Koreanic	1	0	8	4	5	18
Niger-Congo	3	0	17	2	24	46
Romance	21	2	38	5	34	100
Sino-Tibetan	18	0	34	3	15	70
Turkic	5	0	6	0	4	15
Uralic	4	0	10	0	11	25

The data shows that **Strong Verbs** are the main source of difficulty across nearly all language groups, totaling 212 errors. The complex vowel alternations (ablaut) required by these verbs (e.g., become → became/become) consistently place a heavy cognitive load on learners. **Weak Verbs** are the second most variable category with 177 errors, demonstrating that even hybrid paradigms (e.g., pay → paid) frequently trigger systemic confusion. Interestingly, Invariable Verbs constitute a baseline of 100 errors globally, with apparent frequencies in the Romance (n=21), Sino-Tibetan (n=18), and Germanic (n=13) L1 groups. Because these verbs resist standard morphosyntactic changes, learners from diverse linguistic backgrounds share a common tendency to "overgeneralize" these zero-inflection forms, frequently leading to non-target forms such as *broadcasted* or *hurted*. In summary, the global learners data suggests that irregular verb errors are mostly due to common cognitive factors, especially the overgeneralization of regular rules and morphological patterns. This pattern is strongly influenced by the learner's language family. This macro-perspective establishes a clear groundwork for the targeted statistical evaluation of the two most prominent European language families, which follows in the next section.

¹⁶ The "Invariable / Weak" category is used only for the verb "read." While its spelling never changes, its pronunciation shifts in the past and participle forms, making it a mix of both paradigms.

4.1.2. Comparative Analysis of Irregular Verbs: Contrastive Results from Romance vs. Germanic L1 Speakers

After the global error analysis, this section analyzes the comparative performance of learners from two distinct language families: Romance and Germanic. By examining the relationship between native language typologies and irregular verb errors, it explores whether L1 backgrounds influence the specific strategies learners use in the production of English morphology.

In order to comprehend the comparative distribution of error types between Romance and Germanic speakers, the table below provides a clear framework for comparing these two language families.

Table 9. Error Distribution by Language Family

Language Family	Error Type	Count	Percentage (%)
Germanic	Auxiliary Error	5	3.21%
	Incorrect Participle	7	4.49%
	Overgeneralized_ed	45	28.80%
Romance	Auxiliary Error	3	1.92%
	Incorrect Participle	26	16.70%
	Overgeneralized_ed	70	44.90%

As we can infer from Table 9, *overgeneralized_ed* is the predominant error type for both language groups. However, Romance L1 speakers exhibit nearly four times the tendency for *incorrect_participle* errors (16.70%) compared to Germanic speakers (4.49%). This suggests that while both groups use mainly regularization as a default cognitive strategy, Romance learners face specific challenges in producing the phonological or ablaut-based variations required for past participle forms.

In addition to the quantitative distribution, the error tokens in the dataset expose specific morphosyntactic challenges. For instance, the production of "did not took" by a Germanic learner suggests an interference pattern where the auxiliary "did" fails to trigger the required base form, illustrating a potential conflict between their L1 morphological habits and English tense-aspect rules. In comparison, Romance learners often produce incorrect participle errors such as "is took" or "have be." These errors suggest that the issue is not only overgeneralization, but a struggle to properly adapt English "be" or "have" auxiliary constructions to their native Romance aspectual structures.

A Pearson's chi-squared test was performed to determine the statistical significance of the relationship between language family and error type. The analysis generated a χ^2 value of 6.0015 ($df = 2, p = 0.04975$), validating that the variation in error patterns between the two linguistic groups is statistically meaningful ($p < 0.05$).

In order to decide whether error patterns vary in relation to the duration of formal instruction, we examined the distribution of these errors across varying levels of English education of the learners (Figures 3a and 3b).

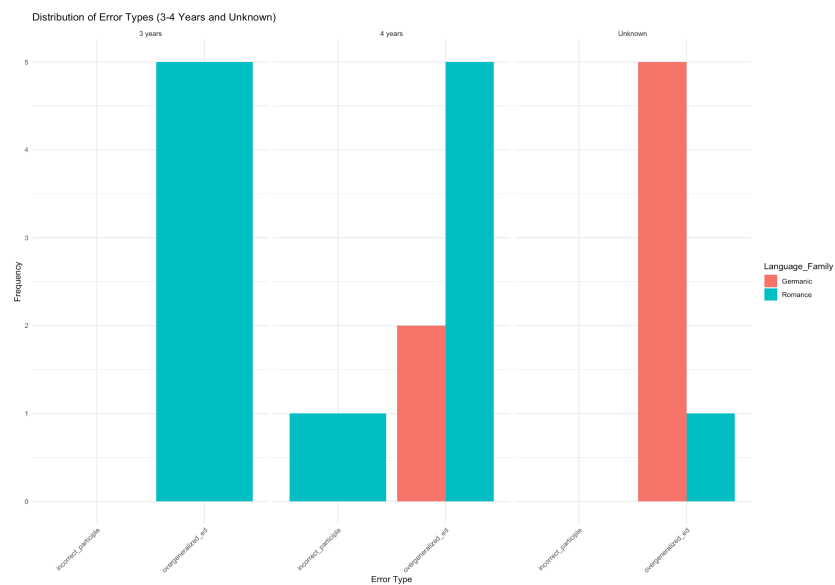


Figure 3a. Error Distribution: Early Stages (3-4 Years and Unknown)

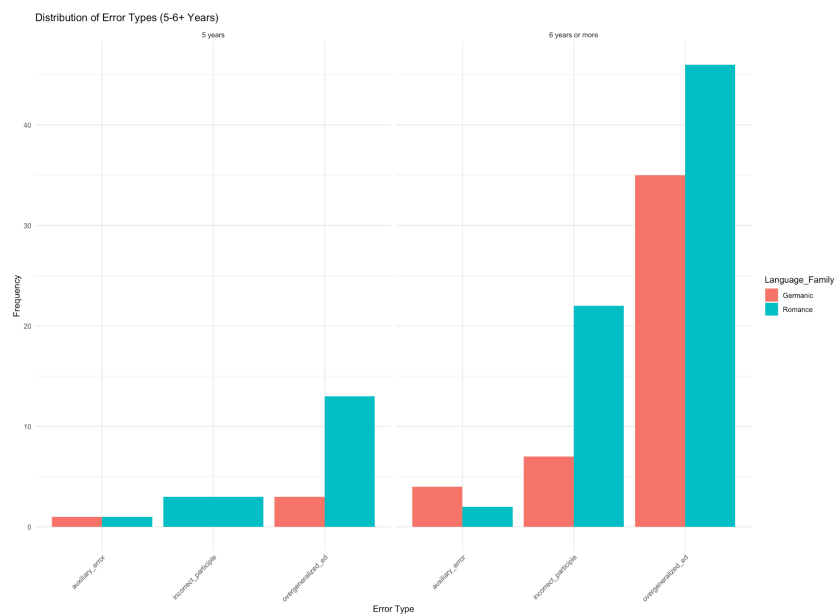


Figure 3b. Error Distribution: Advanced Stages (5-6+ Years)

As seen in both figures, error patterns change as learners spend more time studying English. In the earlier stages (3–4 years and unknown), errors are **mostly** in the form of *overgeneralized_ed*, which shows that students are mostly using simple rule-based strategies in this study. However, as learners exceed five years of study, the errors become more varied, including more *incorrect_participle* and *auxiliary errors*. This shift suggests that as students become more advanced, their errors show that they are trying to understand and use more complex English structures.

Figures above also show that learners continue to use regularized forms of irregular verbs even after many years of study. In the highest duration group (5-6+ years), the frequency of *overgeneralized_ed* still remains high. This implies that when students have trouble remembering irregular verb forms, they still rely on adding the regular suffix as a primary strategy, no matter how many years they have spent learning English.

Finally, we assessed how students with different linguistic backgrounds handle specific types of irregular verbs—Strong, Weak, Suppletive, and Invariable. Table 10 details how often these different types of errors appear in our data.

Table 10. Irregular Verb Type Distribution by Language Family

Language Family	Irregular Verb Type	Error Count
Germanic	Invariable Verb	13
	Strong Verb	21
	Weak Verb	23
Romance	Invariable Verb	21
	Strong Verb	38
	Suppletive Verb	5
	Weak Verb	33
	Weak Verb / Invariable Verb	2

Table 10 shows how often students struggle with “Strong Verbs,” referring to a difficulty that persists regardless of the learner’s native language. L1 Romance students produced more errors in this category (n=38) than their L1 Germanic peers (n=21). This difference suggests that the vowel-based structures of Strong Verbs, such as the *take-took-taken* model, are particularly challenging for Romance learners. While Germanic students may find these patterns more

familiar due to their language's shared history with English, Romance learners face a larger gap between their native systems and English. This often leads to more non-target forms as they attempt to produce the complex vowel changes required by these verbs.

In conclusion, the results show that while regularization is a common strategy for all learners, the way auxiliary and participle errors occur reflects the impact of the learner's native language. Romance speakers, above all, face more difficulty with non-regular morphological variations, reinforcing the fact that a learner's L1 background plays a central role in how they manage the irregularities of English verb morphology.

4.2. Visualizing Error Patterns through MCA Plots

While the descriptive statistics presented in Sections 4.1.1 and 4.1.2 effectively highlight the frequency and distribution of specific error types across both L1 groups and language families, they offer **only a partial view of the implicit** associations between learner variables and morphological performance. Standard quantitative counts treat error categories as independent entities, whereas language acquisition is a multidimensional process where variables often interact. Therefore, this section adopts **Multiple Correspondence Analysis (MCA)** to provide a more integrated perspective on these associations. By placing linguistic error variables (*Irregular_Verb_Type*, *Error_Type*) as “active” variables in a geometric error space, MCA allows us to position learner profiles, such as *Native_Language*, *Language_Family*, *Country*, and *Years_of_English_at_School*, as illustrative variables. This approach enables us to observe not just how often errors occur, but where they cluster within the cognitive map of the learner's interlanguage¹⁷.

The MCA model developed for this study captures 43.2% of the total inertia within the first two dimensions, which represents a considerable proportion for categorical data of this complexity. The high eigenvalues, Dim 1 (0.716) and Dim 2 (0.580), suggest the model's analytical stability and its reliability in differentiating between distinct error profiles.

Dimension 1 (23.9% of total variance) acts as a principal separator of error complexity: on one pole, we observe simple overregularization patterns, while the opposite pole is dominated by complex morphological deviations, such as auxiliary errors and suppletive verb errors.

Dimension 2 (19.3%) appears to capture typological and verb-specific variance. The quality of representation for these categories is confirmed by their Cos^2 values. For instance, “overgeneralized_ed” demonstrates a high Cos^2 value (0.65), indicating that this error type is represented with high accuracy on the map and serves as a central focal point within the error space. It is also worth noting that the eigenvalue of 0.716 for Dimension 1 indicates a high discrimination power regarding error types. The observed gap between Dimension 1 and Dimension 2 implies that the primary source of variance within the ICLE dataset is the constitutive morphological complexity of the verbs, rather than raw frequency of occurrence.

¹⁷ The code used to generate all MCA plots presented in Section 4.2 (Visualizing Error Patterns through MCA Plots) was provided by my supervisor, Mr. Desaguiler.

This structural stability provides a solid analytical basis for interpreting the following proximity maps with a high degree of confidence.

4.2.1. Geometry of Error Categories (Active Map)

The figure below depicts the geometric distribution of active error categories within the computed MCA space. This spatial arrangement explicates that these errors are not randomly distributed; rather, they form distinct clusters that reflect their implicit linguistic properties.

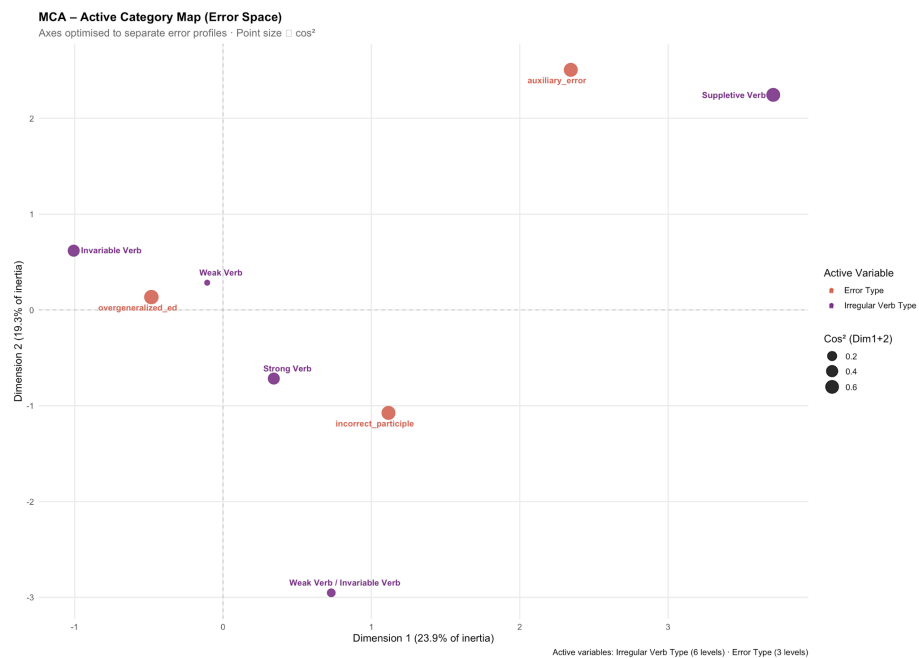


Figure 4. Active Category Map (Error Space)

The figure above shows how active error categories are distributed in the map. The distance between *auxiliary_error* and *incorrect_participle* unveils that they come from different cognitive sources. Intriguingly, the *auxiliary_error* category sits far from the center, unlike the *overgeneralized_ed* category, which is positioned in the neutral middle. Particularly, the gap between *auxiliary_error* and *incorrect_participle* is quite large, about 4 units on the plot. This gap is quite illuminating: it reveals that while *overgeneralized_ed* is a common, default strategy for simplifying morphology, *auxiliary_error* requires a different processing strategy linked to the syntax-morphology interface.

The close proximity of *incorrect_participle* to *Strong Verbs* (approximately 0.9 units) points to a systematic challenge in processing irregular verbs. While *overgeneralized_ed* clusters near the center, acting as a default strategy that learners use regardless of the linguistic context, the *auxiliary_error* category is located in a more complex, peripheral region. This plot visualizes the fact that error types are not a single group; instead, they occupy different cognitive territories.

While *overgeneralized_ed* serves as a “fallback” strategy in the interlanguage, auxiliary and participle errors represent more advanced, yet still inaccurate, attempts to handle the complex Tense-aspect categories of English.

4.2.2. Projecting Learner Profiles

To uncover the underlying associations between learner backgrounds and error categories, we integrated **illustrative variables** into the coordinate space here.

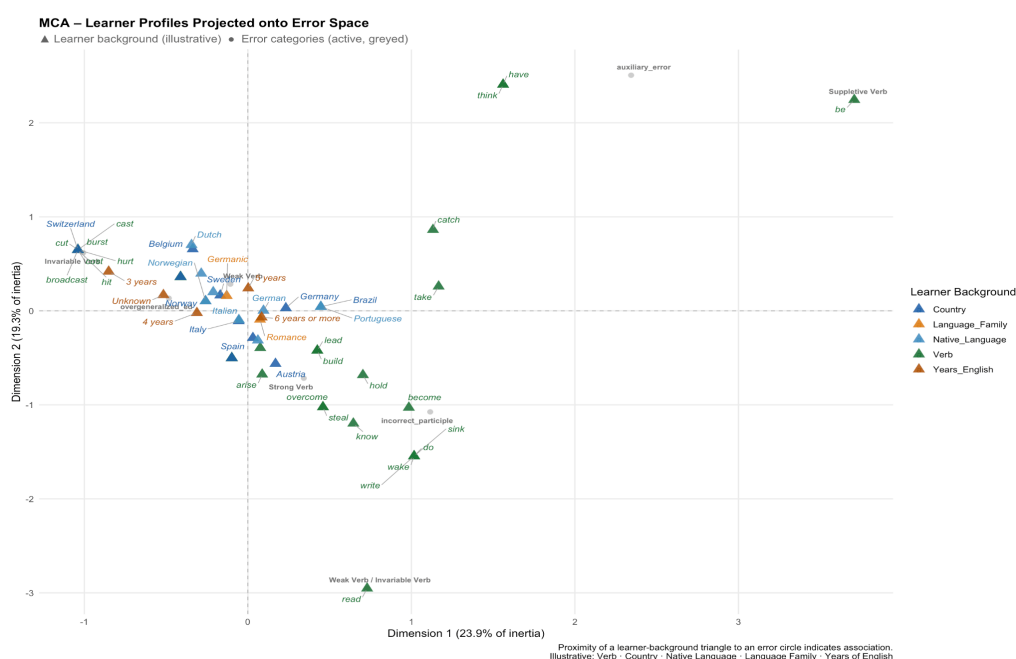


Figure 5: MCA – Learner Profiles Projected onto Error Space

The figure above provides a profound view of how different learner groups relate to these error types. The Romance language family cluster, including Italy and Spain, sits closer to the *incorrect_participle* and *Strong Verb* areas. This illustrates how **Romance** speakers, due to the typological gap from English irregular verbs, face a higher cognitive load when dealing with vowel changes. On the other hand, **Germanic** speakers, such as Germany and the Netherlands, appear more spread out, implying that their errors are less tied to specific verb types and more to general developmental progress.

What is more, the projection of instruction duration of English at an institute reveals systematic trends. The “3-year” group clusters tightly near the *overgeneralized_ed* region, while the “6+ year” group shows a gradual transition into more complex areas, such as *suppletive_verb* errors. Above all, even these advanced learners maintain a central position on the map rather than reaching the extreme periphery. This shows that extended exposure to English does not automatically lead to perfect accuracy; instead, it reorients the learner's error profile from simple,

algorithmic overgeneralizations toward more specific, irregular verb challenges. Most importantly, the fact that advanced learners still associate with the *overgeneralized_ed* cluster validates that regularization persists as a stable feature of interlanguage. This shows a tendency for the fallback to *-ed* suffixation remains a defining characteristic of the learner's profile despite extended exposure, supporting Ellis's (2015) findings on the non-linear development of error profiles in L2 acquisition.

The final symmetric map synthesizes individual learner scores, active error categories, and illustrative background variables, offering a comprehensive view of the interlanguage landscape.

Figure 6. Full Symmetric Map

systematic configurations influenced by both the learner's L1 and the cognitive load requirements of English irregular verbs.

Additionally, the distribution of individual data points contradicts the null hypothesis of uniform error patterns. For example, the position of the Brazilian learner profiles at the edge of the *auxiliary_error* space indicates a divergent cognitive pattern that departs from the general Romance profile. These clusters show that the intersection of Dimension 1 (Complexity) and Dimension 2 (Typology) acts as a coordinate system where a learner's background exercises a systematic attraction toward specific irregular verb challenges. In conclusion, this map suggests a systematic structure in L2 interlanguage, where learner acquisition paths appear to be conditioned by the interplay between typological relationship and morphological difficulty.

4.3. Experimental Results from Bordeaux Montaigne University (2LLIAM222-MATI Classes)

This section is dedicated to the experimental results derived from the questionnaires conducted with 22 undergraduate students at Bordeaux Montaigne University. By analyzing individual error patterns and group-level morphosyntactic performance, this study aims to bridge the gap between static corpus data, as examined within and extended beyond the ICLEv3 framework, and completely spontaneous, controlled language production. The results provide a detailed view of how advanced L2 learners navigate irregular verbal inflections under varying task demands.

4.3.1. Cognitive Load and Task Demands

The experimental design was built on the psycholinguistic assumption that the level of monitoring and processing demand influences the stability of interlanguage systems (Hulstijn, 2001). Test 1 was structured to minimize cognitive load with recognition-based tasks, while Test 2 was designed to maximize load by requiring spontaneous production, thereby creating a "real-life" performance context (Ellis et al., 2009).

According to Schmidt's (2001) Noticing Hypothesis, the transition from explicit grammatical recognition to active recall often reveals systemic weaknesses in the learner's mental lexicon. By comparing these two separate testing formats, this study investigates the processing limits of morphological accuracy. The hypothesis is that the transition from the structured environment of Test 1 to the unguided production of Test 2 will correlate with a higher frequency of non-standard morphosyntactic forms, particularly regarding auxiliary selection and stem vowel modifications (Pinker & Ullman, 2002).

4.3.2. Quantitative Analysis of Morphosyntactic Error Patterns

The global error distribution between the two tests reveals a significant change in learner performance when moving from guided recognition (Test 1) to production-based tasks (Test 2).

The table below¹⁸ summarizes the frequency of error types, highlighting the impact of task-based cognitive load on the morphosyntactic stability of the learner’s interlanguage.

Table 11. Global Distribution of Morphosyntactic Errors by Task Type

Error Category	Test 1 (Frequency)	Test 2 (Frequency)
Auxiliary Error	15	81
Incorrect Participle	77	70
Overgeneralized -ed	22	18
Total	114	169

As shown in the table above, the data reveal different behavioral patterns for each error category under diverse cognitive demands. The most significant finding is the considerable increase in *auxiliary errors* during Test 2, with a 440% increase. This suggests that auxiliary selection (e.g., *did* + base form) may be more vulnerable to processing limitations during spontaneous production, consistent with Hulstijn’s (2002) account of working memory limitations in online processing, whereas performance is more stable in monitored and controlled tasks.

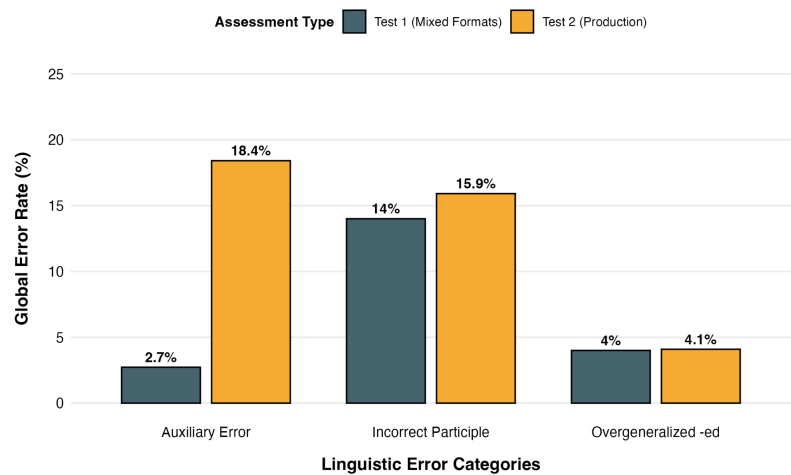


Figure 7. Global Error Distribution by Task Type

The distribution highlights a clear contrast; while auxiliary errors increase under production pressure, other categories follow distinct paths. For instance, incorrect participle errors remain the most frequent category in both tasks (77 in Test 1 vs. 70 in Test 2). Unlike

¹⁸ N = 22 participants. Values represent the absolute frequency of error tokens.

auxiliary errors, the high frequency of participle errors reflects a persistent retrieval failure of irregular forms within the lexical system. According to the Words-and-Rules (WR) model (Pinker & Ullman, 2002), irregular forms are stored in declarative/associative memory as unique entries and must be retrieved rather than formed by rule. Because these forms rely on lexical lookup, learners frequently fail to access them under high processing demands, causing the grammatical system to use instead rule application and produce “regularized” forms.

Lastly, overgeneralized -ed errors show a small decrease in Test 2. This is theoretically relevant; while learners are often expected to rely on regular morphology under stress, the participants displayed a tendency to avoid the *-ed* suffix instead of using other incorrect stem variations or auxiliary avoidance when they could not recall the target irregular form. This points to the difficulty that lies less in knowledge of the regular rule and more in retrieving specific irregular lexical items under pressure.

To provide additional qualitative explanation on these quantitative patterns, Table 12 illustrates representative examples of morphosyntactic deviations identified in the participant data.

Table 12. Representative Examples of Morphosyntactic Deviations by Task

Targeted Form	Learner Deviation	Error Type
made	maded	Overgeneralized -ed
hasn't driven	hasn't drove	Incorrect Participle
didn't think	didn't thought	Auxiliary Error
put	putted	Overgeneralized -ed
has broken	broke	Auxiliary Error
went	goed	Overgeneralized -ed

The classification of these errors clarifies how learners cope with irregular verbs.

Overgeneralized -ed errors (e.g., *maded*, *putted*, *goed*) occur when a learner applies a regular rule to an irregular verb, prioritizing a "shortcut" strategy over retrieving the specific irregular form from memory. **Incorrect Participle errors** (e.g., *hasn't drove*) involve a confusion between verb forms, where the learner uses a simple past form instead of the required past participle. Finally, **Auxiliary errors** (e.g., *didn't thought*) are exemplified by instances of “double marking” of the past tense. These errors suggest that when task demands are high, the learner’s processing system struggles to manage multiple grammatical requirements, which supports Skehan's (2018) framework regarding **Limited Attention Capacity** and the resulting trade-offs in accuracy during spontaneous production.

The results of the analysis are as follows:

$$\chi^2(2) = 36.81, p < 0.001, \text{Cramer's } V = 0.36.$$

The Chi-square analysis indicates a meaningful association between task type and error frequency ($\chi^2(2)=36,81, p<0.001$). This relationship is reinforced by a Cramer's V of 0.36, indicating a strong effect size where the occurrence and distribution of specific error types are largely influenced by the cognitive demands of the production task.

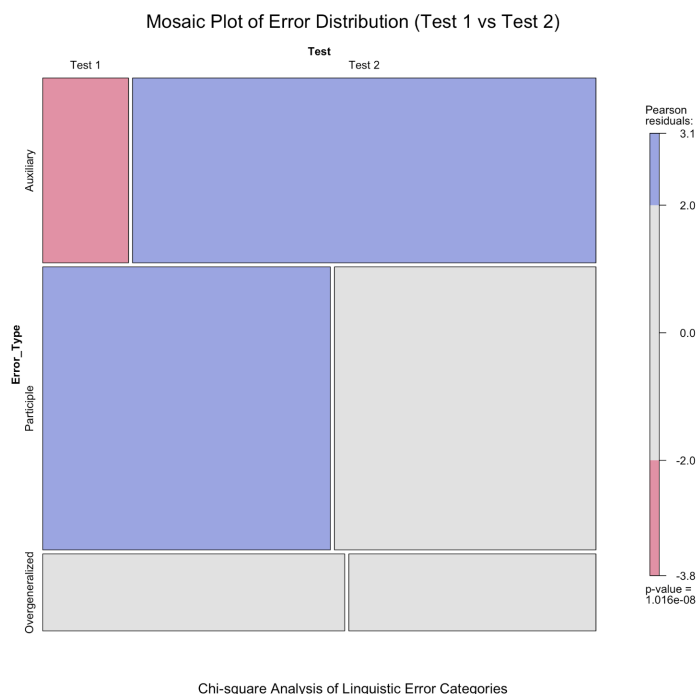


Figure 8. Mosaic Plot of Error Frequencies: Chi-Square Contingency Analysis

As seen in figure, the Pearson residual analysis reveals that the distribution of error types is not randomly distributed between testing conditions. Remarkably, the standardized residual of 6.06 for the *Auxiliary* error category in **Test 2** emphasizes a significant variance from expected frequencies, confirming that these errors are excessively concentrated in the second task. This quantitative distribution functions as the basis for a more detailed analysis of morphological verb categories, which will be addressed in the following section about morphological verb classes.

To extend this analysis, an item-level ranking identifies the specific verbs that triggered the highest frequency of errors across both testing conditions. In **Test 1**, the most error-prone items are found only in the written production section (Section 4). For example, verbs requiring the retrieval of irregular forms, such as think (item X23), drive (X25), take (X24), and sing (X19), account for the highest error counts, with up to 15 instances per item. These results highlight the main source of the "retrieval failure" within the participant pool; even in a

controlled, mixed-format task, these specific irregular stems remained difficult for learners to access in declarative memory.

Table 13. Top 5 Most Error-Prone Verb Items by Task Condition

Test and Item	Target Verb	Error Category	Frequency
Test 1			
X23	Didn't think	Incorrect Participle	15
X25	Hasn't driven	Incorrect Participle	15
X24	Didn't take	Incorrect Participle	14
X22	Didn't pay	Overgeneralized -ed	11
X19	Haven't sung	Incorrect Participle	10
Test 2			
X20	Hadn't had	Auxiliary Error	13
X15	Hadn't drunk	Incorrect Participle	10
X19	Had forgotten	Auxiliary Error	10
X11	Had been	Auxiliary Error	9
X13	Had become	Auxiliary Error	9

The items listed in Table 13¹⁹ reveal an explicit shift in the error distribution between the two tasks. While **Test 1** is dominated by retrieval issues for specific irregular participles, the rankings for **Test 2** are characterized by auxiliary errors in items such as *Hadn't had* (X20), *Had forgotten* (X19), and *Had been* (X11). This item-level transition provides specific evidence for the increase in auxiliary errors reported in Table 11. It suggests that under the high cognitive load of spontaneous production, the learners' processing systems struggle to manage multiple grammatical requirements simultaneously, such as the correct selection of auxiliary verbs and tense markers. This pattern of auxiliary errors in the top-ranked items reinforces Skehan's (2018) framework, where the pressure of active recall during spontaneous production forces a trade-off in accuracy. This quantitative distribution functions as the starting point for a more detailed analysis of morphological verb categories, which will be addressed in the following section.

¹⁹ Item numbers (X#) correspond to the question sequence in the experimental instruments

4.3.3. Distributional Analysis of Morphological Verb Classes

To extend the analysis of the cognitive processing underlying the observed morphosyntactic errors, in this section verbs are categorized into four morphological classes: *Weak*, *Strong*, *Invariable*, and *Suppletive*. Applying the dual-mechanism logic presented in the previous section, the Weak verb category is treated here as the result of the procedural rule system. In contrast, the *Strong*, *Invariable*, and *Suppletive* categories represent irregular forms that are dependent on the lexical retrieval capacity of the declarative system.

The table²⁰ below highlights differences in how error types are distributed across the two tasks, reflecting distinct processing demands.

Table 14. Distribution of Morphological Verb Errors by Test Format

Verb Type	Test 1 (Count)	Test 2 (Count)
Weak	36	2
Strong	65	149
Suppletive	8	15
Invariable	5	3

The morphological error distribution reveals that irregular verb accuracy is closely related to assessment format, rather than being equally distributed across categories. In Test 1, errors are more spread across categories, with Strong verbs being the most affected category (n=65), followed closely by Weak verbs (n=36), which mainly explain minor overgeneralizations. However, the introduction of written production tasks in Test 2 changes importantly the overall error profile. Under free production constraints, errors in Weak verbs drop to near zero (n=2), while the overall error distribution concentrates heavily on Strong verbs (n=149) and, less prominently, Suppletive verbs (n=15). This clear shift of error frequencies indicates that changing the task modality from structured recognition to free production fundamentally reshapes the distribution of learner difficulties.

²⁰ Data represent raw frequency counts of morphological errors categorized by verb type. Test 1 and Test 2 reflect performance across mixed-format and spontaneous production tasks, respectively.

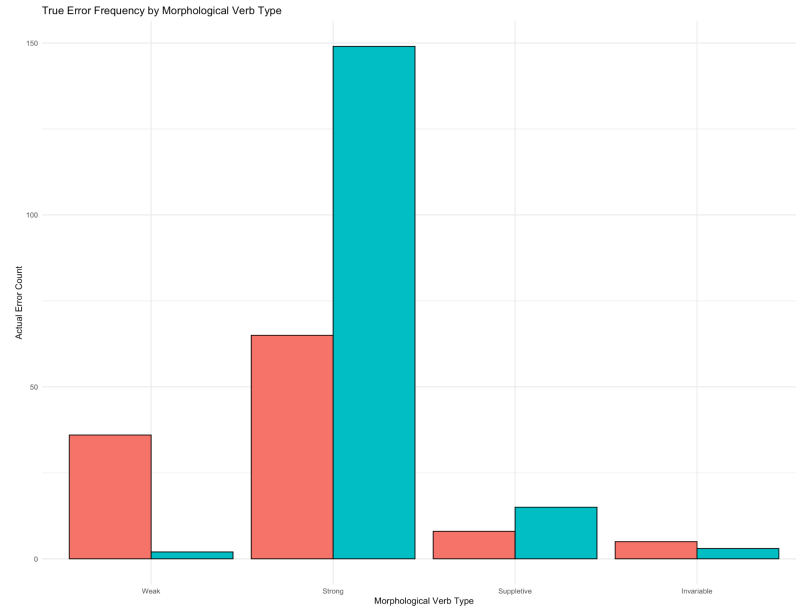


Figure 9. Comparison of Error Frequencies Between Morphological Verb Classes and Test Conditions

As seen in the graph above, Strong verbs exhibit a huge increase in errors, more than doubling from Test 1 ($n=65$) to Test 2 ($n=149$). This visual contrast provides experimental support for the dual-mechanism predictions of the Words-and-Rules framework. Spontaneous production eliminates structured support or recognition cues for learners, strongly engaging the associative lexical memory required to retrieve complex irregular stem changes. The doubling in Suppletive verb errors (8 to 15) further reinforces this "lexical retrieval barrier" during spontaneous production conditions.

Conversely, the near disappearance of "Weak" errors in Test 2 ($n=2$) suggests a different cognitive process: when producing language independently, learners seem to pay more attention to their grammar more carefully, successfully avoiding the overgeneralization errors seen in the structured tasks. In short, Figure 9 shows that while structured formats allow learners to rely on external cues, the pressure of spontaneous production challenges their lexical retrieval capacity. This leads to reduced grammatical accuracy, particularly in complex irregular categories, where the mind struggles to access the correct forms under pressure.

4.3.4. Socio-linguistic Variables: Native Language and Country Profiles

This section examines the influence of socio-linguistic variables, particularly, the participants' native language backgrounds and their countries of birth, on morphosyntactic accuracy. By comparing error distributions across these variables, we aim to determine whether performance patterns are consistent among linguistic groups or if specific **L1-related transfer effects** or **geographic limitations** influence the processing of English morphological verb classes.

The distribution of responses by native language is summarized in Table 15. Within the results obtained from the participants, French-speaking students represent the majority of the dataset, with sub-groupings also covering diverse multilingual profiles (e.g., French-Algerian, French-Russian).

Table 15. Response Frequency by Native Language and Test Condition

Native Language	Test 1 (Correct)	Test 1 (Errors)	Test 2 (Correct)	Test 2 (Errors)
English	19	6	10	10
French	318	82	188	132
French, Algerian	14	11	8	12
French, Portuguese	25	0	20	0
French, Russian	24	1	15	5
French, Spanish	15	10	11	9
German, French	21	4	19	1

The data above²¹ shows that while French-speaking monolinguals make up the majority of our participant pool, their performance exhibits a clear change in error density from Test 1 to Test 2, with errors rising from 82 to 132. This indicates that even within the majority group, the transition between the tasks challenges the morphological processing system. More importantly, however, are the multilingual sub-groupings; specifically, the French-Portuguese participant demonstrates a performance profile where errors remain at 0 during both tests, while the French-Russian participant demonstrates an increase in errors from 1 to 5 during the spontaneous task. This recurring difficulty suggests that the cognitive load associated with managing multiple linguistic systems creates an interference from multiple languages. Unlike the English speaking participant, who maintains relative stability (with errors rising from 6 to 10), these multilingual learners appear to reach a "retrieval limit" in Test 2, where the demand for immediate output reduces their ability to access irregular verb forms accurately. This supports an interpretation based on processing limitations, where the cognitive load of managing multiple linguistic systems makes the retrieval of irregular forms more difficult. This pattern is consistent with the idea that attentional capacity is restricted; when these task demands increase during spontaneous production, accuracy in the lexical retrieval of irregulars decreases (Skehan, 2018).

²¹ Data here represent total frequencies of correct vs. incorrect responses per linguistic background based on the dataset.

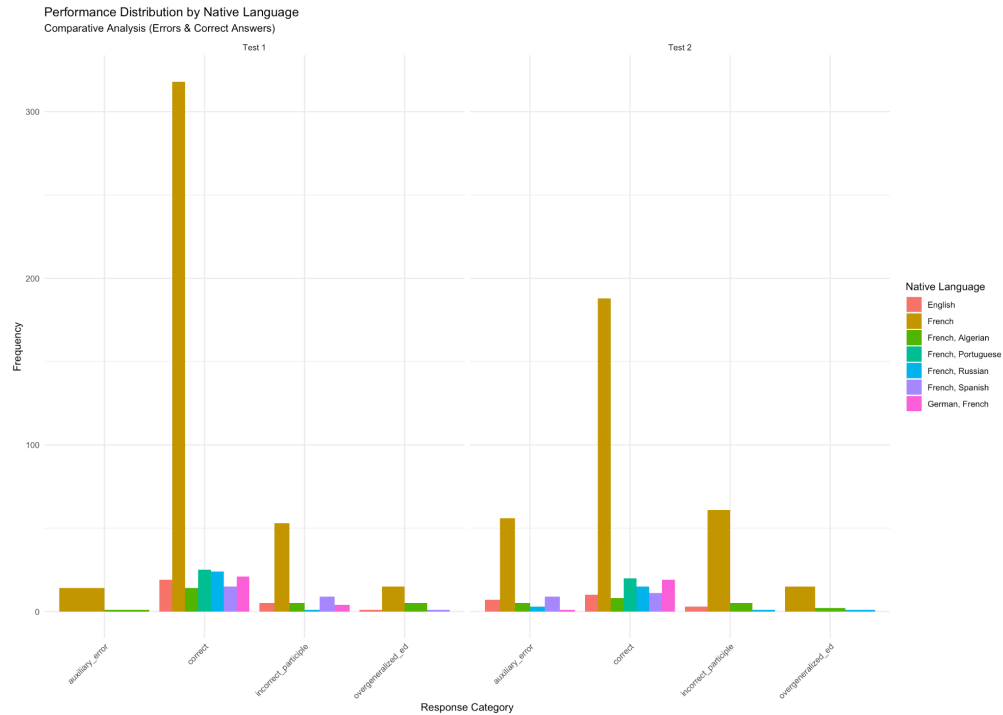


Figure 10. Performance Distribution by Native Language

Visual analysis of participants' performance in relation to their L1s highlights the inconsistent distribution of performance results among test conditions. The main trend is the substantial reduction in "correct" responses and the corresponding growth in "incorrect_participle" and "auxiliary_error" categories during Test 2, particularly within the French-speaking participant group. Overall, this change suggests that the spontaneous production task acts as a stress test for morphological knowledge. The visualization reveals that the previously discussed patterns of regularization are systemic across both monolingual and multilingual groups. This suggests that the time pressure of spontaneous tasks acts as a universal stress test for the declarative memory system, regardless of the learner's L1 background.

To explore the role of external linguistic exposure in morphosyntactic accuracy, participants' performance was also categorized according to their country of birth, which was used as an indicator of the intensity and nature of their English input. The error and correct response frequencies are summarized in Table 16.

Table 16. Error and Correct Response Frequencies by Country

Country	Test 1 (Correct)	Test 1 (Errors)	Test 2 (Correct)	Test 2 (Errors)
England	19	6	10	10
France	347	103	207	153

Germany	21	4	19	1
Luxembourg	24	1	15	5
Portugal	25	0	20	0

The data²², presented in the table above, show a geographic performance gap, pointing to the role of the environment in morphosyntactic stability. Participants born in **France** exhibit a considerable increase in error frequency during Test 2, with errors rising from 103 to 153. This rise becomes more evident when compared to the participant born in **England**, who shows a higher error density in the Test 2 compared to their performance in the Test 1.

The stability observed in other groups, most clearly in Germany and Portugal groups, implies that language exposure is not the only factor involved; instead, the increase in errors of the participants from France appears to reflect a group-specific pattern within the largest participant pool.

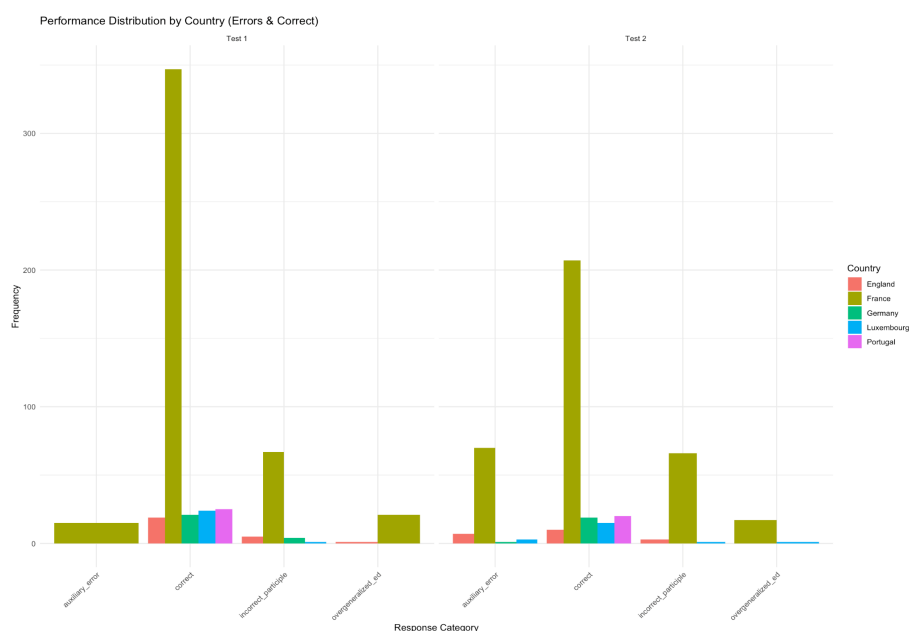


Figure 11. Distribution of Response Types by Country and Test Condition

The figure illustrates the distribution of response categories, including correct responses and specific error types (auxiliary, participle, and -ed overgeneralization), across both tests. While participants from France contribute the largest amount of data, their performance reveals a shift in error density when transitioning from guided recognition (Test 1) to spontaneous production (Test 2). More precisely, the French participants show a noticeable escalation in both auxiliary errors and incorrect participle formations during Test 2. In contrast, the stability

²² Data represent total frequencies of correct versus incorrect responses per geographic location based on the research dataset.

observed in the smaller participant groups from Germany and Portugal indicates that the increased error rates in the French pool are not just a result of the task design but reflect a group specific limitation in lexical retrieval. These findings suggest that when participants are removed from the structured support of Test 1, their reliance on less efficient, associative memory strategies becomes more strong, leading to the observed decrease in morphological accuracy.

4.3.5. Institutional Exposure and Learner Individual Variability

The analysis of morphosyntactic accuracy goes beyond geographic factors and also considers the role of formal institutional exposure. While *country of birth* provides a baseline for linguistic input, individual variability such as length of formal education, serves as an influencing factor in performance. However, it is important to acknowledge a limitation regarding this variable, as the study **did not aim to investigate** the specific geographic location where participants completed their formal education until the end of secondary education. Consequently, it is not possible to determine whether the reported years of English instruction occurred within their country of birth or within France.

To reduce this ambiguity and potential reporting inconsistencies due to the anonymous nature of the data collection, individual data were grouped into two categories: **0–10 years** and **11–14 years**. This grouping strategy was chosen to ensure the reliability of the analysis by minimizing the impact of potential reporting inconsistencies and heterogeneous educational backgrounds. The error and correct response frequencies, categorized by these instructional duration groups, are provided in the table below.

Table 17. Error and Correct Response Frequencies by Years of English Instruction

Test	Years of English	Correct	Auxiliary Error	Incorrect Participle	Overgeneralized -ed
Test 1	0–10 Years	159	8	25	8
Test 1	11–14 Years	277	7	52	14
Test 2	0–10 Years	120	32	36	12
Test 2	11–14 Years	151	49	34	6

The data²³ reveal that the duration of formal exposure does not directly correlate with error reduction in spontaneous production tasks. Although learners within the 11–14 years group demonstrate higher overall accuracy in the mixed-format **Test 1**, this advantage becomes less evident in **Test 2**. This suggests that while institutional exposure facilitates the development of

²³ Data reflect absolute frequency counts.

declarative knowledge, the transition to spontaneous, accurate language production requires qualitatively different cognitive engagement than the current models provide.

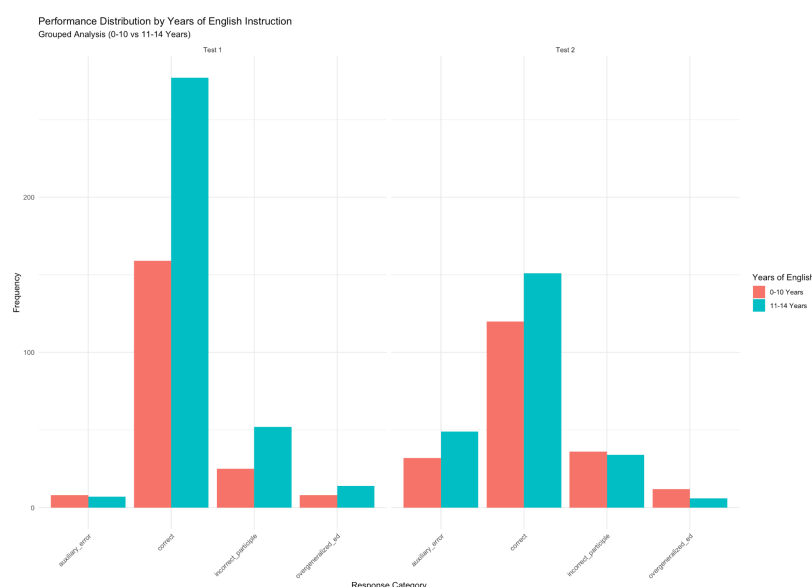


Figure 12. Distribution of Response Types by Years of English Instruction

Following the results presented in Table 17, Figure 12 illustrates the specific error dynamics across instructional groups. A key finding is the sharp increase in auxiliary errors in the 11–14 years group during Test 2. While this group performs better in Test 1, they appear to rely more heavily on memory-based retrieval in the absence of structured support.

On the other hand, the 0–10 years group shows a more stable pattern. The persistence of auxiliary and participle errors in both groups implies that the underlying morphological processing difficulty remains despite longer periods of instruction, indicating a general challenge in developing spontaneous production fluency.

5. Discussions

The present study aimed to examine the dynamic interface between learner L1 background, processing limitations, and morphological accuracy in the production of English irregular verbs. By integrating archival corpus data from the ICLE with experimental data from controlled and spontaneous tasks, this chapter evaluates the factors triggering morphological instability. The results reveal a distinct contrast between guided recognition and spontaneous production, featuring a critical 440% increase in auxiliary errors and a persistent 66.28% overgeneralization rate across all learner groups. These findings suggest that these irregularities are not just learning errors that diminish with classroom practice; instead, they are a strong indicator of a “fossilized plateau” (Saville-Troike & Barto, 2017). The following sections will discuss these results in detail, beginning with the cognitive mechanisms of regularization, moving through the lexical

retrieval barriers and typological influences, and finally addressing the pedagogical implications for advanced language acquisition along with the study's limitations and reflections on future research.

5.1. The Persistence of the -ed Suffix: Regularization as a Universal Cognitive Strategy

L2 learners' tendency to overgeneralize the *-ed* suffix is a recurring feature of interlanguage development, acting more as a cognitive strategy than just a simple error of ignorance. Analysis of the ICLE corpus shows a 66.28% overregularization rate, identifying this as the main morphological fallback strategy for learners. In terms of the Dual-Route Model, this indicates a strong competition between the declarative memory system, tasked with retrieving irregular forms, and the procedural system, which applies standard grammatical rules.

For example, for B2/C1 level learners, difficulty with producing correct irregular forms often points to a systematic retrieval problem. During spontaneous production, the cognitive cost of retrieving complex forms is exceptionally high due to the time pressure. When learners cannot quickly recall these forms from memory, the procedural system relies on the *-ed* suffix, a "winner-take-all" cognitive shortcut that prioritizes lexical meaning over grammatical accuracy. This tendency is closely connected to "Paradigm Leveling," a concept described by Hock (1991), where learners simplify complicated tripartite (A-B-C) systems into more manageable bipartite (A-B-B) structures.

The specific susceptibility of strong verbs, which constituted 212 error occurrences, supports the "Limited Attention Capacity" framework (Skehan, 2018). As shown in Table of Morphological Patterns in Strong Verbs (see table 4a), These verbs often form "weak phonetic links" that learners sacrifice when faced with the high cognitive load of spontaneous tasks. In this current study, this is evidently clear in such errors like "has broke," where learners simplify the irregular pattern to reduce processing pressure. As seen among learners with 11–14 years of English instruction, this overregularization often reaches a "fossilized plateau" (Saville-Troike & Barto, 2017). This suggests that while classroom instruction helps with declarative recognition, it often fails to internalize the complex internal adjustments needed for fluent, spontaneous production.

5.2. The Lexical Retrieval Barrier: Vowel Gradation and Morphological Complexity

The persistent instability of irregular verb production in advanced learners signals a crucial "Lexical Retrieval Barrier." While the *-ed* suffix serves as a highly productive rule, the storage and retrieval of irregular forms require access to a declarative memory system, which is naturally prone to processing constraints. This barrier is most visible in the performance gap between the

two task modalities, where the cognitive cost of retrieving complex, non-rule-based forms often becomes unmanageable.

Strong verbs place a considerably higher cognitive load than weak verbs. As identified in my data, these verbs are especially challenging because they rely on internal vowel changes (Ablaut) to express grammatical information. Unlike weak verbs, which use consistent dental suffixes such as *-t* or *-d* that provide a stable morphological marker, strong verbs require morphological stem modifications. During spontaneous speech, time pressure makes it difficult for the brain to process these internal shifts. This creates “phonetic instability” in the mental lexicon, often resulting in retrieval failures or a “cognitive vacuum” where the learner loses access to the correct form.

In addition, the tripartite (A-B-C) paradigm, such as *drink-drank-drunk*, creates significant interference during memory retrieval. Because past and participle forms are phonologically similar, these stored units compete not only with the regular *-ed* rule but also with each other. My findings underline this “processing competition”: while learners maintained strong accuracy in Test 1, their performance dropped visibly in the spontaneous production tasks of Test 2, where strong verb errors increased from 65 to 149 instances. This increase serves as empirical evidence that even advanced learners often fail to convert their declarative knowledge into automatic output under the “stress test” of spontaneous speech.

This tendency is an attempt to achieve morphological economy. As the brain encounters the processing limits of the lexical retrieval system, it employs “paradigm leveling” to reduce internal competition. By merging the distinction between irregular forms, or by disregarding the irregular stem variations in favor of a regularized pattern, learners minimize the cognitive load needed to maintain lexical forms. To sum up, the high error frequency across the 212 strong verb occurrences in the ICLE corpus shows that the “Lexical Retrieval Barrier” is the critical limitation for advanced L2 learners, as real-time retrieval demands overwhelm their limited cognitive capacity.

5.3. Bridging Data Sources: Task Effects and Processing Pressure

Integrating archival data from the ICLE with performance data from experimental tests provides a solid dual-method approach. This approach is theoretically supported, as it allows us to examine the same cognitive mechanism, the lexical retrieval system, across varying contexts. The difference is that corpus data captures the linguistic product in naturalistic writing, whereas experimental tasks isolate the cognitive process under manipulated performance demands. This synthesis strengthens the argument that interlanguage instability is not just a gap in linguistic knowledge, but a systemic consequence of cognitive constraints during spontaneous performance (Ullman, 2015).

The performance gap between the data sources clarifies a core variance between declarative knowledge and procedural execution. According to Ullman’s (2015) Declarative/Procedural (DP) Model, irregular verbs are idiosyncratic units accessed through the

declarative memory system, while regular *-ed* suffixation is processed through the procedural system. My results show that while learners exhibit high declarative competence in Test 1, where recognition cues enable slow-access retrieval, they experience an observable decrease in procedural performance during Test 2. Specifically, the significant increase in auxiliary errors (from 15 to 81) and the rise in strong verb errors (from 65 to 149) demonstrate that this declarative knowledge rarely reaches the level of unconscious usage required for smooth production.

This trend shows the significant performance limitations rooted in processing Ablaut-based strong verbs. Unlike weak verbs, which allow learners to apply a standard dental suffix, strong verbs cause a major “declarative overhead.” The vowel alternations required for strong verbs must be retrieved as stored item units from declarative memory—a process that is mentally more demanding than applying a procedural rule. Under the time pressure of spontaneous production, as framed by Skehan’s (2018) Limited Attentional Capacity Model, this memory recall cannot handle the procedural system. As a result, learners adopt a “winner-take-all” strategy, using the productive and effortless *-ed* suffix to satisfy communicative demands, selecting processing efficiency over morphological accuracy.

Additionally, the paradox of the “advanced” learners, who have been exposed to English for 11–14 years of formal instruction, emphasizes this cognitive limitation. The discrepancy of the performance fall from Test 2 to Test 1 among 22 participants corresponds to the high overregularization rates detected in the ICLE corpus (66.28%), implying that even at advanced stages, learners reach a state of stabilization, or fossilization, rather than achieving native-level fluency. This lack of progress is likely an outcome of current pedagogical biases; educational programs often tend to focus on explicit knowledge testing, which enhances the rule-based memory but neglects to bridge the gap toward implicit use. The advanced label in this context symbolizes declarative knowledge that is bound to break down during the stress of spontaneous language use.

Therefore, these findings question traditional Grammar-Based Teaching. If the advanced learner’s morphological system remains inconsistent, it is because formal training has not sufficiently engaged the procedural system. From a pedagogical perspective, implicit processing depends on a transition from declarative assessments to production aimed instruction. By increasing the demand for spontaneous oral or written production, learning modules can encourage the interlanguage to surpass stable mental data, supporting the unconscious proficiency required for procedural fluency. This analysis sheds light on the fact that for advanced learners, the main obstacle is not acquiring grammar rules, but mastering their use to produce correct forms under performance pressure.

5.4. Typological Distance and L1 Transfer: Romance–Germanic Contrast

The morphological variability seen in advanced learners is deeply influenced by the typological distance between their native language (L1) and English. In this section, we explore how this

variability between Romance (e.g., French, Italian) and Germanic (e.g., German, Dutch) morphological systems creates specific cognitive constraints on the processing of irregular verbs. By correlating ICLE corpus data with experimental findings, it becomes evident that L1 structural frameworks profoundly determine the learner's internalization strategy, frequently resulting in systematic patterns of negative transfer (Jarvis & Pavlenko, 2008).

The challenge for Romance L1 learners is derived from the functional conflict between their L1's verb-auxiliary systems and the English “Ablaut” system. While Romance languages typically adopt predictable, morphological rules or auxiliary-participle combinations, such as the French *passé composé*, English strong verbs are composed of idiosyncratic stem modifications that lack direct structural equivalents in the Romance linguistic system. This “morphological ambiguity” forces French-speaking learners to process a system where the link between the base form and its past tense is indirect (see Table 5). Thus, under the pressure of spontaneous production, these learners revert to the /-ed/ suffix as a “winner-take-all” strategy, strategically enforcing systemic regularity upon an unpredictable system (Ullman, 2015; see Table 7b; 70.00% overgeneralized -ed rate).

On the other hand, the “Germanic Advantage” conception provides a persuasive explanation for the greater procedural success indicated among learners from Germanic L1 backgrounds. These learners benefit from a common morphological pattern for internal vowel modifications, which effectively lowers the “declarative overhead” required to handle English strong verbs (Ringbom, 2007). Comparative data presented highlights this performance gap: Germanic learners exhibit statistically fewer incorrect participle errors (4.49%) compared to Romance learners (16.70%; see Table 9). This suggests that Germanic speakers maintain an internalized cognitive template for stem changes (Ablaut), allowing them to avoid the declarative-memory adaptive tactics that Romance learners must employ, consequently achieving higher accuracy under processing constraints of spontaneous production (Skehan, 2018).

The error typology characteristic of Romance L1 learners, particularly auxiliary errors, reinforces the cognitive constraints imposed by L1 morphosyntactic divergence. Experimental data conducted to students demonstrates that this difficulty is especially intensified by task modality for nearly all of them: for instance, monolingual French L1 speakers' auxiliary errors increased from 82 in Test 1 to 132 in Test 2 (see Table 15). These errors, generally appearing in the form of “be + V2” or “have + V1,” show us a structural avoidance approach through which learners attempt to assimilate English auxiliaries into Romance aspectual logic (Odlin, 1989). Considering that English strong verbs lack direct structural correspondents in the Romance system, they constitute the primary area of breakdown, as indicated by the rapid increase in strong verb errors from 65 in Test 1 to 149 in Test 2 (see Table 14).

To conclude, these results imply that for Romance L1 learners, the cognitive load of adapting L1-based morphological habits to a Germanic L2 system triggers a substantial exhaustion of cognitive energy. This “typological friction” specifically originates from the incompatibility between Romance synthetic logic, such as the Aux + V3 structures identified in Table 5, and the English simple past system, which requires internal stem changes. The clear

increase in auxiliary errors (a 440% increase; see Table 11) reveals that when learners can no longer exercise conscious oversight through rule retrieval, their accuracy collapses because of the force of this friction. Pedagogically, these results call for a shift regarding “contrastive morphological training.” As 11–14 years of institutional exposure (Table 17) appear insufficient to eliminate these persistent errors, teaching modules should aim to advance beyond rule instruction and emphasize intensive production tasks. Such tasks are necessary to foster the implicit acquisition of Ablaut patterns, intentionally bridging the gap between L1 processing habits and English irregular morphology (Granger, 1998).

5.5. Pedagogical Implications: Beyond Fossilization in the University Classroom

The transition from “active development” (0–10 years) to a “fossilized plateau” (11–14+ years) suggests that learners eventually reach a stage where morphological instability becomes chronic. Research specifies that morphological accuracy tends to plateau after the 10th year, largely due to the fact that institutional instruction predominantly targets declarative knowledge, which is acquired rapidly, whereas procedural knowledge, or “knowing how,” requires extensive exposure and purpose-serving production practice to habitualize correct usage (Ullman, 2015). Accordingly, although learners with 11–14 years of instruction demonstrate higher accuracy in recognition-based tasks (Test 1), their performance noticeably falls during spontaneous production (Test 2; see Table 17 and Figure 12). This mismatch emerges because their procedural system has failed to fully integrate the internal stem modifications characteristic of English strong verbs, exposing a persistent gap between theoretical knowledge and spontaneous production.

These results justify a pedagogical shift from “form-focused” instruction to “task-based” or “output-oriented” methodologies. Paper-based grammar exercises, which characterize English instruction in the first 0–10 years, focus on declarative memory, slowing down the flow of irregular morphology. To break this cycle, undergraduate level syllabuses should be based on tasks where meaning is key and successful outcome is the goal, especially aiming at the proceduralization of irregular forms (Skehan, 2018). Additionally, our assessment methods should go beyond grammar bias by including the stress feature in spontaneous language production both in oral and written forms. Techniques such as post-task transcription are particularly effective, as they push advanced learners to “notice the gap” between their intended output and their fossilized morphological errors.

The “typological friction” identified in the previous section, specifically the interference between Romance language tense logic and the English simple past objective, highlights the need for personalized instruction. Inspired by the concepts established by Granger (1998), instructional programs should be customized to fix specific error patterns encountered by Romance L1 learners. This involves contrastive verb training that explicitly explains the major differences between L1 verb patterns and English irregular verb patterns. Instead of relying only

on “input-centered” instruction, which does not do enough for adults to overcome fossilization, educators should be implementing “pushed output.” This approach forces learners to retrieve accurate forms under pressure, effectively sharpening their awareness of redundant grammatical features (Jarvis & Pavlenko, 2008).

Finally, instructors should help advanced learners get used to the balance between task pressure and performance anxiety. Using pre-task planning effectively lowers the cognitive load, helping learners use complex verb forms instead of relapsing into their L1 habits. In the long term, university syllabuses should redefine “advanced proficiency” to include morphological resilience, the ability to maintain accuracy under pressure, as a standard benchmark. By reconsidering curricula that judge proficiency only by vocabulary size, institutions can help students build the procedural stability needed to pass the “stress test” of spontaneous production without letting them revert to fossilized, overregularized errors.

5.6. Limitations of the Study and Reflections for Future Research

The limitations of this study are mostly caused by the practical realities of a university classroom environment. One of them was the decision not to assign individual student numbers during the experimental phases. While this simplified process, it made individual tracking impossible; naturally, it was impossible to match individual monolingual L1 French students across Test 1 and Test 2, meaning we could only analyze trends at the group level rather than tracing how each student’s performance evolved over time.

The sample size of 22 students provided enough data to identify clear patterns, but it is small for comprehensive statistical generalizations. Future studies with larger groups will be necessary to confirm if these findings hold up across more diverse profiles. At the same time, because the 2LLIAM222 MATI - ORAL course was only for one hour per week, we had to keep our tests under 10 minutes and limited to 25 questions to ensure the best results without rushing. This tight schedule limited our ability to examine every aspect of morphosyntactic performance as deeply as we might have liked.

Also, a notable limitation regarding the background data is the inconsistency in how students reported their “years of English instruction” between the two tests. Because students provided varying information for that question, we could not treat this variable as an exact value for each individual. Instead, we were required to group the participants into two general groups: 0–10 years and 11–14 years of institutional exposure. While this grouping helps minimize the inconsistencies in self-reported data, it potentially hides small progress steps that might occur within these specific year ranges.

On the database side, the ICLE corpus also presented some difficulties. While the dataset contains “date” information related to student profiles at each error occurrence, we encountered inconsistencies where the same learner was associated with mismatching ages across different entries. Due to these issues, using “age” as a reliable variable was not useful and was therefore excluded from our analysis. We also faced technical limitations with the ICLE search interface;

specifically, searching for errors within positive or negative interrogative forms was technically challenging to extract, so we had to leave those out of the study. Lastly, we had initially planned to include a home-based writing assignment to observe how students perform in a more natural setting, such as a task that required them to write paragraphs about a topic that included past tense usage. However, only 5 out of the 22 students submitted it, so in the end, we eliminated this evaluation from the data set and focused entirely on the two main experimental tests.

The findings have paved the way for a more ambitious doctoral research project. For future research, the goal is to advance from this preliminary study to a more comprehensive, long-term design. By assigning individual student IDs, a future study will be able to perform detailed, participant-by-participant research analysis, allowing us to see exactly how individual morphological errors change and progress over the course of a program.

For a future PhD dissertation, I aim to expand the participant number from 22 to 50–100 students, providing the statistical depth needed to formulate more reliable conclusions. Increasing the assessment scope beyond 25 items will also allow for a much more detailed view of irregular verb morphology. Beyond this, I plan to collect both written and spoken production data from these students. Comparing these two modalities will reveal how the process of "internalizing" language differs when a learner moves from a controlled test to spontaneous written and oral production.

Perhaps most importantly, a future extension of this work will also include regular verb morphology. By comparing error patterns in regular versus irregular verbs within a larger, long-term framework, we can better define the true nature of the gap between knowledge and use. This will help us determine if the instability we see with Strong Verbs is a unique problem or part of a wider struggle with L2 morphology. In this next phase of research, my main goal is to look deeper than simply describing fossilization, aiming to map out the specific pedagogical conditions, whether through task-based practice or targeted morphological training, that actually allow advanced learners to move past the plateau.

6. Conclusion

This thesis bridges macro-analytic corpus data from the International Corpus of Learner English (ICLE) with micro-analytic test results from the **2LLIAM222 MATI-ORAL** course at **Université Bordeaux Montaigne**. By combining these diverse sources, this study surpasses traditional error-count methodologies to offer a fresh view on the interlanguage system. The research integrates existing corpus data, which shows the linguistic product of spontaneous writing, with two targeted experimental assessments that aim to measure underlying cognitive processes. The main goal of this work is to rethink how we define proficiency: rather than focusing on fixed grammar skills, this study treats it as "morphological resilience." This approach describes the learner's capacity to retrieve and maintain complex irregular forms, such as tripartite Ablaut distinctions, under the cognitive load of spontaneous production. Findings

demonstrate that at the B2-C1 levels, the learner's interlanguage remains a constant struggle between rule-based proceduralization and lexical memory, where stability is consistently tested and challenged by the shift from recognition to active recall.

6.1. Summary of Findings

The methodological framework of this study is built upon the ICLEv3 corpus, selected for its size and potential to show real spontaneous language production across various L1 backgrounds. To classify the irregular verb errors, we developed a four-way verb classification, **Strong**, **Weak**, **Invariable**, and **Suppletive**, based on conjugations and the cognitive load required for memory retrieval. This was then followed by another classification of three error types: **overgeneralized -ed**, **auxiliary error**, and **incorrect participle**, acting as a clear guide for observing shifts in learner interlanguage. Results reveal a major link of language family background, mostly between Romance and Germanic L1 speakers. Romance L1 speakers faced typological friction, likely resulting from a conflict between their native tense systems and the English Ablaut pattern, which led to higher rates of participle errors. Conversely, Germanic L1 speakers benefited from a mental pattern for stem changes, which acted as positive transfer and reduced the mental effort required for irregular verb retrieval.

The analysis of the Multiple Correspondence Analysis (MCA) maps provides a geometric visualization of these links, highlighting that irregular verb errors are not randomly distributed but create specific mental patterns within the learner's interlanguage space. Dimension 1 of the Active Category map serves as a main way to measure complexity, sorting basic errors from complex challenges found at the syntax-morphology interface, such as auxiliary and suppletive verb errors. Dimension 2 maps the typological division, with the Romance language cluster leaning toward the regions of Strong Verb and Incorrect Participle errors. The Full Symmetric map brings these data points together, suggesting that learners do not change without a pattern through the error space but are consistently attracted to specific challenges based on the interaction between their L1 background and the basic morphological difficulty of the verb type. These maps make it clear that individual differences in background, examined through the framework of Li, Hiver, and Papi (2022), are not just small details but are structural determinants of morphological accuracy.

The tests applied to 22 freshman students at Université Bordeaux Montaigne clarified these results by revealing a large gap between Test 1 (multi-format recognition) and Test 2 (written production). While Test 1 minimized cognitive load through recognition cues, the unguided production demands of Test 2 acted as a “stress test,” causing a 440% increase in auxiliary errors. This gap signifies that declarative knowledge of grammar rules does not easily become automatic under performance pressure, highlighting Ullman (2015)'s observation that proceduralization is slower and less complete in adult learners, who may never reach the same level of procedural stability as native speakers. In addition, grouping participants' “Years of English” information as 0–10 and 11–14 served as an important indicator of morphological

fossilization, suggesting that even learners with over a decade of instruction often reach a developmental plateau. Despite higher accuracy in controlled tasks, these learners frequently had a tendency to use the regular *-ed* suffix as a “fallback strategy” in spontaneous production, supporting the idea that the procedural system often avoids difficult irregular retrieval over simple patterns.

Following these findings, the different kinds of verb errors can be divided into three main groups:

- **Overgeneralized -ed** errors are a basic mental strategy where learners apply a productive procedural rule to verb forms that the learner struggles to recall, seen in forms like "arised," "goed," or "maded". Based on Nick C. Ellis's (2019) Competition Model, these errors occur because of how the brain learns patterns, where the procedural system wins the competition for deciding on the output because the lexical strength of the irregular form is insufficient for a fast retrieval.
- **Auxiliary verb** errors happen due to difficulties at the syntax-morphology interface, often resulting in incorrect double marking such as "didn't thought" or errors in tense consistency. As Ellis (2019) explains, these errors point to the limited attention capacity of the learner, where the simultaneous management of auxiliary selection and lexical retrieval puts too much pressure on the learner's brain to process during timed production.
- **Incorrect participle** errors are due to the mixing of verb tenses, where the distinction between the past simple (V2) and past participle (V3) is not maintained, producing forms like "has became" or "is did". Nick C. Ellis (2019) interprets this as a failure of distributional learning, as the learner fails to distinguish the predictive links between specific auxiliaries and their required morphological forms.

6.2. Pedagogical Implications for ELT

The results of this study have rich implications for English Language Teaching (ELT), particularly at the university level where learners often get stuck in a fossilized plateau. Current syllabuses often prioritize grammar-based teaching, which uses standard grammar tests that mostly target declarative knowledge but fail to foster the procedural stabilization necessary for spontaneous production. To help learners progress further, pedagogical recommendations should be built on focus-on-form, as suggested by Rod Ellis (2015). Unlike unfashionable focus-on-forms, which treat grammar as a list of rules to memorize, focus-on-form integrates attention to linguistic features during tasks where meaning is the goal. This approach ensures that learners do not just memorize irregular patterns but develop the morphological resilience needed to produce them accurately under load.

Instructors should switch to Task-Based Language Teaching (TBLT) from simple memorization tasks, where reaching a real-life goal requires the proceduralization of irregular forms. Paper-based exercises should be used only at the beginning as a quick introduction for declarative recognition, while teachers should emphasize home-based writing assignments and oral production tasks that put students on the spot, such as rapid activities or timed responses. This pressure forces learners to retrieve accurate forms under limits, improving their awareness of morphological gaps. Also, techniques such as post-task transcription and corrective feedback are helpful, as they encourage advanced students to notice the gap between their intended meaning and their fossilized morphological errors. By setting high-level goals that include morphological resilience, institutions can ensure that students develop the procedural stability needed for real-life spontaneous production.

6.3. Theoretical Synthesis

A theoretical synthesis of the study's results uncovers a detailed cognitive framework where morphological development is influenced by the relationship between Ablaut instability, the Fossilization Plateau, and the limits of the Declarative/Procedural (DP) model. The biggest mental challenge for advanced learners is the processing of Strong Verbs, which is based on vowel shifts and creates a heavy declarative overhead. Because these forms are not generated by rules but are kept in memory separately in declarative memory, they are very likely to retrieval failure in quick spontaneous production. This retrieval failure activates a "winner-take-all strategy", as suggested by Ullman's DP model, where the procedural system chooses the easy way of using the regular *-ed* suffix to finish the sentence. This process is similar to the Frequency Effect described by Bybee (2007), where irregular verbs require a high token frequency to resist the copying the regular verb rule. When a verb's frequency in the input is insufficient to maintain a strong lexical representation, it becomes a candidate for overregularization, leading to the "arised" or "drinked" patterns observed in the ICLE dataset.

However, the persistence of these errors among learners with over a decade of instruction suggests the existence of the Fossilization Plateau theorized by Han (2004). Han describes fossilization as a lack of long-term progress of development, where specific interlanguage features differ from target norms in spite of continued exposure. In the case of irregular verbs, this plateau is usually caused by institutional instruction that covers declarative knowledge but lacks real-life practice conditions necessary for proceduralization. Because of this, the learner's system stabilizes at a stage of fixed inconsistency, where they unpredictably produce both target and non-target forms depending on the cognitive load of the task. The perspective of Bybee (2007) suggests that this inconsistency is also influenced by the Conserving Effect, where only high-frequency irregulars are well-learned enough to overcome this declarative challenge. The transition to native-like brain processing is not about knowing more grammar, but about achieving the procedural fluency required to overcome the declarative retrieval barrier altogether.

6.4. Suggestions for Future Research

The present study has provided a first model for understanding morphological instability, but its limitations highlight the need for a more ambitious future PhD research plan. Future research will extend this investigation by expanding the study's size and scope, starting with assigning ID codes to each student to track them over time. This will allow us to see how specific morphological errors progress or fossilize over an entire academic year, avoiding the limits of single tests. Expanding the sample size to 50-100 international students will lead to more reliable statistical generalizations and give more detail for the Multiple Correspondence Analysis modeling. Furthermore, increasing test complexity with a larger set of 25+ questions ranging in difficulty from A1 to C1 levels will reveal a more detailed view of the retrieval limits for lower-frequency irregular paradigms.

To more fully understand the nature of language learning, the research will broaden test types to include True/False acceptability judgments, fill-in-the-blank exercises, and spontaneous production tasks. The main focus of the next phase is to create a collection of dual-modality corpus that gathers both oral and written production data from the same participants. By tracking response time and speed, the study will be able to check how easily the student produces language and reveal how the internalization process differs between different modes of expression. In the long run, this future research will go further than irregular verbs to include regular verb morphology, allowing for a contrastive analysis that can define whether the instability observed in the present study is specific to irregular verbs or a wider problem of L2 morphological acquisition. This detailed dual-modality approach will offer enough detail to fully understand the progress of morphological resilience in the advanced learner.

6.5. Final Reflections

The findings of this thesis are not just an analysis of learner errors; they provide a plan for new teaching methods in how advanced L2 proficiency is defined and measured. The persistent struggle with irregular verb morphology is not a sign of cognitive deficiency, but a normal part of the challenging process involving different memory systems as the learner tries to reach natural, fluent speech. By pointing out the weak spots in grammar, especially how learners struggle with both irregular forms and verb patterns, this research has identified the specific points where instructional practices can be most effective. If the goal of advanced ELT is to equip learners with the ability to deal with language complexities of spontaneous expression, then our teaching methods should adapt from a focus on rule-based accuracy to support morphological resilience. This resilience represents a shift from the "winner-take-all" cognitive strategies of regularization, building a way to manage difficult lexical retrieval even under extreme processing pressure.

As a final matter of fact, this research suggests that the way to achieve morphological fluency is not about learning more and more grammatical rules, but in the frequent practice in

real-life settings of irregular forms that bridges the gap between declarative knowledge and procedural production. The observed fossilization plateau is not an unsolvable problem, but a reminder to use more effective, task-based teaching that challenges the learner to retrieve forms under cognitive load. The transition from "knowing" a verb's forms to "proceduring" them in real-time discourse is the clearest proof of advanced proficiency. As advanced learners move beyond the classroom and face the challenges of professional and academic communication, their ability to handle these irregular forms will continue to be an important piece of their linguistic identity, showing their skill to use "natural usage" found in a language.

Ultimately, the dual-modality results of this research, combining the large ICLE database with our own experimental results, show why it is important to integrate diverse data sources in SLA research. This thesis is a first step toward a wider understanding of morphological acquisition, one that considers how these factors work together: typological background, frequency effects, and neurocognitive constraints. By continuing this work with longer-term tests and reaching out to more students, future research will undoubtedly unlock new strategies for learners to break through the fossilization plateau. Thus, as teachers and researchers, our first goal should be to create a generation of advanced English speakers who develop not only linguistic accuracy but the resilient morphological strength required for fluent, spontaneous interaction in today's world.

7. References

- Anderwald, L. (2009).** *The morphology of English dialects: Verb-formation in non-standard English*. Cambridge University Press.
- Ayoun, D., & Salaberry, M. R. (2008).** Acquisition of English tense–aspect morphology by advanced French instructed learners. *Language Learning*, 58(3), 555–595.
- Bauer, L. (1988).** *Introducing Linguistic Morphology*. Edinburgh University Press.
- Baugh, A. C., & Cable, T. (2013).** *A history of the English language* (6th ed.). Routledge.
- Bybee, J. L. (1985).** *Morphology: A study of the relation between meaning and form*. John Benjamins.
- Bybee, J. L. (2001).** *Phonology and language use*. Cambridge University Press.
- Bybee, J. L. (2007).** *Frequency of use and the organization of language*. Oxford University Press.
- Chomsky, N. (1959).** A Review of B. F. Skinner's Verbal Behavior. *Language*, 35(1), 26–58.
- Corder, S. P. (1974).** Error analysis. In J. P. B. Allen & S. P. Corder (Eds.), *Techniques in applied linguistics* (pp. 122–154). Oxford University Press.
- Desagulier, G. (2017).** *Corpus Linguistics and Statistics with R: Introduction to Quantitative Methods in Linguistics*. Springer.
- Dulay, H. C., Burt, M. K., & Krashen, S. D. (1982).** *Language two*. Oxford University Press.
- Ellis, N. C. (2019).** Essentials of a theory of language cognition. *The Modern Language Journal*, 103, 39–60.
- Ellis, R. (1999).** The study of second language acquisition. Shanghai: Shanghai Foreign Language Education Press.
- Ellis, R., Loewen, S., Elder, C., Erlam, R., Philp, J., & Reinders, H. (2009).** Implicit and explicit knowledge in second language learning, testing and teaching. Multilingual Matters.
- Ellis, R. (2015).** *Understanding Second Language Acquisition* (2nd ed.). Oxford University Press.
- Field, A. (2012).** *Discovering Statistics Using R*. Sage Publications.
- Finegan, E. (2012).** *Language: Its structure and use* (6th ed.). Wadsworth.
- Gass, S. M., & Selinker, L. (2008).** *Second language acquisition: An introductory course* (3rd ed.). Routledge.
- Granger, S. (1998).** *Learner English on computer*. Longman.

- Granger, S., Dupont, M., Meunier, F., Naets, H., & Paquot, M. (2020).** *The International Corpus of Learner English* (Version 3). Centre for English Corpus Linguistics, Université catholique de Louvain.
- Greenacre, M. (2017).** *Correspondence Analysis in Practice* (3rd ed.). Chapman & Hall/CRC.
- Han, Z. (2004).** *Fossilization in adult second language acquisition*. Multilingual Matters.
- Hare, M., & Elman, J. L. (1995).** Learning and morphological change. *Cognition*, 56(1), 61-98.
- Hock, H. H. (1991).** *Principles of historical linguistics*. Mouton de Gruyter.
- Hogg, R. M., & Denison, D. (2006).** *A history of the English language*. Cambridge University Press.
- Hogg, R. M., & Fulk, R. D. (2011).** *A grammar of Old English, Volume 2: Morphology*. A John Wiley & Sons Ltd., Publications.
- Hulstijn, J. H. (2001).** Intentional and incidental second-language vocabulary learning: A reappraisal of elaboration, rehearsal and automaticity. In P. Robinson (Ed.), *Cognition and Second Language Instruction* (pp. 258-286). Cambridge University Press.
- Hulstijn, J. H. (2002).** Towards a unified account of the representation, processing and acquisition of second language knowledge. *Second Language Research*, 18(3), 193–223.
- Husson, F., Lê, S., & Pagès, J. (2017).** *Exploratory Multivariate Analysis by Example Using R* (2nd ed.). Chapman & Hall/CRC.
- Jarvis, S., & Pavlenko, A. (2008).** *Crosslinguistic influence in language and cognition*. Routledge.
- Krashen, S. D. (1985).** *The input hypothesis: Issues and implications*. Longman.
- Ladefoged, P., & Johnson, K. (2015).** *A course in phonetics* (7th ed.). Cengage Learning.
- Lado, R. (1957).** *Linguistics across cultures: Applied linguistics for language teachers*. University of Michigan Press.
- Le Roux, B., & Rouanet, H. (2010).** *Multiple Correspondence Analysis*. SAGE Publications.
- Li, S., Hiver, P., & Papi, M. (Eds.). (2022).** *The Routledge handbook of second language acquisition and individual differences*. Routledge.
- Lightbown, P. M., & Spada, N. (2013).** *How languages are learned* (4th ed.). Oxford University Press.
- Long, M. H. (1996).** The role of the linguistic environment in second language acquisition. In W. C. Ritchie & T. K. Bhatia (Eds.), *Handbook of second language acquisition* (pp. 413–468). Academic Press.
- Lyster, R., & Ranta, L. (1997).** *Corrective feedback and learner uptake: Negotiation of form in communicative classrooms*. *Studies in Second Language Acquisition*, 19(1), 37–66.
- Mackey, A. (2012).** *Input, interaction, and corrective feedback in L2 learning*.

MacWhinney, B. (1987). The Competition Model. In B. MacWhinney (Ed.), *Mechanisms of language acquisition* (pp. 249–308). Lawrence Erlbaum Associates.

Mailhammer, R. (2007). *The Germanic strong verbs: Foundations and development of a new system*. Mouton de Gruyter.

Marcus, G. F., Pinker, S., Ullman, M., Hollander, M., Rosen, T. J., & Xu, F. (1992). Overregularization in Language Acquisition. *Monographs of the Society for Research in Child Development*, 57(4), i-178.

Minkova, D. (2014). *A Historical Phonology of English*. Edinburgh University Press.

Nevalainen, T. (2006). *An introduction to Early Modern English*. Edinburgh University Press.

Odlin, T. (1989). *Language Transfer: Cross-Linguistic Influence in Language Learning*. Cambridge University Press.

Odlin, T. (2003). *Cross-linguistic influence*. In C. J. Doughty & M. H. Long (Eds.), *The handbook of second language acquisition* (pp. 436–486). Blackwell.

Pinker, S. (1999). *Words and rules: The ingredients of language*. Basic Books.

Pinker, S., & Ullman, M. T. (2002). The past and future of the past tense. *Trends in Cognitive Sciences*, 6(11), 456-463.

Quirk, R., Greenbaum, S., Leech, G., & Svartvik, J. (1985). *A comprehensive grammar of the English language*. Longman.

Ringbom, H. (2007). *Cross-linguistic Similarity in Foreign Language Learning*. Multilingual Matters.

Saville-Troike, M., & Barto, K. (2017). *Introducing second language acquisition* (3rd ed.). Cambridge University Press.

Schmidt, R. (1990). The role of consciousness in second language learning. *Applied Linguistics*, 11(2), 129–158.

Schmidt, R. (2001). Attention. In P. Robinson (Ed.), *Cognition and second language instruction* (pp. 3–32). Cambridge University Press.

Selinker, L. (1972). Interlanguage. *International Review of Applied Linguistics*, 10(1–4), 209–231.

Skehan, P. (2018). *Second language task-based performance: Theory, research, assessment*. Routledge.

Skinner, B. F. (1957). *Verbal behavior*. Copley Publishing Group.

Spencer, A. (1991). *Morphological theory: An introduction to word structure in generative grammar*. Blackwell.

Swan, M., & Smith, B. (Eds.). (2001). Learner English: A teacher's guide to interference and other problems (2nd ed.). Cambridge University Press.

Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2), 257–285.

Ullman, M. T. (2015). The declarative/procedural model: A neurobiologically motivated theory of first and second language. In B. VanPatten & J. Williams (Eds.), *Theories in second language acquisition: An introduction* (2nd ed., pp. 135–158). Routledge.

Wickham, H. (2016). ggplot2: Elegant Graphics for Data Analysis (2nd ed.). Springer.

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., ... & Yutani, H. (2019). *Welcome to the Tidyverse*. *Journal of Open Source Software*, 4(43), 1686.

Williams, S., Guijarro-Fuentes, P., & Vulchanova, M. (2022). U-shaped trajectories in an L2 context: Evidence from the acquisition of verb morphology. *Vigo International Journal of Applied Linguistics*, (19), 223–266

8. Appendices

Appendix A

Supplementary Data

This appendix A includes all the datasets and auxiliary materials used for the analyses presented in this thesis. Due to their extensive nature, these resources are placed in a dedicated online repository, securing full transparency and promoting the replication of the study's findings.

Repository Link: https://github.com/burakcolak1999/COLAK_Burak_Thesis_M2_SDL

The repository's contents are structured as follows:

1. Google Forms Data

- **Content:** Contains raw CSV files, visualization charts, the original consent form, analytical scripts, Test 1 and Test 2 related documents, and the primary assessment data for both tests.
- **Role:** Provides the experimental basis for the experimental analysis of Test 1 and Test 2 results.

2. ICLE Corpus

- **Content:** Organized into sub-folders for "All Learners" and "Romance vs. Germanic" language families.
 - **All_Learners:** Includes unified CSV files, numerical data, and processing scripts.
 - **Romance_vs_Germanic:** Features detailed sub-datasets, scripts for comparative analysis, geographical maps, and Multiple Correspondence Analysis (MCA) data.
- **Role:** Acts as the reference corpus, allowing for the examination of linguistic patterns across different native language backgrounds and wider language families.

Appendix B:

R Code for ICLE Corpus Statistical Analysis (All Learners)

This R script performs statistical analysis on the ICLE Corpus all learner irregular verb error dataset. It provides a systematic analytical overview by evaluating the distribution of targeted irregular verb types across language families, while enabling a detailed cross-tabulation of language families by specific error types to identify systematic morphological challenges. Additionally, it performs a micro-analysis to highlight the most error-prone verb forms. The original code remains unchanged to ensure full transparency and reproducibility. For full reproducibility, this script and all associated data files are available in the supplementary online repository linked in Appendix A.

```
# Appendix B: R Code for ICLE Corpus Statistical Analysis (ALL LEARNERS)
# Purpose: This script performs statistical analysis on the ICLE Corpus all learner irregular
verb error dataset.
# It provides a comprehensive overview by evaluating the distribution of targeted
# irregular verb types across language families, while enabling a detailed
# cross-tabulation of language families by specific error types to identify
# systematic morphological challenges. Additionally, it performs a micro-analysis
# to highlight the most error-prone verb forms. The original code remains
# unchanged to ensure full transparency and reproducibility.

# — 1. Necessary Libraries and Data Loading —————
library(dplyr)
library(readr)

# Set working directory
# Note: Ensure your working directory points to the folder containing the data file.
setwd(
  "~/Desktop/COLAK_Burak_Thesis/Data : Irregular Verb Errors/ICLE Errors/All
Learners/CSV"
)

# Read the full dataset
df <- read_delim(
  "irregular_verbs_mca_data_full_list.csv",
  delim = ";",
  skip = 3,
  trim_ws = TRUE
)

# — 2. Global Cross-Tabulation (Raw Frequencies) —————
cat("--- GLOBAL CROSS-TABULATION: ALL LANGUAGE FAMILIES BY ERROR
TYPES ---\n")
```



```

global_ct <- table(df$Language_Family, df$Error_Type)
print(global_ct)

# — 3. Global Row Percentages (Proportional Analysis) —
cat("\n--- ROW PERCENTAGES BY ALL LANGUAGE FAMILIES ---\n")
global_prop <- prop.table(global_ct, margin = 1) * 100
print(round(global_prop, 2))

# — 4. Interaction: Language Families by Irregular Verb Types —
cat("\n--- ALL LANGUAGE FAMILIES BY IRREGULAR VERB TYPES ---\n")
verb_family_ct <- table(df$Language_Family, df$`Irregular Verb Type`)
print(verb_family_ct)

# — 5. Dynamic Micro-Analysis: Top 3 Most Problematic Verbs —
# This code automatically identifies the 3 verbs with the highest total error counts
top_3_verbs <- df %>%
  count(Verb) %>%
  slice_max(n, n = 3) %>%
  pull(Verb)

cat(
  "\n--- ERROR BREAKDOWN FOR TOP 3 MOST FREQUENT ERROR-PRONE
  VERBS:",
  paste(top_3_verbs, collapse = ", "),
  "\n---\n"
)

top_verbs_breakdown <- df %>%
  filter(Verb %in% top_3_verbs) %>%
  count(Verb, Error_Type) %>%
  group_by(Verb) %>%
  mutate(Percentage = (n / sum(n)) * 100)

print(top_verbs_breakdown)

```

Appendix C

R Code for Comparative Analysis (Romance & Language Germanic Families)

This R script performs a comparative statistical analysis on the ICLE Corpus dataset, focusing specifically on the Romance and Germanic language families. It provides a systematic analytical framework by examining the distribution of errors across different language backgrounds and education durations. In addition, it allows for a detailed visualization of error patterns and examines morphological stability via verb-type analysis and inferential statistics (Chi-Square Test) to identify significant differences between these linguistic groups. The original code remains unchanged to ensure full transparency and reproducibility. For full reproducibility, this script and all associated data files are available in the supplementary online repository linked in Appendix A.

```
# Appendix C: R Code for Comparative Analysis (Romance & Germanic Language Families)
# Purpose: This script performs a comparative statistical analysis on the ICLE Corpus
# dataset focusing on Romance and Germanic language families. It provides a
# systematic analytical framework by evaluating the distribution of errors across
# different language backgrounds and education durations. Furthermore, it facilitates
# a detailed visualization of error patterns and examines morphological stability
# through verb-type analysis and inferential statistics (Chi-Square Test) to identify
# significant differences between these linguistic groups. The original code remains
# unchanged to ensure full transparency and reproducibility.

# --- Load necessary libraries ---
library(tidyverse)
library(ggplot2)

# --- Load the dataset ---
df <- read.csv("irregular_verbs_mca_data_romance_germanic.csv", skip = 2, sep = ";")

# --- 1. Comparative Tables ---
cat("--- 4.1.2 Error Distribution by Language Family ---\n")
print(df %>%
  group_by(Language_Family, Error_Type) %>%
  summarise(Count = n(), .groups = 'drop') %>%
  mutate(Percentage = round(Count / sum(Count) * 100, 2)))

cat("\n--- 4.1.2 Error Analysis by Education Duration ---\n")
print(df %>%
  group_by(Years.of.English.at.School, Error_Type) %>%
  summarise(Count = n(), .groups = 'drop'))

# --- 2. Visualization: Group 1 (3-4 years + Unknown) ---
df_group1 <- df %>%
```

```

filter(Years.of.English.at.School %in% c("3 years", "4 years", "Unknown"))

ggplot(df_group1, aes(x = Error_Type, fill = Language_Family)) +
  geom_bar(position = "dodge") +
  facet_wrap(~Years.of.English.at.School) +
  theme_minimal() +
  labs(title = "Distribution of Error Types (3-4 Years and Unknown)",
       x = "Error Type", y = "Frequency") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# --- 3. Visualization: Group 2 (5-6+ years) ---
df_group2 <- df %>%
  filter(Years.of.English.at.School %in% c("5 years", "6 years or more"))

ggplot(df_group2, aes(x = Error_Type, fill = Language_Family)) +
  geom_bar(position = "dodge") +
  facet_wrap(~Years.of.English.at.School) +
  theme_minimal() +
  labs(title = "Distribution of Error Types (5-6+ Years)",
       x = "Error Type", y = "Frequency") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# --- 4. In-depth Verb Type Analysis ---
cat("\n--- 4.1.2 Verb Type Analysis by Language Family ---\n")
print(df %>%
  group_by(Language_Family, Irregular.Verb.Type) %>%
  summarise(Count = n(), .groups = 'drop'))

# --- 5. Statistical Analysis ---
cat("\n--- 4.1.2 Chi-Square Test Result ---\n")
contingency_table <- table(df$Language_Family, df$Error_Type)
print(chisq.test(contingency_table))

```

Appendix D

Multiple Correspondence Analysis – Irregular Verb Errors

This R script performs a Multiple Correspondence Analysis (MCA) to explore the associations between learner profiles and specific error patterns within the ICLE Corpus. By setting linguistic error variables (Irregular Verb Type and Error Type) as active dimensions, the script constructs an optimized error space. Learner background variables (Verb, Country, Native Language, Language Family, and Years of English) are then added into this space as illustrative variables to show systematic patterns and clusters. This methodological approach enables a detailed look at how diverse learner backgrounds correlate with morphological challenges. The original code remains unchanged to ensure full transparency and reproducibility. For full reproducibility, this script and all associated data files are available in the supplementary online repository linked in Appendix A.

```
# =====  
# Appendix D: Multiple Correspondence Analysis – Irregular Verb Errors  
#  
# Your research question:  
# Which learner profiles are associated with which error patterns?  
#  
# Active variables (these construct the geometry — your error space):  
# Irregular_Verb_Type (6 levels), Error_Type (3 levels)  
#  
# Illustrative / supplementary variables (projected post-hoc  
# onto your error space — your learner profile space):  
# Verb, Country, Native_Language, Language_Family,  
# Years_of_English_at_School  
#  
# Rationale for your analysis:  
# By making the linguistic error variables active, you ensure  
# the map axes are optimised to separate error profiles. The  
# learner background categories are then projected into that  
# space, revealing which learner profiles gravitate toward  
# which error clusters — directly answering your research  
# question. Reversing this (background active, errors  
# illustrative) would optimise the map for learner separation  
# and risk burying the error-pattern variation you are  
# interested in.  
# =====  
  
# — 0. Packages  
  
pkgs <- c("FactoMineR", "factoextra", "ggplot2", "ggrepel", "dplyr", "readr")
```

```

for (p in pkgs) {
  if (!requireNamespace(p, quietly = TRUE)) install.packages(p)
}
library(FactoMineR)
library(factoextra)
library(ggplot2)
library(ggrepel)
library(dplyr)
library(readr)

set.seed(42)

# — 1. Load data —————

path <- "irregular_verbs_mca_data_romance_germanic.csv" # replace this fake path with your
the path to you data file on your computer

df_raw <- read.table(path, header = TRUE, sep = ";", skip = 2, check.names = FALSE)

# Standardise column names
names(df_raw) <- gsub("\\s+", "_", trimws(names(df_raw)))
names(df_raw) <- gsub("[^A-Za-z0-9_]", "", names(df_raw))

cat("Columns found:\n"); print(names(df_raw))
cat("\nRows:", nrow(df_raw), "\n")

# — 2. Select & rename relevant columns —————

col_map <- c(
  # Active first
  Irregular_Verb_Type = grep("Irregular", names(df_raw), value = TRUE)[1],
  Error_Type = grep("Error_Type", names(df_raw), value = TRUE)[1],
  # Illustrative second
  Verb = grep("^Verb$", names(df_raw), value = TRUE)[1],
  Country = grep("Country", names(df_raw), value = TRUE)[1],
  Native_Language = grep("Native", names(df_raw), value = TRUE)[1],
  Language_Family = grep("Language_Family", names(df_raw), value = TRUE)[1],
  Years_English = grep("Years", names(df_raw), value = TRUE)[1]
)

cat("\nColumn mapping used:\n"); print(col_map)

df <- df_raw %>%
  select(all_of(col_map)) %>%
  setNames(names(col_map)) %>%

```

```

mutate(across(everything(), ~ factor(trimws(as.character(.)))))

# Drop rows with any NA in active columns
active_vars <- c("Irregular_Verb_Type", "Error_Type")
df <- df %>% filter(if_all(all_of(active_vars), ~ !is.na(.)))

cat("\nFinal dataset dimensions:", nrow(df), "rows x", ncol(df), "cols\n")
cat("\nLevel counts per variable:\n")
for (v in names(df)) cat(" ", v, ":", nlevels(df[[v]]), "levels —",
  paste(levels(df[[v]]), collapse = ", "), "\n")

# — 3. Build the MCA input data-frame —
suppl_vars <- c("Verb", "Country", "Native_Language", "Language_Family",
  "Years_English")

df_mca <- df %>% select(all_of(active_vars), all_of(suppl_vars))
n_active <- length(active_vars)
n_suppl <- length(suppl_vars)
suppl_idx <- (n_active + 1):(n_active + n_suppl)

cat("\nActive (geometry) :", paste(active_vars, collapse = ", "), "\n")
cat("\nIllustrative (projected):", paste(suppl_vars, collapse = ", "), "\n")

# — 4. Run MCA —

set.seed(42)
mca <- MCA(
  df_mca,
  ncp = 5,
  quali.sup = suppl_idx,
  graph = FALSE
)

# — 5. Diagnostics —

cat("\n\n===== Eigenvalues / Variance Explained =====\n")
print(round(mca$eig, 4))

cat("\n\n===== Active category coordinates (Dim 1–2) =====\n")
print(round(mca$var$coord[, 1:2], 4))

cat("\n\n===== Active category contributions (%) =====\n")
print(round(mca$var$contrib[, 1:2], 3))

```

```

cat("\n===== Active category cos2 =====\n")
print(round(mca$var$cos2[, 1:2], 3))

cat("\n===== Illustrative category coordinates (Dim 1–2) =====\n")
print(round(mca$quali.sup$coord[, 1:2], 4))

cat("\n===== Raw rownames of quali.sup coords (used for variable matching)
===== \n")
print(rownames(mca$quali.sup$coord))

# — 6. Build tidy coordinate frames —————
pct <- round(mca$eig[1:2, 2], 1)
x_lab <- paste0("Dimension 1 (", pct[1], "% of inertia)")
y_lab <- paste0("Dimension 2 (", pct[2], "% of inertia)")

## Active categories
act_df <- as.data.frame(mca$var$coord[, 1:2]) %>%
  setNames(c("Dim1", "Dim2")) %>%
  mutate(
    label = rownames(mca$var$coord),
    cos2_sum = rowSums(mca$var$cos2[, 1:2]),
    contrib = rowSums(mca$var$contrib[, 1:2])
  )

cat_to_var <- unlist(lapply(active_vars, function(v) {
  setNames(rep(v, nlevels(df_mca[[v]]), levels(df_mca[[v]]))
}))
act_df$variable <- cat_to_var[act_df$label]

## Illustrative categories
# FactoMineR prefixes supplementary category rownames with "VariableName_level"
# (e.g. "Country_Brazil"). We recover the variable name by matching the longest
# suppl_var name that the rowname starts with, which is robust to any
# level-name collisions and avoids NA from a direct lookup.
ill_df <- as.data.frame(mca$quali.sup$coord[, 1:2]) %>%
  setNames(c("Dim1", "Dim2")) %>%
  mutate(
    label = rownames(mca$quali.sup$coord),
    variable = sapply(label, function(lbl) {
      # Try prefix match "VarName_" first (FactoMineR default)
      matched <- suppl_vars[startsWith(lbl, paste0(suppl_vars, "_"))]
      if (length(matched) > 0) {
        # Pick the longest match to handle names like "Language_Family"
        return(matched[which.max(nchar(matched))])
      }
    })
  )

```

```

    }
    # Fallback: check if the label equals a known level across suppl vars
    for (v in suppl_vars) {
      if (lbl %in% levels(df_mca[[v]])) return(v)
    }
    return(NA_character_) # should not occur; flagged for inspection
  })
) %>%
filter(!is.na(variable)) # safety: drop any unmatched rows rather than show NA in legend

# Diagnostic: print the label → variable mapping to verify no NA slipped through
cat("\nIllustrative label -> variable mapping:\n")
print(ill_df %>% select(label, variable) %>% arrange(variable))

## Individual (row) scores
ind_df <- as.data.frame(mca$ind$coord[, 1:2]) %>%
  setNames(c("Dim1", "Dim2"))

# — 7. Colour palettes —————
# Active: linguistic error variables — warm tones
active_palette <- c(
  "Irregular_Verb_Type" = "#762a83", # purple
  "Error_Type"         = "#d6604d"  # red-orange
)

# Illustrative: learner background variables — cool tones
suppl_palette <- c(
  "Verb"           = "#1b7837", # green
  "Country"        = "#2166ac", # dark blue
  "Native_Language" = "#4393c3", # mid blue
  "Language_Family" = "#e08214", # orange
  "Years_English"  = "#b35806"  # amber
)

# — 8. Plot 1 – Active (error) categories only —————
p1 <- ggplot(act_df, aes(x = Dim1, y = Dim2, colour = variable)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey70", linewidth = 0.4) +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey70", linewidth = 0.4) +
  geom_point(aes(size = cos2_sum), alpha = 0.88) +
  geom_text_repel(
    aes(label = label),
    size      = 3.4,
    max.overlaps = 20,
    segment.color = "grey60",

```



```

    segment.size = 0.3,
    fontface     = "bold",
    box.padding  = 0.5,
    point.padding = 0.3
  ) +
  scale_colour_manual(
    values      = active_palette,
    name        = "Active Variable",
    na.translate = FALSE,
    labels      = c("Irregular_Verb_Type" = "Irregular Verb Type",
                    "Error_Type"         = "Error Type")
  ) +
  scale_size_continuous(range = c(2.5, 7), name = "Cos2 (Dim1+2)") +
  labs(
    title   = "MCA – Active Category Map (Error Space)",
    subtitle = "Axes optimised to separate error profiles · Point size ∝ cos2",
    x       = x_lab,
    y       = y_lab,
    caption = "Active variables: Irregular Verb Type (6 levels) · Error Type (3 levels)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title     = element_text(face = "bold", size = 14),
    plot.subtitle  = element_text(colour = "grey40"),
    legend.position = "right",
    panel.grid.minor = element_blank()
  )

ggsave("mca_plot1_active_error_space.pdf", p1, width = 10, height = 7)
ggsave("mca_plot1_active_error_space.png", p1, width = 10, height = 7, dpi = 300)
cat("\nPlot 1 saved.\n")

```

— 9. Plot 2 – Learner background projected onto error space —

This is the key plot for the research question:

"Which learner profiles cluster near which error types?"

```

p2 <- ggplot() +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey70", linewidth = 0.4) +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey70", linewidth = 0.4) +

  # Error categories greyed in background (reference geometry)
  geom_point(data = act_df,
    aes(x = Dim1, y = Dim2),
    colour = "grey65", size = 2.5, shape = 16, alpha = 0.55) +
  geom_text_repel(data = act_df,
    aes(x = Dim1, y = Dim2, label = label),

```

```

    colour    = "grey45",
    size       = 2.8,
    fontface   = "bold",
    max.overlaps = 15,
    segment.color = "grey80",
    segment.size = 0.2,
    box.padding = 0.4) +

# Learner background categories foregrounded as triangles
geom_point(data = ill_df,
  aes(x = Dim1, y = Dim2, colour = variable),
  shape = 17, size = 4, alpha = 0.9) +
geom_text_repel(data = ill_df,
  aes(x = Dim1, y = Dim2, label = label, colour = variable),
  size       = 3.4,
  fontface   = "italic",
  max.overlaps = 20,
  segment.color = "grey60",
  segment.size = 0.3,
  box.padding = 0.5,
  point.padding = 0.3) +

scale_colour_manual(values = suppl_palette, name = "Learner Background",
  na.translate = FALSE) +

labs(
  title   = "MCA – Learner Profiles Projected onto Error Space",
  subtitle = "▲ Learner background (illustrative) ● Error categories (active, greyed)",
  x       = x_lab,
  y       = y_lab,
  caption = paste0(
    "Proximity of a learner-background triangle to an error circle indicates association.\n",
    "Illustrative: Verb · Country · Native Language · Language Family · Years of English"
  )
) +
theme_minimal(base_size = 12) +
theme(
  plot.title    = element_text(face = "bold", size = 14),
  plot.subtitle = element_text(colour = "grey40"),
  legend.position = "right",
  panel.grid.minor = element_blank()
)

ggsave("mca_plot2_learner_profiles_on_error_space.pdf", p2, width = 11, height = 7.5)
ggsave("mca_plot2_learner_profiles_on_error_space.png", p2, width = 11, height = 7.5, dpi =
300)
cat("Plot 2 saved.\n")

```

```

# — 10. Plot 3 – Full combined map —————
p3 <- ggplot() +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey70", linewidth = 0.4) +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey70", linewidth = 0.4) +

  # Individual scores (open circles)
  geom_point(data = ind_df,
    aes(x = Dim1, y = Dim2),
    colour = "grey85", size = 1.8, shape = 1) +

  # Active error categories – filled circles, sized by cos²
  geom_point(data = act_df,
    aes(x = Dim1, y = Dim2, colour = variable, size = cos2_sum),
    shape = 16, alpha = 0.92) +
  geom_text_repel(data = act_df,
    aes(x = Dim1, y = Dim2, label = label, colour = variable),
    size = 3.3,
    fontface = "bold",
    max.overlaps = 25,
    segment.color = "grey60",
    segment.size = 0.3,
    box.padding = 0.5,
    point.padding = 0.3) +

  # Illustrative learner-background categories – filled triangles
  geom_point(data = ill_df,
    aes(x = Dim1, y = Dim2, fill = variable),
    colour = "grey25", shape = 24, size = 3.8, alpha = 0.88) +
  geom_text_repel(data = ill_df,
    aes(x = Dim1, y = Dim2, label = label, colour = variable),
    size = 3.0,
    fontface = "italic",
    max.overlaps = 25,
    segment.color = "grey70",
    segment.size = 0.25,
    box.padding = 0.4,
    point.padding = 0.3) +

  scale_colour_manual(
    values = c(active_palette, suppl_palette),
    name = "Variable",
    na.translate = FALSE # suppress any residual NA level from the legend
  ) +
  scale_fill_manual(values = suppl_palette, guide = "none") +

```

```

scale_size_continuous(range = c(2.5, 7), name = "Cos2 (active)") +
labs(
  title = "MCA – Full Symmetric Map",
  subtitle = "● Active error categories ▲ Illustrative learner-background categories ○ Individuals",
  x = x_lab,
  y = y_lab,
  caption = paste0(
    "Active (error space): Irregular Verb Type · Error Type\n",
    "Illustrative (learner profiles): Verb · Country · Native Language · Language Family · Years of English"
  )
) +
theme_minimal(base_size = 12) +
theme(
  plot.title = element_text(face = "bold", size = 14),
  plot.subtitle = element_text(colour = "grey40", size = 9),
  legend.position = "right",
  panel.grid.minor = element_blank()
)

ggsave("mca_plot3_full_map.pdf", p3, width = 12, height = 8)
ggsave("mca_plot3_full_map.png", p3, width = 12, height = 8, dpi = 300)
cat("Plot 3 saved.\n")

```

```

# — 11. Export numerical results —————
write_csv(
  act_df %>% select(label, variable, Dim1, Dim2, cos2_sum, contrib),
  "mca_active_coordinates.csv"
)
write_csv(
  ill_df %>% select(label, variable, Dim1, Dim2),
  "mca_illustrative_coordinates.csv"
)
write_csv(
  as.data.frame(mca$eig) %>%
    mutate(dim = paste0("Dim", row_number())) %>%
    select(dim, everything()),
  "mca_eigenvalues.csv"
)
cat("CSV files saved.\n")
cat("\nAll done! Check your working directory for the output files.\n")

print(p1)
print(p2)

```

```
print(p3)
```

Appendix E

Test 1 and Test 2 Analysis

This R script offers a full comparative statistical analysis of the dataset, focusing on the performance results from Test 1 (Mixed Formats) and Test 2 (Production). It builds a systematic analytical framework by labelling raw student responses into three main linguistic error categories: auxiliary error, incorrect participle error, and overgeneralization -ed error. The script includes data cleaning, frequency distribution analysis, and the generation of academic visualizations to illustrate error patterns. Additionally, it performs cross-tabulation of error types against demographic variables, such as native language, age, country, and duration of English instruction, and examines morphological stability through verb-type categorization. To identify statistically significant differences between the assessment types, the script also runs inferential statistics, such as the Chi-Square test, along with Cramer's V for effect size calculation and standardized Pearson residuals. The original code remains unchanged to ensure full transparency and reproducibility. For full reproducibility, this script and all associated data files are available in the supplementary online repository linked in Appendix A.

```
#  
=====
```

```
# Appendix E: Test 1 and Test 2 Analysis  
#  
=====
```

```
library(dplyr)  
library(stringr)  
  
# --- 1. PROCESS TEST 1 (Columns 5 to 29 contain X1 to X25) ---  
test1_cleaned <- test1_data  
  
# Automatically trim whitespace from all character columns  
test1_cleaned[] <- lapply(test1_cleaned, function(x) if(is.character(x)) trimws(x) else x)  
  
test1_cleaned <- test1_cleaned %>%  
  mutate(  
    # Multiple-Choice (X1 to X14) -> Columns 5 to 18  
    X1_err = case_when(.[[5]] == "maded" ~ "overgeneralized_ed", TRUE ~ "correct"),  
    X2_err = "correct",  
    X3_err = case_when(.[[7]] == "writed" ~ "overgeneralized_ed", TRUE ~ "correct"),  
    X4_err = "correct",  
    X5_err = case_when(.[[9]] %in% c("caughted", "caughted") ~ "overgeneralized_ed", TRUE  
    ~ "correct"),
```

```

X6_err = case_when(.[[10]] == "finded" ~ "overgeneralized_ed", TRUE ~ "correct"),
X7_err = "correct",
X8_err = case_when(.[[12]] == "have been" ~ "auxiliary_error", TRUE ~ "correct"),
X9_err = case_when(.[[13]] == "had have" ~ "auxiliary_error", TRUE ~ "correct"),
X10_err = case_when(.[[14]] == "has came" ~ "auxiliary_error", TRUE ~ "correct"),
X11_err = case_when(.[[15]] == "boughted" ~ "overgeneralized_ed", TRUE ~ "correct"),
X12_err = case_when(.[[16]] == "loosen" ~ "incorrect_participle", TRUE ~ "correct"),
X13_err = case_when(.[[17]] == "ate" ~ "incorrect_participle", TRUE ~ "correct"),
X14_err = case_when(.[[18]] == "saw" ~ "incorrect_participle", TRUE ~ "correct"),

# Grammaticality Judgment (X15 to X18) -> Columns 19 to 22
X15_err = case_when(.[[19]] == "Correct" ~ "overgeneralized_ed", TRUE ~ "correct"),
X16_err = case_when(.[[20]] == "Correct" ~ "incorrect_participle", TRUE ~ "correct"),
X17_err = case_when(.[[21]] == "Incorrect" ~ "incorrect_participle", TRUE ~ "correct"),
X18_err = case_when(.[[22]] == "Incorrect" ~ "incorrect_participle", TRUE ~ "correct"),

# Fill-in-the-Blanks (X19 to X25) -> Columns 23 to 29
X19_err = case_when(.[[23]] %in% c("Didn't sang", "Haven't sang", "Haven't sing") ~
"incorrect_participle", TRUE ~ "correct"),
X20_err = case_when(.[[24]] == "Didn't cutted" ~ "overgeneralized_ed", .[[24]] ==
"Haven't cut" ~ "auxiliary_error", TRUE ~ "correct"),
X21_err = case_when(.[[25]] == "Didn't putted" ~ "overgeneralized_ed", TRUE ~
"correct"),
X22_err = case_when(.[[26]] %in% c("Didn't payed", "Haven't payed") ~
"overgeneralized_ed", .[[26]] == "Didn't paid" ~ "incorrect_participle", TRUE ~ "correct"),
X23_err = case_when(.[[27]] == "Didn't thought" ~ "incorrect_participle", TRUE ~
"correct"),
X24_err = case_when(.[[28]] %in% c("Didn't took", "Haven't took", "Haven't take", "Didn't
taken") ~ "incorrect_participle", TRUE ~ "correct"),
X25_err = case_when(.[[29]] == "Hasn't driven" ~ "overgeneralized_ed", .[[29]] %in%
c("Didn't drove", "Hasn't drive", "Hasn't drove") ~ "incorrect_participle", .[[29]] == "Didn't
drive" ~ "auxiliary_error", TRUE ~ "correct")
)

# --- 2. PROCESS TEST 2 (Columns 5 to 24 contain X1 to X20) ---
test2_cleaned <- test2_data

test2_cleaned[] <- lapply(test2_cleaned, function(x) if(is.character(x)) trimws(x) else x)

test2_cleaned <- test2_cleaned %>%
  mutate(
    X1_err = "correct", # Column 5
    X2_err = case_when(str_detect(.[[6]], "payed") ~ "overgeneralized_ed", str_detect(.[[6]],
"paid") ~ "incorrect_participle", TRUE ~ "correct"),
    X3_err = case_when(str_detect(.[[7]], "Wroted") ~ "overgeneralized_ed", str_detect(.[[7]],
"Written") ~ "incorrect_participle", TRUE ~ "correct"),

```

```

X4_err = case_when(str_detect(.[[8]], "Goed") ~ "overgeneralized_ed", TRUE ~
"correct"),
X5_err = case_when(.[[9]] %in% c("Puted", "Putted") ~ "overgeneralized_ed", TRUE ~
"correct"),
X6_err = case_when(str_detect(.[[10]], "comed") ~ "overgeneralized_ed",
str_detect(.[[10]], "came") ~ "incorrect_participle", TRUE ~ "correct"),
X7_err = case_when(.[[11]] == "Been" ~ "incorrect_participle", .[[11]] == "Had been" ~
"auxiliary_error", TRUE ~ "correct"),
X8_err = case_when(.[[12]] == "Builded" ~ "overgeneralized_ed", TRUE ~ "correct"),
X9_err = case_when(str_detect(.[[13]], "sayed") ~ "overgeneralized_ed", str_detect(.[[13]],
"said") ~ "incorrect_participle", TRUE ~ "correct"),
X10_err = case_when(.[[14]] == "Chosed" ~ "overgeneralized_ed", .[[14]] == "Choose" ~
"incorrect_participle", TRUE ~ "correct"),
X11_err = case_when(.[[15]] %in% c("Has been", "Was") ~ "auxiliary_error", TRUE ~
"correct"),
X12_err = case_when(.[[16]] %in% c("Had fled", "Had flight", "Had fly", "Had flough",
"Had flew") ~ "incorrect_participle", .[[16]] %in% c("Flied", "Flyed") ~ "overgeneralized_ed",
. [[16]] %in% c("Flew", "Has flown", "Flow") ~ "auxiliary_error", TRUE ~ "correct"),
X13_err = case_when(.[[17]] %in% c("Had became", "Has became") ~
"incorrect_participle", .[[17]] %in% c("Has become", "Became") ~ "auxiliary_error", TRUE ~
"correct"),
X14_err = case_when(.[[18]] %in% c("Has broke", "Had broke") ~ "incorrect_participle",
. [[18]] == "Broke" ~ "auxiliary_error", TRUE ~ "correct"),
X15_err = case_when(.[[19]] %in% c("Haven't drank", "Hadn't drank", "Haven't drink",
"Hadn't drink", "Didn't drank") ~ "incorrect_participle", .[[19]] %in% c("Haven't drank",
"Didn't drink") ~ "auxiliary_error", TRUE ~ "correct"),
X16_err = case_when(.[[20]] == "Have understand" ~ "incorrect_participle", .[[20]] %in%
c("Understood", "Had understood") ~ "auxiliary_error", TRUE ~ "correct"),
X17_err = case_when(.[[21]] %in% c("Had rose", "Had rise", "Had rosen") ~
"incorrect_participle", .[[21]] %in% c("Has risen", "Had risen") ~ "overgeneralized_ed",
. [[21]] == "Has risen" ~ "auxiliary_error", TRUE ~ "correct"),
X18_err = case_when(.[[22]] %in% c("Haven't spoke", "Didn't spoken", "Didn't spoke") ~
"incorrect_participle", .[[22]] %in% c("Didn't speak", "Hadn't spoken") ~ "auxiliary_error",
TRUE ~ "correct"),
X19_err = case_when(.[[23]] == "Had forgot" ~ "incorrect_participle", .[[23]] %in%
c("Have forgotten", "Forgot", "Had forget") ~ "auxiliary_error", TRUE ~ "correct"),
X20_err = case_when(.[[24]] %in% c("Didn't had", "Hadn't have") ~ "incorrect_participle",
. [[24]] %in% c("Hasn't had", "Didn't have") ~ "auxiliary_error", TRUE ~ "correct")
)

```

```

# --- 3. VERIFICATION ---

```

```

cat("\n--- PIPELINE COMPLETED SUCCESSFULLY ---\n")
print(paste("Test 1 Cleaned Columns:", ncol(test1_cleaned)))
print(paste("Test 2 Cleaned Columns:", ncol(test2_cleaned)))

```



```

#
=====

# ERROR TYPE FREQUENCY CALCULATIONS
#
=====

library(tidyr)

cat("\n=====\\n")
cat("  TEST 1: GLOBAL ERROR DISTRIBUTION  ")
cat("\n=====\\n")

test1_cleaned %>%
  select(ends_with("_err")) %>%
  pivot_longer(cols = everything(), names_to = "Question", values_to = "Error_Type") %>%
  filter(Error_Type != "correct") %>%
  group_by(Error_Type) %>%
  summarise(
    Count = n(),
    Percentage = round((n() / (nrow(test1_cleaned) * 25)) * 100, 2)
  ) %>%
  arrange(desc(Count)) %>%
  print()

cat("\n=====\\n")
cat("  TEST 2: GLOBAL ERROR DISTRIBUTION  ")
cat("\n=====\\n")

test2_cleaned %>%
  select(ends_with("_err")) %>%
  pivot_longer(cols = everything(), names_to = "Question", values_to = "Error_Type") %>%
  filter(Error_Type != "correct") %>%
  group_by(Error_Type) %>%
  summarise(
    Count = n(),
    Percentage = round((n() / (nrow(test2_cleaned) * 20)) * 100, 2)
  ) %>%
  arrange(desc(Count)) %>%
  print()

#
=====
=====

```

GLOBAL ERROR VISUALIZATION

#

```
library(ggplot2)
```

```
library(tidyr)
```

```
library(dplyr)
```

1. Prepare and combine the summary data

```
t1_summary <- test1_cleaned %>%
```

```
  select(ends_with("_err")) %>%
```

```
  pivot_longer(cols = everything(), values_to = "Error_Type") %>%
```

```
  filter(Error_Type != "correct") %>%
```

```
  group_by(Error_Type) %>%
```

```
  summarise(Count = n(), Total_Tasks = nrow(test1_cleaned) * 25) %>%
```

```
  mutate(Percentage = (Count / Total_Tasks) * 100, Test = "Test 1 (Mixed Formats)")
```

```
t2_summary <- test2_cleaned %>%
```

```
  select(ends_with("_err")) %>%
```

```
  pivot_longer(cols = everything(), values_to = "Error_Type") %>%
```

```
  filter(Error_Type != "correct") %>%
```

```
  group_by(Error_Type) %>%
```

```
  summarise(Count = n(), Total_Tasks = nrow(test2_cleaned) * 20) %>%
```

```
  mutate(Percentage = (Count / Total_Tasks) * 100, Test = "Test 2 (Production)")
```

```
combined_summary <- bind_rows(t1_summary, t2_summary)
```

2. Create the Academic Bar Plot

```
error_plot <- ggplot(combined_summary, aes(x = Error_Type, y = Percentage, fill = Test)) +
```

```
  geom_bar(stat = "identity", position = position_dodge(0.8), width = 0.7, color = "black", size
```

```
  = 0.3) +
```

```
  geom_text(aes(label = paste0(round(Percentage, 1), "%")),
```

```
    position = position_dodge(0.8), vjust = -0.5, size = 3.5, fontface = "bold") +
```

```
  scale_fill_manual(values = c("Test 1 (Mixed Formats)" = "#4A6572", "Test 2 (Production)" =
```

```
  "#F9AA33")) +
```

```
  scale_x_discrete(labels = c("auxiliary_error" = "Auxiliary Error",
```

```
    "incorrect_participle" = "Incorrect Participle",
```

```
    "overgeneralized_ed" = "Overgeneralized -ed")) +
```

```
  labs(
```

```
    x = "Linguistic Error Categories",
```

```
    y = "Global Error Rate (%)",
```

```
    fill = "Assessment Type"
```

```
  ) +
```

```
  theme_minimal(base_family = "sans") +
```

```
  theme(
```

```

axis.title.x = element_text(face = "bold", size = 12, margin = margin(t = 12)),
axis.title.y = element_text(face = "bold", size = 12, margin = margin(r = 12)),
axis.text = element_text(size = 10, color = "black"),
legend.position = "top",
legend.title = element_text(face = "bold", size = 10),
panel.grid.major.x = element_blank(),
panel.grid.minor = element_blank(),
plot.margin = margin(15, 15, 15, 15)
) +
ylim(0, 25)

# Display the plot
print(error_plot)

# Save the plot in high definition for the thesis document
# ggplot2 will save it directly to your current working directory
ggsave("global_error_distribution_plot.png", plot = error_plot, width = 8, height = 5, dpi =
300)

#
=====

=====
# QUESTION-LEVEL ERROR RANKINGS
#
=====

=====

cat("\n=====\\n")
cat("  TEST 1: TOP 5 MOST ERROR-PRONE QUESTIONS ")
cat("\n=====\\n")

test1_cleaned %>%
  select(ends_with("_err")) %>%
  pivot_longer(cols = everything(), names_to = "Question", values_to = "Error_Type") %>%
  filter(Error_Type != "correct") %>%
  group_by(Question, Error_Type) %>%
  summarise(Count = n(), .groups = 'drop') %>%
  arrange(desc(Count)) %>%
  head(5) %>%
  print()

cat("\n=====\\n")
cat("  TEST 2: TOP 5 MOST ERROR-PRONE QUESTIONS ")
cat("\n=====\\n")

```

```

test2_cleaned %>%
  select(ends_with("_err")) %>%
  pivot_longer(cols = everything(), names_to = "Question", values_to = "Error_Type") %>%
  filter(Error_Type != "correct") %>%
  group_by(Question, Error_Type) %>%
  summarise(Count = n(), .groups = 'drop') %>%
  arrange(desc(Count)) %>%
  head(5) %>%
  print()

#
=====

# EMPIRICAL ANALYSIS: PERFORMANCE BY NATIVE LANGUAGE (TEST 1 & TEST
2)
#
=====

library(dplyr)
library(tidyr)
library(ggplot2)

# 1. PREPARE UNIFIED DATA (Native Language)
lang_data_master <- bind_rows(
  test1_cleaned %>% select(Native_Language, ends_with("_err")) %>%
    pivot_longer(ends_with("_err"), values_to = "Response_Type") %>% mutate(Test = "Test
1"),
  test2_cleaned %>% select(Native_Language, ends_with("_err")) %>%
    pivot_longer(ends_with("_err"), values_to = "Response_Type") %>% mutate(Test = "Test
2")
)

# 2. CONSOLE OUTPUT: APA 7 READY TABLES
cat("\n--- TABLE: PERFORMANCE BY NATIVE LANGUAGE (TEST 1 & 2) ---\n")
lang_data_master %>%
  group_by(Test, Native_Language, Response_Type) %>%
  summarise(Count = n(), .groups = 'drop') %>%
  pivot_wider(names_from = Response_Type, values_from = Count, values_fill = 0) %>%
  print()

# 3. VISUALIZATION: PERFORMANCE BY NATIVE LANGUAGE
ggplot(lang_data_master, aes(x = Response_Type, fill = Native_Language)) +
  geom_bar(position = "dodge") +
  facet_wrap(~Test) +

```

```

theme_minimal() +
labs(title = "Performance Distribution by Native Language",
      subtitle = "Comparative Analysis (Errors & Correct Answers)",
      x = "Response Category",
      y = "Frequency",
      fill = "Native Language") +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

#
=====

# EMPIRICAL CROSS-TABULATION FOR AGE GROUPS (TEST 1 & TEST 2)
#
=====

cat("\n=====\\n")
cat("  TEST 1: ERROR TYPES BY AGE")
cat("\n=====\\n")

test1_cleaned %>%
  select(Age, ends_with("_err")) %>%
  pivot_longer(cols = ends_with("_err"), names_to = "Question", values_to = "Error_Type")
%>%
  filter(Error_Type != "correct") %>%
  group_by(Age, Error_Type) %>%
  summarise(Error_Count = n(), .groups = 'drop') %>%
  pivot_wider(names_from = Error_Type, values_from = Error_Count, values_fill = 0) %>%
  print()

cat("\n=====\\n")
cat("  TEST 2: ERROR TYPES BY AGE")
cat("\n=====\\n")

test2_cleaned %>%
  select(Age, ends_with("_err")) %>%
  pivot_longer(cols = ends_with("_err"), names_to = "Question", values_to = "Error_Type")
%>%
  filter(Error_Type != "correct") %>%
  group_by(Age, Error_Type) %>%
  summarise(Error_Count = n(), .groups = 'drop') %>%
  pivot_wider(names_from = Error_Type, values_from = Error_Count, values_fill = 0) %>%
  print()

```

```

#
=====

# EMPIRICAL CROSS-TABULATION FOR COUNTRIES (TEST 1 & TEST 2)
#
=====

library(dplyr)
library(tidyr)
library(ggplot2)

# 1. PREPARE UNIFIED DATA (All responses: Errors + Correct)
master_data <- bind_rows(
  test1_cleaned %>% select(Country, ends_with("_err")) %>%
    pivot_longer(ends_with("_err"), values_to = "Response_Type") %>% mutate(Test = "Test
1"),
  test2_cleaned %>% select(Country, ends_with("_err")) %>%
    pivot_longer(ends_with("_err"), values_to = "Response_Type") %>% mutate(Test = "Test
2")
)

# 2. GENERATE SINGLE MASTER GRAPH
ggplot(master_data, aes(x = Response_Type, fill = Country)) +
  geom_bar(position = "dodge") +
  facet_wrap(~Test) +
  theme_minimal() +
  labs(title = "Performance Distribution by Country (Errors & Correct)",
    x = "Response Category",
    y = "Frequency",
    fill = "Country") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# 3. CONSOLE OUTPUT (APA 7 Table Data)
cat("\n--- SUMMARY TABLE FOR APA 7 (MASTER DATA) ---\n")
master_data %>%
  group_by(Test, Country, Response_Type) %>%
  summarise(Count = n(), .groups = 'drop') %>%
  pivot_wider(names_from = Response_Type, values_from = Count, values_fill = 0) %>%
  print()

#
=====

# PERFORMANCE BY YEARS OF ENGLISH (0-10 & 11-14 YEARS)

```

```

#
=====

library(dplyr)
library(tidyr)
library(ggplot2)

# 1. PREPARE UNIFIED DATA AND APPLY CATEGORICAL GROUPING
# We clean 'Years_English' (remove "years" text) to ensure accurate grouping
master_data_years <- bind_rows(
  test1_cleaned %>% select(Years_English, ends_with("_err")) %>%
    pivot_longer(ends_with("_err"), values_to = "Response_Type") %>% mutate(Test = "Test
1"),
  test2_cleaned %>% select(Years_English, ends_with("_err")) %>%
    pivot_longer(ends_with("_err"), values_to = "Response_Type") %>% mutate(Test = "Test
2")
) %>%
# Clean text to numeric: "13 years" becomes 13
mutate(Years_English = as.numeric(gsub("[^0-9]", "", Years_English))) %>%
mutate(Years_Group = case_when(
  Years_English <= 10 ~ "0-10 Years",
  Years_English > 10 ~ "11-14 Years",
  TRUE ~ "Unknown"
))

# 2. CONSOLE OUTPUT: APA 7 READY TABLES
cat("\n--- TABLE: PERFORMANCE BY YEARS OF ENGLISH (GROUPED) ---\n")
master_data_years %>%
  group_by(Test, Years_Group, Response_Type) %>%
  summarise(Count = n(), .groups = 'drop') %>%
  pivot_wider(names_from = Response_Type, values_from = Count, values_fill = 0) %>%
  print()

# 3. VISUALIZATION: PERFORMANCE BY YEARS OF ENGLISH (GROUPED)
ggplot(master_data_years, aes(x = Response_Type, fill = Years_Group)) +
  geom_bar(position = "dodge") +
  facet_wrap(~Test) +
  theme_minimal() +
  labs(title = "Performance Distribution by Years of English Instruction",
    subtitle = "Grouped Analysis (0-10 vs 11-14 Years)",
    x = "Response Category",
    y = "Frequency",
    fill = "Years of English") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

```

#
=====

# PARTICIPANT-LEVEL ERROR SUMMARY
#
=====

cat("\n=====\\n")
cat("  TEST 1: AVERAGE ERRORS PER PARTICIPANT")
cat("\n=====\\n")

test1_cleaned %>%
  mutate(Total_Errors = rowSums(select(., ends_with("_err")) != "correct")) %>%
  summarise(
    Average_Errors = mean(Total_Errors),
    Median_Errors = median(Total_Errors),
    Max_Errors_by_Single_Student = max(Total_Errors)
  ) %>%
  print()

cat("\n=====\\n")
cat("  TEST 2: AVERAGE ERRORS PER PARTICIPANT")
cat("\n=====\\n")

test2_cleaned %>%
  mutate(Total_Errors = rowSums(select(., ends_with("_err")) != "correct")) %>%
  summarise(
    Average_Errors = mean(Total_Errors),
    Median_Errors = median(Total_Errors),
    Max_Errors_by_Single_Student = max(Total_Errors)
  ) %>%
  print()

#
=====

# MORPHOLOGICAL ERROR DISTRIBUTION
#
=====

library(dplyr)
library(tidyr)

```



```

library(ggplot2)

# 1. PROCESS FUNCTION (Gathers only the actual processed error columns)
process_test_corrected <- function(cleaned_df, test_name, mapping_logic) {
  cleaned_df %>%
    # Select only the error recoded columns
    select(ends_with("_err")) %>%
    # Convert to long format
    pivot_longer(cols = everything(), names_to = "Q_Num", values_to = "Status") %>%
    # Create an index for mapping (1 to 25 for T1, 1 to 20 for T2)
    group_by(Q_Num) %>%
    mutate(Q_Idx = as.numeric(str_extract(Q_Num, "\\d+"))) %>%
    ungroup() %>%
    # Map to morphological type
    mutate(verb_type = mapping_logic(Q_Idx)) %>%
    mutate(Test = test_name)
}

# 2. MAPPING LOGICS (Same as your original logic)
map_t1 <- function(idx) {
  case_when(
    idx %in% c(1,4,5,9,11,12,22,23) ~ "Weak",
    idx %in% c(2,3,6,7,10,13,14,15,16,17,18,19,24,25) ~ "Strong",
    idx %in% c(20,21) ~ "Invariable",
    idx %in% c(8) ~ "Suppletive",
    TRUE ~ "Strong"
  )
}

map_t2 <- function(idx) {
  case_when(
    idx %in% c(8) ~ "Weak",
    idx %in% c(1,2,3,6,9,10,12,13,14,15,16,17) ~ "Strong",
    idx %in% c(5) ~ "Invariable",
    idx %in% c(4, 7, 11) ~ "Suppletive",
    TRUE ~ "Strong"
  )
}

# 3. RUN UPDATED PIPELINE
all_data_corrected <- bind_rows(
  process_test_corrected(test1_cleaned, "Test 1", map_t1),
  process_test_corrected(test2_cleaned, "Test 2", map_t2)
)

# 4. GENERATE TRUE ERROR TABLE

```

```

# This will count ONLY actual errors (excluding "correct" responses)
table_data_output <- all_data_corrected %>%
  filter(Status != "correct") %>% # Filters out the real correct answers
  group_by(verb_type, Test) %>%
  summarise(Error_Count = n(), .groups = 'drop') %>%
  pivot_wider(names_from = Test, values_from = Error_Count, values_fill = 0) %>%
  mutate(verb_type = factor(verb_type, levels = c("Weak", "Strong", "Suppletive",
"InvARIABLE"))) %>%
  arrange(verb_type)

cat("\n--- TRUE MORPHOLOGICAL ERROR DISTRIBUTION TABLE ---\n")
print(table_data_output)

# 5. PLOT TRUE ERROR DATA
plot_data_corrected <- all_data_corrected %>%
  filter(Status != "correct")

ggplot(plot_data_corrected, aes(x = factor(verb_type, levels=c("Weak", "Strong",
"Suppletive", "InvARIABLE")),
  fill = Test)) +
  geom_bar(position = "dodge", color = "black", size = 0.3) +
  theme_minimal() +
  labs(title = "True Error Frequency by Morphological Verb Type",
    x = "Morphological Verb Type", y = "Actual Error Count")

#
=====
=====
# INDEPENDENT CHI-SQUARE FOR TEST 1 VS TEST 2
#
=====
=====

# Adjusted matrix: 3 rows (Error Types) x 2 columns (Test 1 and Test 2)
global_error_matrix <- matrix(
  c(15, 81, # Auxiliary Error
    77, 70, # Incorrect Participle
    22, 18), # Overgeneralized -ed
  nrow = 3, byrow = TRUE,
  dimnames = list(c("Auxiliary", "Participle", "Overgeneralized"), c("Test1", "Test2")))
)

cat("\n===== \n")
cat(" VALID INDEPENDENT CHI-SQUARE (TEST 1 VS TEST 2)")
cat("\n===== \n")

```

```

# Run the Chi-Square test and store the result
chi2_result <- chisq.test(global_error_matrix)
print(chi2_result)

#
=====

# CRAMER'S V (EFFECT SIZE) CALCULATION
#
=====

chi2_stat <- chi2_result$statistic      # Chi-squared statistic value
n_total  <- sum(global_error_matrix)    # Total sample size (N)
min_dim  <- min(nrow(global_error_matrix) - 1, ncol(global_error_matrix) - 1) # min(r-1,
c-1)

# Cramer's V Formula
cramers_v <- sqrt(chi2_stat / (n_total * min_dim))

cat("\n=====\\n")
cat("  CRAMER'S V (EFFECT SIZE)")
cat("\n=====\\n")
cat("Cramer's V:", round(cramers_v, 4), "\\n")

# Interpretation based on df = 1 (min(r-1, c-1) = 1)
cat("Interpretation (df = 1): ")
if (cramers_v < 0.1) {
  cat("Negligible / Very small effect\\n")
} else if (cramers_v < 0.3) {
  cat("Small effect\\n")
} else if (cramers_v < 0.5) {
  cat("Medium effect\\n")
} else {
  cat("Large effect\\n")
}

#
=====

# STANDARDIZED PEARSON RESIDUALS
#
=====

```

```

cat("\n=====\\n")
cat("  STANDARDIZED PEARSON RESIDUALS")
cat("\n=====\\n")

# Extract standardized residuals from the chi2_result object
std_residuals <- chi2_result$stdres
print(round(std_residuals, 4))

cat("\nNote: Residuals > 2 or < -2 indicate statistically significant deviations\\n")
cat("from the expected frequencies (alpha = 0.05).\\n")

# install.packages("vcd") # Commented out to prevent reinstalling every time

# Load necessary library for mosaic plots
library(vcd)

# 1. Define the matrix with named dimensions to get the correct axis titles
global_error_matrix <- matrix(
  c(15, 81, # Auxiliary Error
    77, 70, # Incorrect Participle
    22, 18), # Overgeneralized -ed
  nrow = 3, byrow = TRUE,
  dimnames = list(
    Error_Type = c("Auxiliary", "Participle", "Overgeneralized"), # Named 'Error_Type' instead
    of just a list
    Test = c("Test 1", "Test 2") # Named 'Test' instead of just a list
  )
)

# 2. Create the Mosaic Plot
mosaic(global_error_matrix,
  shade = TRUE,
  legend = TRUE,
  main = "Mosaic Plot of Error Distribution (Test 1 vs Test 2)",
  sub = "Chi-square Analysis of Linguistic Error Categories")

```

Appendix F

Test 1 Questions

Irregular Verbs Usage Test 1

19/05/2026, 00:14

Irregular Verbs Usage Test 1

Hello, my name is Burak COLAK. I am a second-year (M2) Master's student in Sciences du Langage at Bordeaux Montaigne University.

I have designed this test to support my thesis titled "Irregular Verb Errors Among International Students," specifically for students in the course L1 - 2LILAM222 MATI - ORAL.

This test consists of 25 questions focusing on the use of irregular verbs in their V2 (Past Simple) and V3 (Past Participle) forms.

This test will take approximately 10 minutes to complete. Please answer all questions individually and do not use any external help.

Thank you for your participation.

* Indicates required question

Student Background

Please answer the following questions briefly using your real personal information. Your responses will remain anonymous and will only be used for research purposes.

1. How old are you? *

2. Which country were you born in? *

3. What is your native language? (It could be more than one language) *

4. How long have you been learning English at school? *

SECTION 1: Past Simple

This section tests your knowledge of the simple past tense of irregular verbs. You need to choose the **correct past form of the verb** to complete each sentence.

5. 1. Yesterday, I ____ a very important decision. *

4 pc

Mark only one oval.

- ☐ made
- ☐ maded
- ☐ didn't made
- ☐ have make

6. 2. I ____ my assignment last night. *

4 pc

Mark only one oval.

- ☐ doed
- ☐ did
- ☐ didt
- ☐ didn't did

7. 3. She ____ a letter to her friend. *

4 pc

Mark only one oval.

- ☐ writed
☐ written
☐ wrote
☐ wroted

8. 4. They ____ a new house. *

4 pc

Mark only one oval.

- ☐ built
☐ buildied
☐ builted
☐ didn't built

9. 5. He ____ the ball very rapidly. *

4 pc

Mark only one oval.

- ☐ catchied
☐ caught
☐ caughtied
☐ didn't caught

10. 6. He ____ the answer quickly. *

4 pc

Mark only one oval.

- ☐ has find
- ☐ didn't found
- ☐ found
- ☐ finded

11. 7. She ____ a beautiful picture. *

4 pc

Mark only one oval.

- ☐ drewed
- ☐ drew
- ☐ drawn
- ☐ didn't drewed

12. 8. They ____ kind to me. *

4 pc

Mark only one oval.

- ☐ beed
- ☐ have been
- ☐ didn't be
- ☐ were

13. 9. We ____ a great time yesterday. *

4 pc

Mark only one oval.

- ☐ haved
- ☐ had have
- ☐ had
- ☐ didn't had

14. 10. He ____ very early from school this morning.

4 pc

Mark only one oval.

- ☐ has came
- ☐ comed
- ☐ came
- ☐ didn't came

SECTION 2: Past Participle

This section focuses on **past participles**. You must select the correct past participle form of each verb

15. 11. I have ____ a new phone. *

4 pc

Mark only one oval.

- ☐ buyed
- ☐ boughted
- ☐ bought
- ☐ had bought

16. 12. She has ____ her keys. *

4 pc

Mark only one oval.

- ☐ losed
- ☐ lost
- ☐ losted
- ☐ losen

17. 13. They have already ____ the dinner. *

4 pc

Mark only one oval.

- ☐ eaten
- ☐ ated
- ☐ ate
- ☐ eated

18. 14. He has ____ the movie before. *

4 pc

Mark only one oval.

- ☐ seed
- ☐ saw
- ☐ seen
- ☐ had seen

SECTION 3: Error Recognition

At this section, you need to decide whether the past participles used at each sentence is **correct** or **incorrect**.

19. 15. She has begun the task. *

4 pc

Mark only one oval.

- ☐ Correct
☐ Incorrect

20. 16. I swum yesterday. *

4 pc

Mark only one oval.

- ☐ Correct
☐ Incorrect

21. 17. He has broken the window. *

4 pc

Mark only one oval.

- ☐ Correct
☐ Incorrect

22. 18. We have never flown to London. *

4 pc

Mark only one oval.

- ☐ Correct
☐ Incorrect

SECTION 4: Production

Write the correct negative form of the verb in parentheses. Some sentences require the **past simple negative**, while others require the **past participle negative**.

23. 19. We ____ (sing) the song just now. * 4 pc

24. 20. Last month, I ____ (cut) my finger while chopping vegetables. * 4 pc

25. 21. Recently, she ____ (put) her focus on her studies. 4 pc

26. 22. Two days ago, we ____ (pay) for the tickets because the concert was free. 4 pc

27. 23. Last week, he ____ (think) the answer was correct. 4 pc

28. 24. Yesterday, I ____ (take) the bus. 4 pc

29. 25. He ____ (drive) the car yet. * 4 pc

Thank You!

Thank you for participating and contributing to the findings section of my thesis.

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Appendix F.5

Test 2 Questions

Irregular Verbs Usage Test 2

19/05/2026, 00:08

Irregular Verbs Usage Test 2

Hello, my name is Burak COLAK. I am a second-year (M2) Master's student in Sciences du Langage at Bordeaux Montaigne University.

I have designed this second test to support my thesis titled "Irregular Verb Errors Among International Students," specifically for students in the course L1 - 2LILAM222 MATI - ORAL.

This test consists of 20 questions focusing on the use of irregular verbs in their V2 (Past Simple) and V3 (Past Participle) forms.

This test will also take approximately 10 minutes to complete. Please answer all questions individually without using any external help.

Thank you for your participation.

* Indicates required question

Student Background

Please answer the following questions briefly using your real personal information. Your responses will remain anonymous and will only be used for research purposes.

1. How old are you? *

2. Which country were you born in? *

3. What is your native language? (It could be more than one language) *

https://docs.google.com/forms/u/0/d/1qVzb19RK4fDKUPe5OS5swyFg_X3UYiZeKkAputqvIJE/printform

Page 1 of 6

4. How long have you been learning English at school? *

SECTION 1: Past Simple (V2)

Fill in the blanks with the correct **Past Simple (V2)** form of the verb in parentheses. Pay close attention to the context to decide whether a **positive** or **negative** form is required.

5. 1. Yesterday, I _____ (**see**) a very interesting documentary about linguistics on TV. * 5 pc

6. 2. We _____ (**not / pay**) for the parking ticket yesterday because the machine was broken. * 5 pc

7. 3. The student _____ (**write**) her entire essay in just two hours last night. * 5 pc

8. 4. My friends _____ (**go**) to the cinema last Friday, but I stayed at home. * 5 pc

9. 5. Last week, the professor _____ (**put**) the new reading list on the university portal. 5 pc

10. 6. They _____ (**not / come**) to the party last night because they were feeling * 5 pc
ill.

11. 7. I _____ (**be**) very surprised when I heard the news yesterday morning. * 5 pc

12. 8. Two days ago, the technician _____ (**build**) a new server for the * 5 pc
department.

13. 9. He _____ (**not / say**) anything when the teacher asked about the error in * 5 pc
class.

14. 10. For his final project, he _____ (**choose**) a very difficult topic last month. * 5 pc

SECTION 2: Perfect Tenses (V3 - Present & Past Perfect)

Fill in the blanks with the correct **Past Participle (V3)** form of the verb in parentheses. Use **Present Perfect (have/has + V3)** or **Past Perfect (had + V3)** based on the time markers.

15. 11. Alicia was happy because she _____ (**be**) to that museum many times * 5 pc
before.

16. 12. The pilot was confident because he _____ (**fly**) that specific route before. * 5 pc

17. 13. By the time Katy graduated, she _____ (**become**) fluent in three languages. * 5 pc

18. 14. Look! Someone _____ (**break**) the window of the classroom. * 5 pc

19. 15. We were very thirsty because we _____ (**not / drink**) any water since the morning. * 5 pc

20. 16. All of the students _____ (**understand**) all the grammar rules we studied so far. * 5 pc

21. 17. The sun _____ already _____ (**rise**) by the time we started our morning walk. * 5 pc

22. 18. I _____ (**not / speak**) to my coordinator about the Erasmus application yet. * 5 pc

23. 19. By the time I reached the airport, I realized I _____ (**forget**) my passport. * 5 pc

24. 20. He was tired because he _____ (**not / have**) any rest during the whole day. * 5 pc

Thank You!

Thank you for participating and contributing to the findings section of my thesis.

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Google Forms

Appendix G

Research Consent Form & Reference Letter by Alison May Lagarde

RESEARCH CONSENT FORM

This study is a research project titled "**Irregular Verb Errors Among International Students – ICLE Corpus Based Study**" and aims to **identify error analysis in irregular verb usage among L1 students**.

The study is conducted by **Burak Colak**, an **M2 student in Sciences du Langage at Bordeaux Montaigne University**, under the supervision of **Prof. Guillaume Desagulier**, and its results will provide insights into the **linguistic development and error patterns of students**.

- **Voluntary Participation:** Your participation in this study is based on a voluntary basis.
- **Purpose and Method:** In line with the purpose of the study, participants will be asked to complete two **Google Form questionnaires**; data will be collected in **written form** through **multiple-choice** and **short-answer** questions designed to assess grammar proficiency.
- **Confidentiality:** You do not have to provide any information that would reveal your identity; participants' names will be kept strictly confidential.
- **Data Usage:** Data will only be used for scientific purposes (Methodology and Findings sections of the thesis) and will not be shared without permission.
- **Data Protection:** Data will be protected through **anonymization**. Only **age, country of birth, native language, total years of English study**, and linguistic errors will be processed as data. Data will be stored for 5 years and then destroyed.
- **Withdrawal:** You may leave the study at any time if you feel uncomfortable. In such a case, your data will be destroyed.

You may direct your questions to **Burak Colak** from the **Sciences du Langage** department at **Bordeaux Montaigne University** via his official university email.

Researcher Name: Burak Colak

Email: burak.colak@etu.u-bordeaux-montaigne.fr

Consent Statement: I agree to the use of the information I provide for scientific purposes, knowing that I participate in this study entirely of my own free will and that I can leave the study at any time.

Date:

Participant Name/Surname:

Signature:

Written Reference for Burak Colak (Alison May Lagarde)

During Semester 2, Burak was involved in the teaching of LCE1 Langue Orale classes and demonstrated a strong commitment to both his academic and professional development. As part of his Master's research project, he conducted a study involving student questionnaires. He approached this work thoughtfully and professionally, ensuring that students were well informed and comfortable throughout the process.

Alongside his research activities, Burak gained valuable teaching experience within the classroom. He initially supported lessons through the delivery of starter activities before gradually taking responsibility for planning and teaching entire sessions. This progression reflected both his growing confidence and his ability to apply feedback constructively in order to develop his teaching practice.

Burak demonstrated strong communication and interpersonal skills throughout his work with students. He was able to create a positive and supportive learning environment in which students felt comfortable participating in discussions and expressing their ideas. His lessons encouraged active engagement, and students responded positively to the range of topics he introduced and facilitated. He showed an ability to adapt his teaching approach to the needs of the group, maintaining student interest while encouraging meaningful interaction in the target language.

Throughout his time working with me, Burak consistently displayed professionalism and reliability. He communicated effectively with me as a colleague, was well prepared for his teaching responsibilities, and approached new challenges with enthusiasm and a willingness to learn. I am confident that Burak possesses the academic ability, professional skills, and personal qualities necessary to succeed in future employment or further study.

Bordeaux 17th June 2026

Alison May Lagarde

Signature 